Collectively Big, Locally Small

Formalizing Small Data in Healthcare and Learning Privately Across Siloed Hospitals

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Health systems produce data at an enormous rate, yet most clinical learning problems are small. A rare disease registry holds a few hundred patients, an intensive care unit sees a handful of the outcome it wants to predict, a regional hospital owns a fraction of the cases needed to validate a model, and the data that would close the gap sit in other institutions that cannot legally or practically share them. "Small data" is invoked constantly in the literature but is rarely defined in a way that can be measured, compared, or reasoned about. This survey and position paper does four things. First, it gives a task-relative formal definition of small data through an effective sample size and a smallness index that combine learning-theoretic sample complexity with the clinical sample-size criteria used in prediction modeling. Second, it introduces a new two-axis taxonomy. The first axis classifies ten forms of scarcity in healthcare by the term of the smallness index they degrade, grouped into ontic, epistemic, temporal-contextual and custodial scarcity. The second axis classifies every remedy by the provenance of the information it injects (pooling, proxy, prior, unlabeled data, procurement, re-expression), and we prove that re-expression, which includes augmentation and synthetic data, cannot add information about the target that was not already present in the data or in the generator. Third, we code a corpus of 290 works on both axes, check it against a systematic search of 5,204 database records reported to PRISMA 2020, and use it to review the solution families across imaging, electronic health records, omics and tabular data, to summarize the quantitative evidence, and to locate where it is thin. Only 18 of the 290 works are multi-institution studies, all of them on pooling, and temporal obsolescence and measurement poverty are barely addressed by any channel, in the corpus and in the wider literature alike. Fourth, for sparse networks of hospitals that operate as data silos, we derive a privacy-preserving hierarchical borrowing scheme that runs by gossip over the existing collaboration graph without a central server. A companion paper turns it into a method that, without tuning, was more accurate than federated and personalized learning tuned on other networks on cancer networks held out from its development, and on a public benchmark the most accurate in some settings but not in others. We show that, in expectation over sites, it adds nearly the same number of "virtual patients", $\sigma^2/(\tau^2 + M)$, to every site, so that the smallest hospitals gain the most, that differential privacy enters only through the population-level estimate, and that the gossip phase incurs no additional privacy cost, and we summarize eighteen simulations and a first test on real data from four hospitals, reported in full in a companion paper, against one-shot and iterative federated baselines and under adversarial conditions. We close with a research agenda.

CCS Concepts: • **Computing methodologies** → **Machine learning**; • **Applied computing** → **Health informatics**; • **Security and privacy** → *Privacy protections.*

Additional Key Words and Phrases: small data, healthcare, effective sample size, taxonomy, federated learning, decentralized learning, differential privacy, hierarchical Bayes, data silos, rare diseases, foundation models, synthetic data

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1 INTRODUCTION

The standard story of machine learning in medicine is a story about scale. Large labeled collections of retinal photographs, skin lesions and chest radiographs produced models that matched specialists on narrow tasks [Rajpurkar et al. 2022; Topol 2019], and the broader field learned that performance tends to grow predictably with data [Halevy et al. 2009; Hestness et al. 2017; Kaplan et al. 2020; Sun et al. 2017]. The story is accurate for the problems it describes. It is not representative of the problems most clinicians and health systems actually face.

Consider what a typical clinical question looks like from the inside. Rare diseases are individually rare but collectively common: the Orphanet database lists more than six thousand of them, affecting an estimated 3.5% to 5.9% of the population [Nguengang Wakap et al. 2020], and most have cohorts of tens or hundreds of patients worldwide [Banerjee et al. 2023; Schaefer et al. 2020]. Prediction models in common conditions are routinely built on a few dozen outcome events, which is why the clinical literature has spent three decades arguing over events per variable [Peduzzi et al. 1996; Riley et al. 2019b; van Smeden et al. 2019]. During the first months of COVID-19, hundreds of models were developed from small, local, convenience samples, and systematic reviews found nearly all of them to be at high risk of bias and none fit for clinical use [Roberts et al. 2021; Wynants et al. 2020]. Even large systems turn small at the point of deployment: a proprietary sepsis model used in hundreds of hospitals achieved an area under the curve of 0.63 on external validation and missed two thirds of sepsis cases [Wong et al. 2021], and convolutional networks for pneumonia learned to recognize the hospital that produced an image rather than the disease [Zech et al. 2018].

These failures share a structure. Health data are abundant in aggregate and scarce where they are needed: for the specific outcome, in the specific subgroup, at the specific site, in the current period, with labels. They are also fragmented. Every hospital holds a slice of the population under its own legal, ethical and technical constraints [Cohen and Mello 2018; European Parliament and Council of the European Union 2016; Price and Cohen 2019], and the European Health Data Space regulation [European Parliament and Council of the European Union 2025] is a recent recognition that this fragmentation is itself a policy problem. The consequence is a paradox that motivates this paper: *healthcare is collectively big and locally small.*

The term "small data" appears in social science [Kitchin and Lauriault 2015], in personal health [Estrin 2014; Hekler et al. 2019], in statistics [Faraway and Augustin 2018], and in hundreds of applied papers. It is almost always used informally, as a synonym for "fewer samples than we would like". Three consequences follow. Methods are compared across problems whose difficulty is not comparable. Remedies that cannot work in a given setting are applied anyway, as when synthetic data is expected to compensate for a cohort that does not contain the relevant information. And the specific case of many small, isolated institutions, which is the default organizational form of healthcare, is treated either as a federated learning problem with a central server that often does not exist, or not at all.

Contributions. This paper is both a survey and a position paper. Its contributions are as follows.

(1) **A unifying, task-relative definition.** We define small data relative to a task, through an *effective sample size* n^{eff} that accounts for clustering, missing labels, label noise, distribution shift and outcome rarity, and a *smallness index* $\sigma = n^{\star}/n^{\text{eff}}$ that compares it with the sample size the task requires (Section 2). The definition brings existing quantities (Kish's design effect,

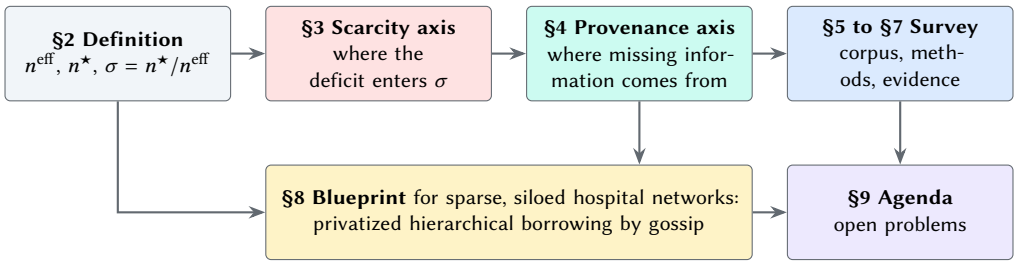


Fig. 1. Structure of the paper. The definition of Section 2 produces the two axes of the taxonomy, which serve as the coding scheme of the survey and shape the design of the blueprint.

the sample complexity of learning with label noise, and the sample-size criteria of clinical prediction models) into one place, decomposes the shortfall by cause so that each factor points to a remedy, and extends it to data split across custodians (*siloed smallness*), which no existing notion covers (Table 2). It is a diagnostic, not a new estimator: in a test on 218 hospitals it did not predict the gain from collaboration better than raw size (Section 2.5).

(2) **A new taxonomy.** We propose the Scarcity-Provenance taxonomy (Sections 3 and 4). Its first axis classifies ten forms of scarcity in healthcare by the term of σ they degrade. Its second axis classifies remedies by where the missing information comes from. We prove a "no free lunch" result for re-expression (Proposition 4.1) that explains formally why augmentation and synthetic data cannot by themselves solve a small-data problem.

(3) **A coded survey.** We classify a curated corpus of 290 works on both axes (Section 5; the full coding is in the online supplement), review the solutions channel by channel (Section 6), and compare where the literature has actually worked with where the taxonomy says it could, which exposes specific empty cells (Section 7).

(4) **A blueprint for siloed hospital networks.** For sparse networks of hospitals without a trusted central server, we derive a privacy-preserving hierarchical borrowing scheme with closed-form risk, explicit privacy cost and convergence on sparse graphs (Section 8), argue that it is a principled default for this setting, and summarize its evaluation in eighteen simulations and its evaluation on real cancer networks held out from its development (Section 8.7). The method for realistic model heads, its full privacy analysis and the evaluation are developed in a companion paper [Simões et al. 2026a]. The ingredients are standard; what is new is their combination for serverless hospital networks, and its evaluation against federated and personalized methods tuned as a consortium could tune them.

(5) **A research agenda** with concrete open problems (Section 9).

Scope and method. We cover supervised and semi-supervised prediction from clinical data: imaging, structured electronic health records (EHR), clinical text, omics and tabular registries. We do not cover causal effect estimation from small trials in depth, although several of our results apply to it. How the corpus was assembled and coded, and how a systematic database search reported to PRISMA 2020 complements it, is described in Section 5. Existing surveys cover individual families, such as federated learning in healthcare [Nguyen et al. 2022; Rieke et al. 2020; Xu et al. 2021], few-shot learning [Wang et al. 2020], meta-learning [Hospedales et al. 2022], imperfect medical datasets [Cheplygina et al. 2019; Tajbakhsh et al. 2020], informed machine learning [von Rueden et al. 2023] and synthetic data [Chen et al. 2021a; Jordon et al. 2022]. Among the surveys we found,

Table 1. Main notation.

Symbol	Meaning
$\mathcal{T} = (P, \mathcal{P}, \mathcal{H}, \ell, \varepsilon, \delta)$	learning task: target distribution, family of distributions the learner must handle, hypothesis class, loss, tolerance, confidence
n, n^{eff}	nominal number of records; effective sample size (Definition 2.2)
$n^{\star}_{\mathcal{T}}$	sample complexity of $\mathcal{T}$ (Definition 2.1)
$\sigma_{\mathcal{T}}(\mathcal{D})$	smallness index $n^{\star}_{\mathcal{T}}/n^{\text{eff}}(\mathcal{D})$ (Definition 2.3)
DEFF, ρ_L, η, ρ_S, π	design effect, labeled fraction, label noise rate, shift efficiency, outcome prevalence
K, $\mathcal{G} = (V, E)$, W	number of hospitals, collaboration graph, gossip matrix
θ_k, $\hat{\theta}_k$, $v_k = \sigma^2/n_k$	parameter of site k, its local estimate, and the variance of that estimate
μ, τ^2, M	population mean of site parameters, between-site heterogeneity, mean squared error of the estimate of μ
(ϵ, δ)	differential privacy parameters (written ϵ to distinguish them from the tolerance ε)

none defines small data formally, none organizes remedies by information provenance, and none treats the sparse, serverless hospital network as a first-class setting.

How the parts connect. The paper is built as a single argument, shown in Figure 1. We first make smallness measurable (Section 2). The terms of that measure tell us *why* a dataset is small, which gives the scarcity axis of the taxonomy (Section 3). Knowing why points to *where* the missing information could come from, which gives the provenance axis (Section 4). The two axes then serve as the coding scheme for a corpus of 290 works (Section 5), which organizes the survey of existing methods (Section 6) and the reading of their evidence (Section 7). The gaps that this reading exposes, together with the definition of siloed smallness from Section 2, determine the design of the blueprint (Section 8), and what remains unsolved becomes the agenda (Section 9).

2 WHAT IS SMALL DATA? A FORMAL ACCOUNT

The paradox of the introduction can only be studied once it can be measured. This section turns "small" from an adjective into a quantity, and that quantity is the thread that runs through the rest of the paper.

2.1 Existing notions

Notions of small data fall into five groups. The first is *absolute*: a dataset is small if it has few records, often with arbitrary cut-offs such as a few hundred or a few thousand. This ignores the task. One thousand patients is a lot for estimating a mean blood pressure and very little for a 50-predictor model of a 2% outcome. The second is *relational*, as in Kitchin and Lauriault's contrast between small data, collected for a specific question with controlled sampling, and big data, which is exhaustive, fine-grained and collected as a by-product [Kitchin and Lauriault 2015]. Estrin's "n of 1" [Estrin 2014] and the small data paradigm of Hekler et al. [2019] belong here, stressing individual-level inference rather than population scale. Faraway and Augustin [2018] show that a small designed sample can beat a large biased one, which is a statement about information, not about size. The third group is *criterion-based* and comes from clinical prediction research: a dataset is adequate if it contains at least ten events per candidate predictor [Peduzzi et al. 1996], or more defensibly if it meets the criteria of Riley et al. [2019b], which bound overfitting, optimism in apparent fit and imprecision of the intercept [Riley et al. 2020; van Smeden et al. 2019].

The fourth group is *empirical*: it measures, rather than predicts, how performance depends on size. Learning curves fitted to pilot data, typically an inverse power law, extrapolate the sample size

Table 2. Notions of small data compared on six properties. ✓: has the property; ○: partially; ✗: does not. Task: the verdict is relative to the task (model and outcome). Model class: the models it applies to. Quality: accounts for clustering, missing or noisy labels or distribution shift. Cause: says why the data are small. Sites: addresses data split across custodians. Pre-fit: computable without fitting the model.

Notion	Task	Model class	Quality	Cause	Sites	Pre-fit	Main strength or limitation
Absolute record count	✗	any	✗	✗	✗	✓	simple; ignores the task
Events per variable [Peduzzi et al. 1996]	○	regression	✗	✗	✗	✓	counts the minority outcome; the threshold of 10 is a rule of thumb [van Smeden et al. 2019]
Riley et al. criteria [Riley et al. 2019b]	✓	regression	✗	✗	✗	✓	principled targets on overfitting and precision; needs an anticipated R^2
Learning curves [Balki et al. 2019; Figueroa et al. 2012]	✓	any	○	✗	✗	✗	measures the actual model on the actual data; needs pilot data and training
Small sample size effects [Raudys and Jain 1991; Zhang and Ling 2018]	✓	specific classifiers	✗	✗	✗	✓	ratio of examples to dimension or degrees of freedom
Prior effective sample size [Morita et al. 2008]	○	parametric Bayesian	✗	✗	✗	✓	expresses a prior in patients; says nothing about the dataset
Relational [Hekler et al. 2019; Kitchin and Lauriault 2015]	✗	none	○	○	✗	✓	names purpose and sampling design; gives no number
Smallness index σ (ours)	✓	any (through $n^\star$)	✓	✓	✓	✓	decomposes the loss by cause; inherits the error of the proxy for $n^\star$ and of each factor estimate; does not measure the model as a learning curve does

a classifier needs to reach a target accuracy [Figueroa et al. 2012], and they are the most common way of determining sample size in medical imaging studies [Balki et al. 2019]. In statistical pattern recognition, the small sample size problem is the degradation of a classifier when the number of training examples is small relative to the dimension, with recommended ratios of examples per class to features that depend on the classifier [Raudys and Jain 1991]; materials science has made the same observation that the effect of data size is mediated by the degrees of freedom of the model [Xu et al. 2023; Zhang and Ling 2018]. Small samples also corrupt the estimate of performance itself: cross-validation with feature selection outside the loop is strongly optimistic at small n [Vabalas et al. 2019]. The fifth group expresses information as a number of patients without reference to a dataset size: the *effective sample size of a prior* is the number of observations whose information the prior is worth [Morita et al. 2008], the same currency as the "virtual patients" of Section 8.

Table 2 compares these notions with ours on six properties. The criterion-based notions are quantitative and relative to the task, but tied to regression and blind to how the data were collected; learning curves measure the actual model, which no formula can do, but only after data have been gathered and models trained, and they do not say why the data are small; the relational notions name the cause but give no number. No existing notion addresses data that are sufficient in aggregate but split across custodians. Our definition does not replace these quantities. It uses the Riley criteria, or any other estimate of sample complexity, as its numerator; it adds a denominator that records where the information of the nominal records was lost, so that each loss points to a cause and a remedy (Sections 3 and 4); and it extends to several custodians.

2.2 Setup and notation

A learning *task* is a tuple $\mathcal{T} = (P, \mathcal{P}, \mathcal{H}, \ell, \varepsilon, \delta)$, where P is a distribution over $\mathcal{X} \times \mathcal{Y}$ describing the target population at the point of use, $\mathcal{P} \ni P$ is the family of distributions that are consistent with what is known about the task before seeing data, $\mathcal{H}$ a hypothesis class, ℓ a loss bounded in $[0, 1]$, and (ε, δ) the required excess risk and confidence. The risk of h under Q is $R_Q(h) = \mathbb{E}_{(x,y)\sim Q}\, \ell(h(x), y)$, and $R = R_P$. The family $\mathcal{P}$ is needed because P itself is unknown: a learner that knew P could output the best hypothesis without any data, so sample complexity is only meaningful as a guarantee that holds for every distribution the learner cannot rule out. Notation used throughout is collected in Table 1.

Definition 2.1 (Sample complexity of a task). The sample complexity $n^\star_{\mathcal{T}}$ is the smallest n for which there is a learner A such that, for every $Q \in \mathcal{P}$ and $\mathcal{D} \sim Q^n$ independent and identically distributed (i.i.d.) and fully and correctly labeled,

$$\Pr_{\mathcal{D}\sim Q^n}\left[R_Q(A(\mathcal{D})) - \inf_{h\in\mathcal{H}} R_Q(h) \le \varepsilon\right] \;\ge\; 1 - \delta.$$

$n^\star$ is rarely known exactly, but it has usable proxies. For binary classification with the 0-1 loss, $\mathcal{P}$ the set of all distributions (agnostic learning) and $\mathcal{H}$ of VC dimension d, $n^\star = \Theta\big((d + \log(1/\delta))/\varepsilon^2\big)$ [Shalev-Shwartz and Ben-David 2014], and data-dependent versions replace d with Rademacher complexity [Bartlett and Mendelson 2002]. For logistic and survival prediction models, the criteria of Riley et al. give closed forms. They are not sample complexities in the sense of Definition 2.1 but regression-specific surrogates, which give the size at which a model with p parameters meets targets on shrinkage, optimism and the precision of the intercept; we use them as proxies for $n^\star$ when the task is a clinical prediction model, and σ then inherits the error of the proxy. Their first criterion, which targets a global shrinkage factor S (typically 0.9) for a model with p candidate parameters and anticipated Cox-Snell R^2_{CS}, is

$$n^\star \;\ge\; \frac{p}{(S-1)\,\ln\!\big(1 - R^2_{\mathrm{CS}}/S\big)} \qquad \text{[Riley et al. 2019b].} \tag{1}$$

Either way, $n^\star$ grows with model complexity and with the precision demanded, and for binary outcomes it is driven by the number of events rather than the number of patients.

2.3 Effective sample size

Clinical records are not i.i.d., not all labeled, not all correctly labeled, and not drawn from the population where the model will be used. Each of these reduces the information a record carries.

Definition 2.2 (Effective sample size). Let $\mathcal{D}$ contain n records, and fix a target quantity $\psi(P)$, such as a model parameter or the risk of a fixed model, with an estimator $\hat{\psi}$ that is efficient under i.i.d. sampling from P. The effective sample size $n^{\mathrm{eff}}(\mathcal{D})$ is the number of i.i.d., correctly labeled records drawn from P for which $\hat{\psi}$ would have the same variance as the best estimator of $\psi(P)$ that can be computed from $\mathcal{D}$. To first order, when the mechanisms below act independently,

$$n^{\mathrm{eff}} \;=\; n \cdot \underbrace{\frac{1}{\mathrm{DEFF}}}_{\text{clustering}} \cdot \underbrace{\rho_L}_{\text{labels}} \cdot \underbrace{(1-2\eta)^2}_{\text{label noise}} \cdot \underbrace{\rho_S}_{\text{shift}}, \tag{2}$$

where $\mathrm{DEFF} = 1 + (\bar{m} - 1)\,\mathrm{ICC}$ is the design effect for $\bar{m}$ records per patient with intra-class correlation ICC [Kish 1965]; ρ_L is the fraction of records with labels missing completely at random; $\eta < 1/2$ is a symmetric label-noise rate, whose effect on sample complexity scales as $(1-2\eta)^{-2}$ [Angluin and Laird 1988] (for the prevalence of a binary label the exact factor is $(1-2\eta)^2\,\pi(1-\pi)/\tilde{\pi}(1-\tilde{\pi}) \le$

$(1-2\eta)^2$, where $\tilde{\pi}$ is the prevalence of the noisy labels); and $\rho_S = (\sum_i w_i)^2/(n\sum_i w_i^2)$ is Kish's efficiency of the importance weights $w_i = p_P(x_i)/p_{\mathcal{D}}(x_i)$ needed to transport the sample to P [Kish 1965]. The weights exist only if P is absolutely continuous with respect to the distribution that generated $\mathcal{D}$; where P puts mass on inputs that $\mathcal{D}$ cannot contain, no weighting recovers the information, and the effective sample size for that region is zero. For a binary outcome with prevalence π, the information about the decision boundary is controlled by the minority class, $n_{\text{ev}}^{\text{eff}} = n^{\text{eff}} \cdot \min(\pi, 1-\pi)$.

Equation (2) is deliberately simple. Each factor is a standard, estimable quantity, and each can be replaced by a more refined estimator (for instance, the fraction of missing information under multiple imputation) without changing the structure. The factors multiply because each is a ratio of variances under an independent mechanism; when mechanisms interact, the product is an approximation. None of the factors is new, and Equation (2) should be read as a first-order heuristic: it is meant to compare datasets and to show where information is lost, not to replace a variance computed for a specific estimator. What it adds is a single place where the clinical and the learning-theoretic notions of sample size meet, relative to a task.

Definition 2.3 (Smallness index). The smallness index of a dataset $\mathcal{D}$ for task $\mathcal{T}$ is $\sigma_{\mathcal{T}}(\mathcal{D}) = n^{\star}_{\mathcal{T}}/n^{\text{eff}}(\mathcal{D})$. The dataset is *small for* $\mathcal{T}$ when $\sigma_{\mathcal{T}}(\mathcal{D}) > 1$. We call the regimes $\sigma \le 1$ *sufficient*, $1 < \sigma \le 10$ *strained*, $10 < \sigma \le 100$ *small*, and $\sigma > 100$ *extreme*.

The thresholds are conventions, chosen so that each regime corresponds to a qualitatively different strategy. In the strained regime, regularization, simpler models and careful validation usually close the gap. In the small regime, information has to be imported from outside the dataset. In the extreme regime, only strong priors or pooling at a scale beyond any single institution will help.

Example 2.4 (An ICU deterioration model). A hospital has $n = 2{,}000$ ICU stays from patients with on average $\bar{m} = 1.6$ stays each and ICC $= 0.5$, so DEFF $= 1.3$. Seventy percent of stays have an adjudicated outcome, adjudication disagrees with a reference standard in 10% of cases (which we treat as a symmetric noise rate $\eta = 0.1$), and the case mix differs from the deployment population with Kish efficiency 0.6. By Equation (2), $n^{\text{eff}} \approx 414$, and at an outcome prevalence of 8% the dataset contains about 33 effective events. For a model with $p = 20$ parameters and an anticipated R^2_{CS} of 15% of its maximum, Equation (1) gives $n^{\star} \approx 2{,}707$. Hence $\sigma \approx 6.5$: a dataset that looks moderate is strained, close to small. Figure 2 shows the decomposition.

Where the notions disagree. The notions of Table 2 give four different verdicts on the dataset of Example 2.4. By record count, 2,000 stays is a respectable single-site cohort. By events per variable it is small: 160 events for 20 parameters is 8, below the usual 10. By the Riley criteria it is short by about a quarter: 2,000 records against the 2,707 required. By $\sigma \approx 6.5$ it is strained, close to small, because the Riley calculation treats every record as an independent, correctly labeled draw from the deployment population, which these are not. The difference between the last two, a factor of 4.8, is the quality deflation of Equation (2), and its decomposition is what σ adds: adjudicating the 30% of unlabeled stays would recover a factor of 1.43, correcting the noisy labels a factor of 1.56, and reweighting or collecting data closer to the deployment case mix a factor of 1.67, whereas collecting more stays of the same kind would only grow n. A learning curve could check the verdict on the model actually trained, but only after the data are collected and labeled.

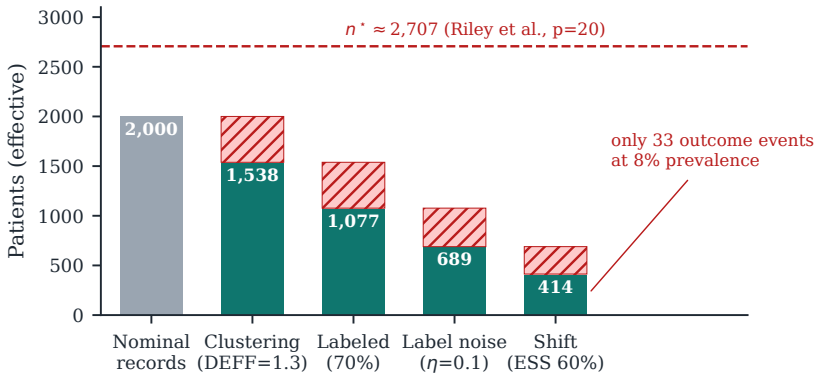


Fig. 2. Deflation of a nominal sample of 2,000 ICU stays into an effective sample size (Example 2.4). Hatched segments show the information lost to each mechanism of Equation (2). The dashed line is the sample size required by the first criterion of Riley et al. [2019b] for a 20-parameter model at 8% prevalence.

2.4 Siloed smallness

The definitions so far concern one dataset. Healthcare adds a second level: data that would be sufficient together are held separately.

Definition 2.5 (Siloed smallness). Let $\mathcal{D}_1, \ldots, \mathcal{D}_K$ be held by K custodians and let $\mathcal{D}_\cup$ be their union. The task $\mathcal{T}$ is *siloed-small* if $\sigma_\mathcal{T}(\mathcal{D}_k) > 1$ for every k while $\sigma_\mathcal{T}(\mathcal{D}_\cup) \le 1$, where $n^{\text{eff}}(\mathcal{D}_\cup)$ is computed after accounting for the heterogeneity across custodians (the shift factor ρ_S of Equation (2) applied to the mixture). The ratio $\Gamma = \min_k \sigma_\mathcal{T}(\mathcal{D}_k)/\sigma_\mathcal{T}(\mathcal{D}_\cup)$ is the *custodial gap*.

A siloed-small task can be solved in principle without collecting a single new patient; the obstacle is access. A task that is small even for $\mathcal{D}_\cup$ is *intrinsically small*, and no amount of collaboration among the existing custodians will solve it. Rare disease work often sits between the two. The distinction matters because the remedies are different, and because heterogeneity between custodians can make $n^{\text{eff}}(\mathcal{D}_\cup)$ much smaller than $\sum_k n^{\text{eff}}(\mathcal{D}_k)$. Section 8 makes this precise.

2.5 Does the index predict who gains?

A definition can be useful without being a better predictor, and we tested the stronger claim directly, under a plan committed before any number was computed (online supplement). For each of the 218 hospitals of 19 real networks (the five held-out and twelve design TCGA networks, Arizona Medicare and UCI Heart Disease of Section 8), we computed σ with the Riley criteria as $n^\star$, and its raw size, events and events per variable, and correlated each with the hospital's gain from hierarchical borrowing over working alone. In these data $n^{\text{eff}} = n$, since every record is a different patient with an adjudicated label and the target is the hospital's own population, so the test can show only whether the task-relativity of σ adds predictive value.

It does not (Table 3). None of the four quantities predicts the gain: every Spearman correlation is between -0.11 and $+0.11$, pooled or within networks. The difference D between the absolute correlation of σ and that of the best other quantity is -0.054 at $\epsilon = 1$ (95% interval -0.107 to -0.003) and -0.049 at $\epsilon = 10$ (-0.108 to -0.001), so σ is, if anything, slightly worse than raw size. Which hospital gains is decided less by how small it is than by how much it resembles the others, which none of these quantities measures. On the six tables of a synthetic-data study [Simões et al. 2026b],

where σ ranges from 0.8 to 9.5, the larger tables gained more from synthetic rows (Spearman +0.83 for size, −0.60 for σ), consistent with Proposition 4.1; with six tables this is descriptive only.

We therefore claim no predictive advantage for σ. Its value is that of a diagnostic: it is relative to the task, it decomposes the loss of information by cause so that each factor points to a remedy, and it extends to data split across custodians, which no other notion of Table 2 does.

Table 3. Spearman correlation of each quantity with a hospital's gain from hierarchical borrowing over working alone (one minus the ratio of held-out log-losses), 218 hospitals of 19 networks, pooled and averaged within the 18 networks with at least five hospitals. A predictor of the gain should correlate positively (σ) or negatively (the others). D: absolute correlation of σ minus the largest absolute correlation of the others, with a 95% cluster-bootstrap interval over networks. Per-hospital values are in the online supplement.

	$\epsilon = 1$		$\epsilon = 10$	
Predictor of the gain	pooled	within networks	pooled	within networks
Smallness index σ	+0.04	−0.01	+0.06	+0.02
Patients n	−0.10	−0.02	−0.11	−0.05
Events	−0.00	+0.03	−0.03	−0.02
Events per variable	−0.01	+0.03	−0.04	−0.02
D [95% CI]	−0.054 [−0.107, −0.003]		−0.049 [−0.108, −0.001]	

2.6 Why healthcare is a special case

Small data exist in every field, but five properties make healthcare unusual. *Stakes are asymmetric*: errors fall on patients, so the tolerance ε is small and the required $n^\star$ is large. *Labels are expensive and delayed*: ground truth often needs specialist review, biopsy, or months of follow-up [Willemink et al. 2020]. *Heterogeneity is structural*: devices, coding practices, referral patterns and populations differ across sites in ways that models detect and exploit [Finlayson et al. 2021; Gichoya et al. 2022; Zech et al. 2018]. *Re-identification risk is high*: Rocher et al. estimate that 99.98% of Americans would be correctly re-identified in any dataset using 15 demographic attributes [Rocher et al. 2019], which undermines de-identification as a basis for sharing [Price and Cohen 2019; Sweeney 2002]. And *subgroups matter*: performance in a minority subgroup is a clinical and ethical requirement, not a secondary metric [Chen et al. 2021b; Obermeyer et al. 2019]. Each subgroup has its own, smaller n^{eff}.

From measurement to diagnosis. The smallness index tells us *how* small a dataset is for a task. It does not yet tell us *why*, and the why matters: a low ρ_L, a small π and a large custodial gap all raise σ, but each calls for a different response. The next section uses the structure of σ itself to classify the causes.

3 THE SCARCITY AXIS: WHERE THE DEFICIT COMES FROM

The first axis of our taxonomy answers the question *why is this dataset small?* Existing classifications tend to mix causes (rare disease), symptoms (class imbalance) and remedies (few-shot learning). We instead classify scarcity by the term of the smallness index it degrades. This makes the classes operational, because each one points to a specific quantity that can be estimated. They are not mutually exclusive. Rarity of condition (O1) and event sparsity (O2) both lower the effective number of events, contextual mismatch (T2) and custodial fragmentation (C1) usually arrive together, and a single cohort can be label-poor and high-dimensional at once. A dataset therefore has a *profile* over the ten classes, and that profile is more informative than any single label. This is also how we code the literature in Section 5: a work can target several classes.

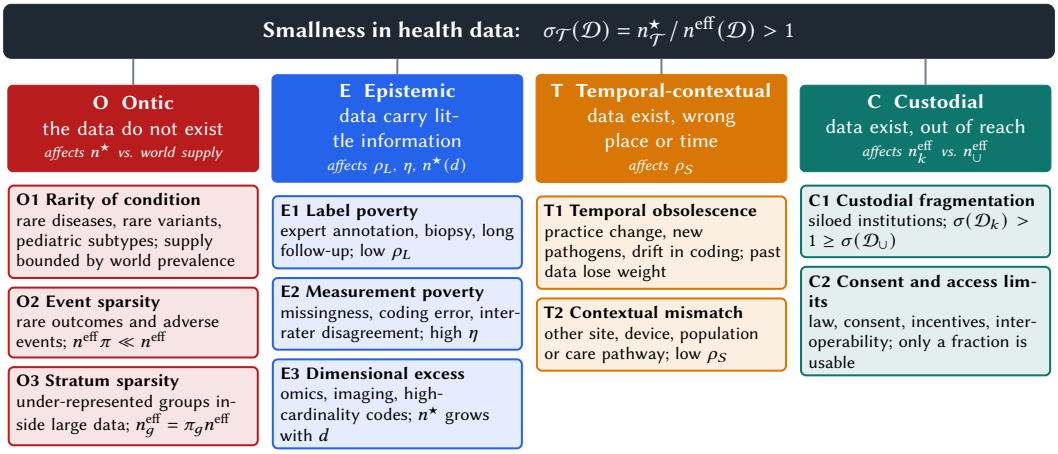


Fig. 3. The scarcity axis of the Scarcity-Provenance taxonomy. Families are defined by the term of the smallness index they degrade; the ten classes are the forms these take in healthcare.

3.1 Ontic scarcity (O): the data do not exist

Ontic scarcity is a property of the world, not of the dataset. More effort spent on collection or access will not remove it, only reduce it.

O1, rarity of condition. For a disease with worldwide prevalence q, the total supply of cases is bounded by qN_{world}, and in practice by the fraction diagnosed and recorded. When $n^{\star}$ approaches that bound, the task is intrinsically small in the sense of Definition 2.5. This is the situation of many of the conditions cataloged by Orphanet [Nguengang Wakap et al. 2020], and the reason machine learning in rare diseases relies so heavily on transfer, knowledge and international pooling [Banerjee et al. 2023; Schaefer et al. 2020].

O2, event sparsity. The condition may be common, but the outcome of interest is rare: in-hospital cardiac arrest, a severe adverse drug reaction, a post-operative complication. The number of patients is large but the effective number of events $n^{\text{eff}}\pi$ is small, which is what drives the sample-size criteria [Peduzzi et al. 1996; Riley et al. 2019b]. Oversampling or reweighting changes the decision threshold, not the information.

O3, stratum sparsity. Within a large dataset, a subgroup g with share π_g has $n_g^{\text{eff}} = \pi_g n^{\text{eff}}$. Clinically meaningful subgroups (by ancestry, sex, age band, comorbidity or disease subtype) are often small even when the whole is not. Failures concentrated in such strata have been called hidden stratification [Oakden-Rayner et al. 2020], and the related fairness failures are well documented [Chen et al. 2021b; Obermeyer et al. 2019].

3.2 Epistemic scarcity (E): the data carry little information

Here the patients exist and are recorded, but each record says less about the target than it could.

E1, label poverty. Unlabeled images, notes and signals are cheap, but labels need specialist time, invasive confirmation, or follow-up long enough for the outcome to occur [Willemink et al. 2020]. Label poverty acts through ρ_L. It is the scarcity class for which semi-supervised, self-supervised and weakly supervised learning were developed [Cheplygina et al. 2019; Tajbakhsh et al. 2020].

E2, measurement poverty. Missing values that are not missing at random, billing codes that reflect reimbursement more than pathology, and disagreement between raters all reduce the

information per record. In Equation (2) this acts through η and through the missing-information fraction.

E3, dimensional excess. Genomic, transcriptomic and imaging data have $d \gg n$. In this case $n^\star$ grows with d, often faster than any realistic cohort, unless strong structural assumptions such as sparsity are imposed [Bühlmann and van de Geer 2011; Tibshirani 1996]. Dimensional excess differs from the other classes in that smallness can be created by the choice of representation, and therefore partly removed by changing it.

3.3 Temporal-contextual scarcity (T): the data are about somewhere else

T1, temporal obsolescence. Treatment guidelines, coding systems, assays and pathogens change. Data collected before a change are partly irrelevant afterwards, so the usable window is shorter than the archive. The early COVID-19 literature is an extreme case [Roberts et al. 2021; Wynants et al. 2020].

T2, contextual mismatch. A model trained at one site, scanner or population meets a different distribution at deployment [Finlayson et al. 2021; Koh et al. 2021]. In Equation (2) this is the factor ρ_S, which can be close to zero when supports barely overlap. Learning theory makes the same point through the divergence term in domain adaptation bounds [Ben-David et al. 2010]. External validation studies show the effect directly [Debray et al. 2015; Wong et al. 2021; Zech et al. 2018].

3.4 Custodial scarcity (C): the data are out of reach

C1, custodial fragmentation. The information exists and is relevant, but is distributed across institutions that do not pool it. This is the silo problem formalized in Definition 2.5, and the setting of Section 8.

C2, consent and access limits. Even inside one institution, only part of the data can be used for a given purpose, because of consent scope, legal basis, governance review, or simply the absence of interoperable extraction [Cohen and Mello 2018; European Parliament and Council of the European Union 2016; Mandel et al. 2016]. Common data models such as OMOP and the OHDSI network [Hripcsak et al. 2015; Overhage et al. 2012] and the FAIR principles [Wilkinson et al. 2016] attack the interoperability side of C2.

Using the scarcity profile. Table 4 summarizes the ten classes with the quantity to estimate for each and the characteristic failure when it is ignored. We suggest that studies report this profile alongside n^{eff} and σ. It takes a few lines and would make results across small-data papers far easier to compare.

From diagnosis to remedy. A scarcity profile identifies which factor of σ is degraded, but not how to repair it. Every repair has to bring in information about the target from somewhere, and the possible sources are limited. The second axis of the taxonomy, introduced next, classifies remedies by that source.

4 THE PROVENANCE AXIS: WHERE THE MISSING INFORMATION COMES FROM

The second axis of the taxonomy classifies remedies, and it is the axis along which the rest of the paper is organized. Existing surveys group methods by technique ("generative models", "transfer learning", "federated learning"). Technique is the wrong primitive for small data, because two techniques that look different can draw on the same source of information, and two that look alike can draw on different ones. A GAN trained on the local cohort and a GAN pretrained on a million public images are both "synthetic data", but only the second brings anything new. We therefore classify remedies by the *provenance* of the information they add.

Table 4. Scarcity classes, the quantity that measures each, and the characteristic failure when it is ignored.

Class	What to estimate	Typical failure if ignored
O1 Rarity of condition	world supply qN_{world} vs. $n^\star$	optimistic results from a single centre that do not replicate
O2 Event sparsity	effective events $n^{\text{eff}}\pi$	overfitting, miscalibrated risks [Riley et al. 2019b]
O3 Stratum sparsity	n_g^{eff} for each relevant group g	hidden stratification, subgroup harm [Oakden-Rayner et al. 2020]
E1 Label poverty	labeled fraction ρ_L, label delay	models trained only on the easy, labeled cases
E2 Measurement poverty	noise rate η, missingness mechanism	learning the documentation process, not the disease
E3 Dimensional excess	d/n^{eff}, intrinsic dimension	spurious signatures, unstable feature selection
T1 Temporal obsolescence	time-sliced performance	silent decay after deployment
T2 Contextual mismatch	Kish efficiency ρ_S, site-held-out error	shortcut learning, external failure [Zech et al. 2018]
C1 Custodial fragmentation	custodial gap Γ	many under-powered single-site models
C2 Consent and access limits	usable fraction per purpose	selection bias in the usable subset

4.1 A conservation principle

Let θ denote the unknown quantity we want to learn: a parameter, a regression function, or a decision rule for the target population. Let $\mathcal{D}$ be the local data and let Z denote any external resource used by a method, such as a pretrained network, a knowledge base, or a published risk score.

Proposition 4.1 (No free lunch for re-expression). *Let $\tilde{\mathcal{D}} = g(\mathcal{D}, Z, U)$ be any dataset produced by a (possibly randomized) procedure g, where U is randomness independent of $(\theta, \mathcal{D}, Z)$. Then*

$$I(\theta; \mathcal{D}, \tilde{\mathcal{D}}) \;\le\; I(\theta; \mathcal{D}, Z).$$

In particular, if Z is independent of $(\theta, \mathcal{D})$, or absent, then $I(\theta; \mathcal{D}, \tilde{\mathcal{D}}) = I(\theta; \mathcal{D})$. Moreover, for any loss L and any estimator $\hat{\theta}(\mathcal{D}, \tilde{\mathcal{D}})$, the randomized estimator $(\mathcal{D}, Z, U) \mapsto \hat{\theta}(\mathcal{D}, g(\mathcal{D}, Z, U))$ has the same risk at every value of θ. Hence the minimax and Bayes risks attainable from $(\mathcal{D}, \tilde{\mathcal{D}})$ are no smaller than those attainable from $(\mathcal{D}, Z)$ with randomized estimators, and, for a loss convex in the estimate, no smaller than those attainable from $(\mathcal{D}, Z)$ without randomization.

Proof. Since U is independent of $(\theta, \mathcal{D}, Z)$ and $\tilde{\mathcal{D}}$ is a function of $(\mathcal{D}, Z, U)$, the chain $\theta \to (\mathcal{D}, Z) \to (\mathcal{D}, Z, \tilde{\mathcal{D}})$ is Markov. By the data processing inequality [Cover and Thomas 2006], $I(\theta; \mathcal{D}, Z, \tilde{\mathcal{D}}) \le I(\theta; \mathcal{D}, Z)$, and by the chain rule $I(\theta; \mathcal{D}, \tilde{\mathcal{D}}) \le I(\theta; \mathcal{D}, Z, \tilde{\mathcal{D}})$. For the second claim, $I(\theta; \mathcal{D}, Z) = I(\theta; \mathcal{D}) + I(\theta; Z \mid \mathcal{D})$ and $I(\theta; Z \mid \mathcal{D}) \le I(\theta, \mathcal{D}; Z) = 0$ when Z is independent of $(\theta, \mathcal{D})$. The reverse inequality $I(\theta; \mathcal{D}, \tilde{\mathcal{D}}) \ge I(\theta; \mathcal{D})$ holds trivially. For the risk statement, the two estimators coincide as functions of $(\mathcal{D}, Z, U)$, so their risks agree for every θ, and the set of estimators based on $(\mathcal{D}, \tilde{\mathcal{D}})$ is contained in the set of randomized estimators based on $(\mathcal{D}, Z)$. For a convex loss, replacing a randomized estimator $\delta(\mathcal{D}, Z, U)$ by $\mathbb{E}_U\, \delta(\mathcal{D}, Z, U)$ does not increase the risk at any θ, by Jensen's inequality. □

The proposition is elementary, but its consequences are not widely appreciated in the applied literature. The information statement treats θ as random; the risk statement holds for every fixed θ and every loss, and does not rely on lower bounds such as Fano's inequality. Together they say that no augmentation, oversampling or synthetic generation can lower the minimax or Bayes risk for θ below what $(\mathcal{D}, Z)$ allows. Re-expression can still help in practice, and often does, for three reasons that the proposition makes explicit: the generator or augmentation family carries prior knowledge Z (rotation invariance in histology, anatomical plausibility in a pretrained generator); the downstream learner may use the re-expressed data more efficiently, which is a computational rather than an informational gain; or the synthetic data are used for a purpose other than learning θ, such as software testing. What re-expression cannot do is replace the missing patients.

4.2 Six provenance channels

Write the posterior for the parameter of site k as

$$p(\theta_k \mid \mathcal{D}_k, \mathcal{S}) \;\propto\; \underbrace{p(\mathcal{D}_k \mid \theta_k)}_{\text{likelihood: local data}} \;\cdot\; \underbrace{p(\theta_k \mid \mathcal{S})}_{\text{prior: everything else}}, \tag{3}$$

where $\mathcal{S}$ denotes all information other than the local labeled sample, and the local data are assumed to depend on $\mathcal{S}$ only through θ_k. Every remedy for small data does one of two things: it enlarges the likelihood term by adding observations that bear on θ_k, or it sharpens the prior term by bringing in information from $\mathcal{S}$. The six channels below exhaust the possible sources by construction, because they answer three questions about any input to a method other than $\mathcal{D}_k$. Is it data? If not, it is knowledge (P3). If it is data, was it newly acquired for the task? If so, it is procurement (P5). If it is existing data, does it carry labels for the same task and population? Labeled data for the same task held by other custodians is pooling (P1); data from a different task or population, labeled or not, is proxy (P2); unlabeled inputs from the target population are P4. A method whose only input is $\mathcal{D}_k$ itself, transformed, is re-expression (P6). Figure 4 shows the six channels.

P1 Pooling. Labeled data for the same task from other custodians: data sharing, consortia, distributed statistics, federated and decentralized learning. Information crosses a custodial boundary, so pooling carries a privacy cost, and heterogeneity between custodians limits what it can deliver.

P2 Proxy. Data from related but different tasks or populations: transfer learning, multi-task and meta-learning, self-supervised pretraining on external data, foundation models. The cost is relevance: the gain depends on how close the proxy is to the target [Ben-David et al. 2010; Tripuraneni et al. 2020].

P3 Prior. Non-data knowledge: physiology, pharmacology, clinical guidelines, ontologies, causal structure, published effect sizes, expert elicitation. It enters as constraints, priors or architecture.

P4 Unlabeled. Unlabeled data from the target population: semi-supervised and self-supervised learning, weak supervision. Unlabeled inputs are informative about θ only when the input distribution and the conditional of the label share parameters, which theory links to the causal direction of the task [Schölkopf et al. 2012].

P5 Procurement. New labeled observations obtained by deliberate action: prospective collection, active learning, targeted annotation. This is the only channel that increases the local likelihood term with fresh, relevant information.

P6 Re-expression. Transformations of existing data: augmentation, resampling, synthetic data, imputation. By Proposition 4.1 it adds no information beyond $(\mathcal{D}, Z)$; whatever it contributes comes from the prior knowledge in Z.

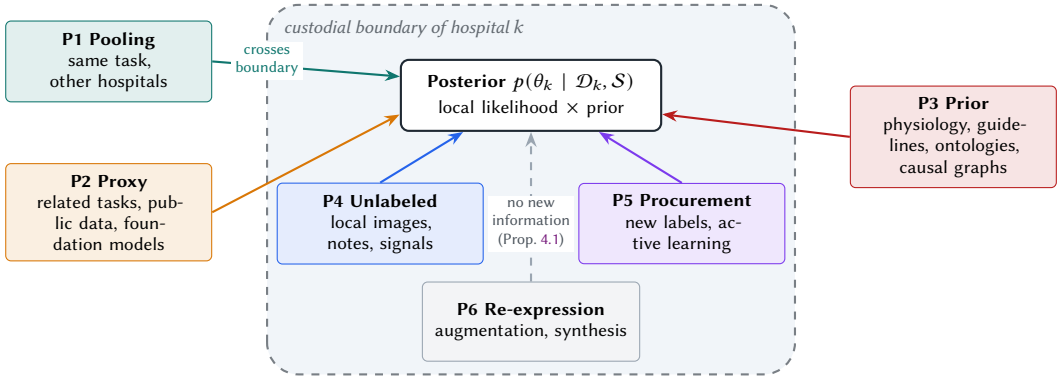


Fig. 4. The provenance axis. Each remedy draws information from one of six channels. Only pooling necessarily crosses the custodial boundary; only procurement adds fresh labeled observations from the target; re-expression adds no information about θ_k beyond what the local data and the generator already contain.

Two attributes of each channel matter for system design. The first is whether information crosses a custodial boundary, which determines the privacy cost: only P1 crosses one by necessity, and P2 does so when the proxy data belong to another custodian. The second is the source of bias: heterogeneity for P1, relevance for P2, misspecification for P3, the cluster or manifold assumption for P4, selection for P5 and generator bias for P6.

4.3 Matching scarcity to provenance

The two axes together give a matrix (Figure 5) that answers a practical question: given the scarcity profile of a problem, which channels can in principle help? The entries are our judgment, informed by the mechanisms of Section 4 and by the literature, graded from none to strong. They are claims about potential, not measurements, and Section 7 sets them against where the coded corpus has actually worked (Figure 9). Three patterns stand out. Ontic scarcity (O1 to O3) is addressed most strongly by pooling at a scale beyond the local site and by priors; procurement helps with event and stratum sparsity but is bounded by the world supply of cases for rare conditions. Custodial scarcity (C1, C2) is addressed by pooling far more than by any other channel, which is why privacy-preserving pooling is the central technical problem for siloed healthcare. And re-expression is weak against every class, which follows from Proposition 4.1 and matches the evidence.

The matrix is a set of claims about what can work in principle. The next three sections test it against what has actually been built and measured: Section 5 describes how we assembled and coded the literature, Section 6 reviews the methods channel by channel, and Section 7 asks what the reported results, and the corpus as a whole, do and do not support.

5 REVIEW METHODOLOGY

The taxonomy gives us a vocabulary. To use it as a survey instrument, we need a body of literature coded in that vocabulary and an honest account of how it was assembled. This section provides both, following the guidelines of Kitchenham and Charters [2007] for literature reviews and of Wohlin [2014] for snowballing, and using PRISMA 2020 [Page et al. 2021] as the reporting target.

5.1 Research questions

The review addresses five questions, each answered in a specific part of the paper.

RQ1 How can smallness be defined so that it is task-relative and measurable? (Section 2)

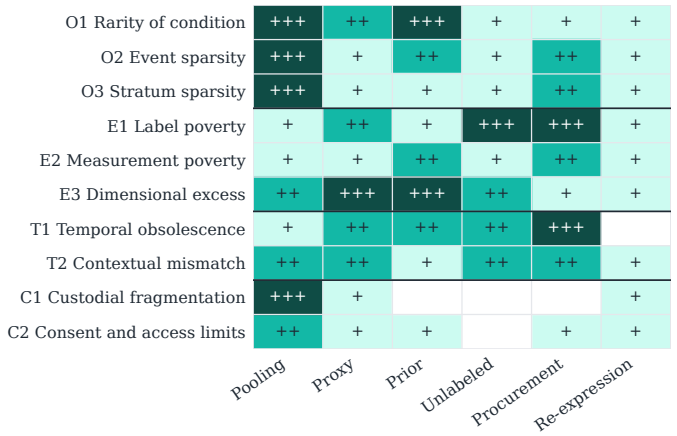


Fig. 5. The Scarcity-Provenance matrix. Rows are the ten scarcity classes of Figure 3; columns are the six provenance channels of Figure 4. Entries grade how strongly a channel can address a class (blank: not at all; + weak; ++ moderate; + + + strong). Grades are the authors' judgment of potential; observed coverage in the coded corpus is shown in Figure 9.

RQ2 Which remedies exist for each scarcity class, and through which provenance channel do they import information? (Section 6)

RQ3 At what level are these remedies supported by evidence: theory, a single dataset, or data held by several institutions? (Section 7)

RQ4 Which combinations of scarcity class and channel that the taxonomy deems promising are under-studied? (Section 7)

RQ5 Which approach fits a sparse network of hospitals that cannot share records and has no trusted coordinator? (Section 8)

5.2 Corpus construction

The corpus was assembled in two stages. The *seed* set contained the foundational work for each provenance channel, the existing surveys listed in Section 1, and the meta-research on clinical prediction models and medical imaging AI. The *expansion* stage applied backward and forward snowballing [Wohlin 2014] from the seed set, with the inclusion criteria below, until new candidates stopped adding channel and modality combinations not yet covered. Every record was checked against the publisher's page or a bibliographic index, and records whose details we could not confirm were dropped.

A work was included if it (I1) proposes, evaluates or reviews a way of learning from limited, fragmented or poorly labeled data, and either addresses health data directly or is the method that health applications build on; or (I2) documents, in health data, a failure or risk caused by scarcity, by distribution shift or by sharing. Works were excluded if they (X1) are textbooks, regulations or reporting guidelines; (X2) concern scaling with abundant data; or (X3) provide general learning theory not tied to a specific channel. Excluded works are still cited where they support an argument. Of the 329 works identified in this way, 39 were excluded as background (X1: 18, X2: 5, X3: 16), which leaves a coded corpus of 290 works published between 1956 and 2025, 204 of them since 2019.

Table 5. Coding scheme applied to every work in the corpus.

Attribute	Values	Rule
Primary channel	P1 to P6, none	the channel through which most of the claimed information gain enters
Secondary channel	P1 to P6, empty	a second channel that is essential to the method, not incidental
Scarcity classes	O1 to C2, multi-label	the classes the work states it targets or documents, judged from its setting
Modality	imaging, EHR, tabular, text, omics, signals, multimodal, general	the data the work is evaluated on; general for methods not specific to health
Contribution type	method, theory, study, multi-site, resource, review, evidence	multi-site only if data come from two or more independent institutions

5.3 Coding scheme

Each work was coded on the five attributes of Table 5. Channels and scarcity classes are those of Sections 3 and 4. A work has exactly one primary channel, the one through which most of its information gain is claimed, and at most one secondary channel; 48 of the 257 works that propose or study a remedy have one. Works that document a problem without proposing a remedy (33) are coded with the channel "none". Scarcity classes are multi-label and were assigned from the problem the work states it addresses and from its evaluation setting, not from what the method could do in principle. Contribution type records the primary contribution. A work is coded as a multi-site study only when its main contribution is training or validating on data held by two or more independent institutions; method papers that also report an external validation remain methods, and simulated partitions of a single dataset count as single-dataset evidence. The full coding is listed in the online supplement and released with the paper as a CSV file together with the scripts that produce every count and figure in Section 7.

Data extraction. For the works evaluated on health data, we also extracted the datasets, their size and number of sites, and the headline result, from each work's abstract or own page, each field supported by a verbatim quote (Section 7.2).

5.4 Systematic search

The snowballed corpus was complemented by a systematic database search, executed on 8 October 2026 and reported following PRISMA 2020 [Page et al. 2021] (Figure 6). Its purpose is to test the corpus, not to replace it: we ask how much of the corpus a reproducible search recovers, and whether the coverage profile and the gaps of Section 7 hold in the wider literature.

Sources and query. PubMed was searched through the NCBI E-utilities, and medRxiv and the arXiv records indexed by Europe PMC through the Europe PMC API, from 2010 to the search date, in titles and abstracts. Europe PMC indexes only the life-science categories of arXiv, so computer-science preprints that are not also published in an indexed venue are not covered. Scopus, Web of Science, IEEE Xplore and the ACM Digital Library require subscriptions or API keys and were not searched; the scripts accept their exports, which are deduplicated and screened with the rest. The query, adapted to each source's syntax (both spellings replace the inner wildcard of decentrali*ed, which neither source supports), was:

```
("small data" OR "small sample*" OR "limited data" OR "data scarcity" OR "few-shot"
OR "low-resource" OR "rare disease*" OR "label-efficient" OR "data-efficient" OR
federated OR decentrali*ed OR "privacy-preserving") AND (health* OR clinical OR
medic* OR hospital* OR patient*) AND ("machine learning" OR "deep learning" OR
"transfer learning" OR "synthetic data" OR "data augmentation" OR "active learning"
```

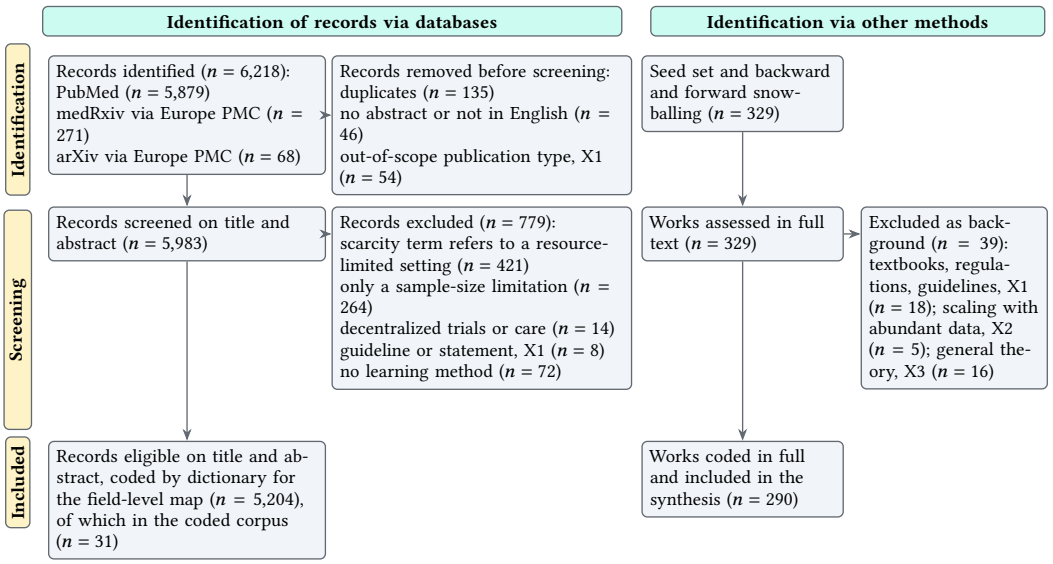


Fig. 6. PRISMA 2020 flow diagram. Left: the database search; its eligible records are coded by dictionary for the field-level map of Section 5.4 and are not part of the coded synthesis. Right: the seed set and snowballing of Section 5, which give the 290 works coded in full and analyzed in Sections 6 and 7.

```
OR "semi-supervised" OR "self-supervised" OR "foundation model*" OR "hierarchical
model*")
```

Screening. Records were deduplicated by DOI and normalized title, and titles and abstracts were screened against criteria I1, I2 and X1 to X3 by explicit rules, so that the screening can be rerun exactly. The rules exclude records whose only scarcity term refers to a resource-limited setting rather than to scarce data, to a sample-size limitation stated in passing, or to decentralized trials or care; records that are guidelines or statements; and records with no learning method. A single reviewer checked the rules on 200 records, drawn in two random samples from both sides of the decision. Two rules that misfired in these checks were removed, and the final rules reached a precision of 63% and a negative predictive value of 94% against the reviewer's judgments. The screening is therefore deliberately sensitive: about a third of the eligible records are false positives, which the field-level analysis below tolerates but a synthesis would not. Eligible records were not assessed in full text.

How much of the corpus the search finds. Of the 201 corpus works that were published since 2010 and evaluated on health data, 144 are indexed in the searched sources, and the query retrieved 31 of them (22%). The missed works are mostly evidence of type I2, such as studies of external validation failure, hidden stratification or sample size, which describe a scarcity problem without the remedy vocabulary of the query, and methods that report a gain on a small dataset without naming scarcity. Broadening the vocabulary does not solve this: a version of the query with 39 additional scarcity terms and 16 additional method terms returned about 76,000 PubMed records and still found only about half of the corpus. This is the pattern that Greenhalgh and Peacock [2005] found for complex evidence, where protocol-driven searches located about a third of the relevant sources and snowballing most of the rest, and it is why the coded synthesis rests on the snowballed corpus while the search serves as a check on it.

The field-level map. The 5,204 eligible records were coded with dictionaries that encode the definitions of Sections 3 and 4. On the 147 corpus works with an abstract in Europe PMC, the dictionary coder agreed with the manual primary channel in 72% of cases (Cohen's $\kappa = 0.65$), and recovered 63% of the manually assigned scarcity classes. In the field, pooling is the most frequent primary channel (34%), followed by proxy (20%) and re-expression (12%); prior (3%), unlabeled (5%) and procurement (1%) are rare, and 26% of records match no channel. Pooling holds a similar share of the corpus (102 of 290 works), so its dominance is a property of the literature and not only of our selection. Cell counts of the two maps are moderately correlated (Spearman $\rho = 0.55$), and 7 of the 10 gap cells of Figure 9 also fall in the bottom third of the field-level map: the temporal-obsolescence row and procurement for measurement poverty are as empty in the wider literature as in the corpus. Three gaps are not confirmed. Dimensional excess addressed by pooling is well populated in the field, mostly by federated and privacy-preserving genomic studies, and dimensional excess addressed by unlabeled data moderately so; both are under-covered by the corpus rather than by the literature. Stratum sparsity addressed by pooling also appears well populated, but most of its records match only the generic term bias, and the records that explicitly borrow strength across strata are almost all trial designs, such as basket trials and Bayesian subgroup analyses, which are outside the scope of this review (Section 1). Finally, 4,086 of the eligible records were published in 2023 or later, which confirms that the decline after 2022 in Figure 8 reflects the corpus and not the field.

5.5 Threats to validity

Four limitations qualify everything in the next two sections. First, *the coded corpus is curated, not exhaustive*. It was built to cover every cell of the taxonomy and to support the argument of Section 8, and counts describe the corpus, not the field. The systematic search of Section 5.4 bounds this limitation: it confirms the dominance of pooling and seven of the ten gaps, and it identifies three cells and the most recent years as under-covered by the corpus. Second, *coding was done by a single coder*, the first author, and inter-rater agreement has not been measured; the agreement with an independent dictionary coder ($\kappa = 0.65$) is a check of consistency, not a substitute. Third, the primary channel is a judgment about where a method's gain comes from, and some works, foundation models trained with self-supervision for example, sit between two channels. Fourth, the search covered PubMed, medRxiv and the life-science part of arXiv but not the subscription databases or the computer-science part of arXiv, and its eligible records were screened by rules checked on a sample rather than read in full, so the field-level map is a check on proportions, not a second corpus.

6 EXISTING SOLUTIONS, ORGANIZED BY PROVENANCE

We now review solution families in the order of the provenance channels of Section 4. For each family we describe the mechanism, the scarcity classes of Figure 3 it addresses, the healthcare evidence, and the conditions under which it fails. Pooling receives the most space, because it is the only strong channel against custodial scarcity and therefore the foundation of Section 8. Table 7 gives a compact overview. By primary channel, the corpus holds 102 pooling, 60 proxy, 25 prior, 23 unlabeled, 15 procurement and 32 re-expression works, plus 33 that document a problem without proposing a remedy; the online supplement lists and codes every one of them.

6.1 How the main remedies work

Before the review by channel, we state the mechanism of the canonical family of each channel in a common notation (Figure 7, Table 6). Site k holds $\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^{n_k}$, its empirical risk is $F_k(\theta) = n_k^{-1} \sum_{i \in \mathcal{D}_k} \ell(f_\theta(x_i), y_i)$, and $N = \sum_k n_k$.

P1 Pooling. Federated averaging [McMahan et al. 2017] minimizes the pooled risk $F(\theta) = \sum_k (n_k/N) F_k(\theta)$ without moving records. In round t each site runs a few local gradient steps from the current model, $\theta_k^{t+1} = \theta^t - \eta \sum_s \nabla F_k(\cdot)$, and a server averages,

$$\theta^{t+1} = \sum_{k=1}^{K} \frac{n_k}{N} \theta_k^{t+1}. \tag{4}$$

Personalized variants keep one model per site and tie it to the shared one, as in Ditto [Li et al. 2021a]: $\min_{\theta_k} F_k(\theta_k) + \frac{\lambda}{2}\|\theta_k - \theta^*\|^2$, with λ to be tuned. Privacy is added on top. A randomized algorithm $\mathcal{A}$ is (ϵ, δ)-differentially private [Dwork et al. 2006; Dwork and Roth 2014] if, for any two datasets $\mathcal{D}, \mathcal{D}'$ that differ in one patient and every set of outputs S,

$$\Pr[\mathcal{A}(\mathcal{D}) \in S] \;\le\; e^{\epsilon} \Pr[\mathcal{A}(\mathcal{D}') \in S] + \delta. \tag{5}$$

The Gaussian mechanism achieves it by releasing $g(\mathcal{D}) + \mathcal{N}(0, s^2 I)$ with $s = \Delta\sqrt{2\ln(1.25/\delta)}/\epsilon$ for $\epsilon < 1$, where Δ is the largest change one patient can cause in g; DP-SGD clips each patient's gradient to bound Δ and composes the cost over steps [Abadi et al. 2016]. Secure aggregation [Bonawitz et al. 2017] reveals only the sum: every pair of sites shares a random mask $r_{kj} = -r_{jk}$, each sends $x_k + \sum_j r_{kj}$, and the masks cancel in $\sum_k x_k$.

P2 Proxy. Transfer learning starts from parameters θ^{src} fitted on a large related source and fine-tunes on the target, often with a penalty that keeps the solution near the start [Li et al. 2018; Pan and Yang 2010]:

$$\hat{\theta} = \arg\min_{\theta} \; F_k(\theta) + \lambda \, \|\theta - \theta^{\text{src}}\|^2. \tag{6}$$

Self-supervised pretraining creates θ^{src} without labels. Contrastive learning [Chen et al. 2020a] pulls the embeddings z_i, z_j of two augmented views of one input together and pushes other inputs away, $\ell_{ij} = -\log\!\big(e^{\text{sim}(z_i,z_j)/T} / \sum_{m \neq i} e^{\text{sim}(z_i,z_m)/T}\big)$; masked autoencoders reconstruct hidden patches instead [He et al. 2022].

P3 Prior. Bayesian inference combines the likelihood with a prior, $p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)\, p(\theta)$ [Gelman et al. 2013]. In the hierarchical normal model that Section 8 builds on, $\theta_k \sim \mathcal{N}(\mu, \tau^2)$ and $\hat{\theta}_k \mid \theta_k \sim \mathcal{N}(\theta_k, v_k)$, and the posterior mean shrinks the local estimate toward the population,

$$\mathbb{E}[\theta_k \mid \hat{\theta}_k] = \alpha_k \, \hat{\theta}_k + (1 - \alpha_k)\, \mu, \qquad \alpha_k = \frac{\tau^2}{\tau^2 + v_k}, \tag{7}$$

the empirical Bayes form of Stein's estimator [Efron and Morris 1973; Stein 1956]. The smaller the site, the larger v_k and the more it borrows. Knowledge that is not data enters the same way, as a prior on parameters, as a constraint from a mechanistic model [Raissi et al. 2019], or as structure from an ontology [Choi et al. 2017a].

P4 Unlabeled. Semi-supervised learning adds a term on the n_U unlabeled inputs. Pseudo-labelling [Lee 2013] and FixMatch [Sohn et al. 2020] train on the model's own confident predictions,

$$\mathcal{L} = F_k(\theta) + \lambda \, \frac{1}{n_U} \sum_{u} \mathbb{1}\big[\max p_\theta(y \mid a_w(u)) \ge c\big] \; \ell\big(f_\theta(a_s(u)),\, \hat{y}_u\big), \qquad \hat{y}_u = \arg\max_y p_\theta(y \mid a_w(u)), \tag{8}$$

where a_w and a_s are weak and strong augmentations and c a confidence threshold.

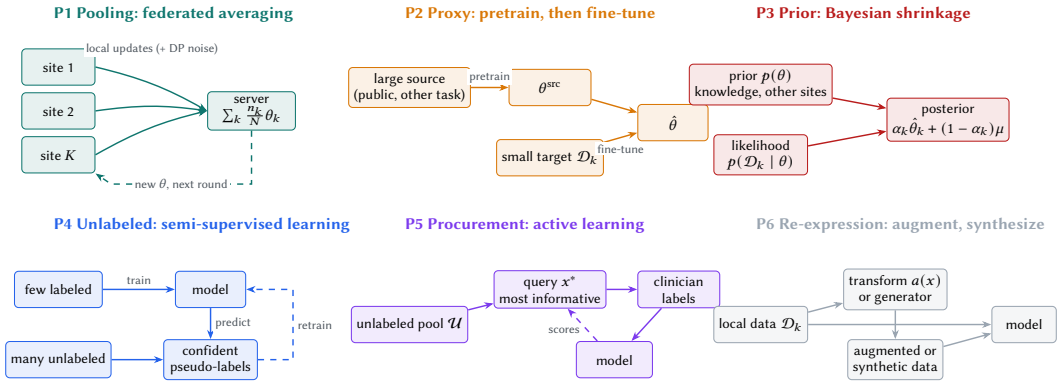


Fig. 7. How the canonical family of each provenance channel works (Section 6.1). P1: sites update a shared model locally and a server averages the updates (Equation (4)); serverless variants replace the server by neighbour averaging. P2: a model fitted on a large source is fine-tuned on the small target (Equation (6)). P3: the prior and the local likelihood combine into a posterior that shrinks small sites toward the population (Equation (7)). P4: the model labels unlabeled inputs it is confident about and retrains on them (Equation (8)). P5: the model chooses which inputs a clinician labels next (Equation (9)). P6: local data are transformed or resynthesized; the added samples carry no new information about the target.

P5 Procurement. Active learning spends a labelling budget where it is most informative [Settles 2009]. With a pool $\mathcal{U}$ of unlabeled candidates, it queries

$$x^* = \arg\max_{x \in \mathcal{U}} \mathrm{I}\big[y; \theta \mid x, \mathcal{D}_k\big], \tag{9}$$

the mutual information between the label and the parameters (BALD), estimated in deep networks by Monte Carlo dropout [Gal et al. 2017]; predictive entropy is the simpler alternative.

P6 Re-expression. Augmentation minimizes the risk over transformed copies, $\mathbb{E}_{(x,y)}\mathbb{E}_a\, \ell(f_\theta(a(x)), y)$, with transformations chosen to preserve the label [Shorten and Khoshgoftaar 2019], or over interpolations such as mixup, $\tilde{x} = \lambda x_i + (1-\lambda)x_j$, $\tilde{y} = \lambda y_i + (1-\lambda)y_j$ [Zhang et al. 2018]. Synthetic data come from a generator fitted to the same data, for example a denoising diffusion model trained with $\mathbb{E}_{t,x_0,\varepsilon}\|\varepsilon - \varepsilon_\phi(x_t, t)\|^2$ [Ho et al. 2020] or a GAN [Goodfellow et al. 2014]; minority classes can also be oversampled by interpolation [Chawla et al. 2002]. Because the generator is fitted to $\mathcal{D}_k$, these samples add no information about θ_k that $\mathcal{D}_k$ did not contain (Proposition 4.1); they change how it is used.

6.2 P1 Pooling: bringing the same task from other custodians

Centralized pooling and common data models. The simplest form of pooling is a shared repository. Public datasets such as MIMIC-III [Johnson et al. 2016] have been central to the field precisely because they removed custodial scarcity for one population. Distributed networks built on common data models, such as OHDSI on the OMOP model [Hripcsak et al. 2015; Overhage et al. 2012], keep data local but harmonize their structure so that the same analysis can be run everywhere. Harmonization reduces C2 scarcity but does not by itself allow joint model fitting.

Distributed statistics. Long before federated learning, biostatistics developed ways to fit models across sites without moving records. Meta-analysis combines site-level estimates, and random-effects meta-analysis [DerSimonian and Laird 1986] explicitly models between-site heterogeneity, a point we return to in Section 8. Individual participant data meta-analysis is more powerful but

Table 6. Canonical mechanism of each provenance channel: what it optimizes or computes, what it needs, and its main cost in a network of small hospitals.

Channel	Canonical family	Needs	Main cost
P1 Pooling	federated averaging [McMahan et al. 2017], personalized FL [Li et al. 2021a], DP [Dwork et al. 2006]	other custodians with the same task; a server or a communication graph	privacy noise and rounds of communication; one shared model can hurt atypical sites
P2 Proxy	fine-tuning [Pan and Yang 2010], self-supervised pretraining [Chen et al. 2020a; He et al. 2022]	a large related source	gains shrink as the source drifts from the target
P3 Prior	Bayesian and hierarchical models [Gelman et al. 2013]	a credible prior or mechanistic model	a wrong prior biases every site toward it
P4 Unlabeled	pseudo-labelling [Lee 2013], FixMatch [Sohn et al. 2020]	many unlabeled inputs from the target	confirmation bias when early predictions are wrong
P5 Procurement	active learning [Gal et al. 2017; Settles 2009]	a clinician who can label on request	labelling time; the first model must be good enough to choose
P6 Re-expression	augmentation [Shorten and Khoshgoftaar 2019], mixup [Zhang et al. 2018], generative models [Ho et al. 2020]	only the local data	no new information about the target (Proposition 4.1)

needs record-level access [Riley et al. 2010]. Distributed regression exchanges sufficient statistics or gradients: GLORE fits logistic regression across sites by exchanging Newton-Raphson summaries [Wu et al. 2012]; DataSHIELD wraps such procedures in disclosure controls [Gaye et al. 2014]; ODAL and related one-shot algorithms use surrogate likelihoods to obtain near-pooled estimates with one round of communication [Duan et al. 2020a; Jordan et al. 2019]; and the Personal Health Train infrastructure sends analyses to data stations rather than data to analyses [Beyan et al. 2020]. Jochems et al. [2016] demonstrated a distributed model across hospitals in radiotherapy without any data leaving them. These methods are well understood, often exact, and auditable. Their limitation is model class: they work best for low-dimensional parametric models.

Federated learning. Federated averaging [McMahan et al. 2017] trains a shared model by alternating local gradient steps with server-side averaging. Healthcare adopted it quickly [Kaissis et al. 2020; Nguyen et al. 2022; Rieke et al. 2020; Xu et al. 2021]. Early work on institutional cyclical training [Chang et al. 2018] and on brain tumor segmentation showed that federated models could reach 99% of the quality of a model trained on centrally pooled data [Sheller et al. 2020]. EXAM, trained across 20 institutions to predict oxygen needs of COVID-19 patients, improved average AUC by 16% and generalisability by 38% over single-site models [Dayan et al. 2021]. The largest study to date, across 71 sites and 6,314 glioblastoma patients, improved delineation of the surgically targetable tumor by 33% over a model trained on public data [Pati et al. 2022]. FLamby provides seven cross-silo healthcare benchmarks with natural splits, which makes such claims testable [Ogier du Terrail et al. 2022].

The central difficulty is heterogeneity. When site distributions differ, local updates drift apart and the averaged model can be worse than local ones [Hsieh et al. 2020; Zhao et al. 2018]. Remedies include proximal regularization (FedProx) [Li et al. 2020c], control variates (SCAFFOLD) [Karimireddy et al. 2020], site-specific normalization (FedBN) [Li et al. 2021b], frequency-space augmentation for domain generalization (FedDG) [Liu et al. 2021], minimax objectives that protect the worst-off site [Mohri et al. 2019], and ensemble distillation [Lin et al. 2020]. Medical versions of these ideas harmonize amplitude spectra and local drift across scanners [Jiang et al. 2022], close the gap between federated and centralized segmentation on unseen sites [Xu et al. 2022], adapt across acquisition sites in multi-site fMRI [Li et al. 2020a], and combine self-supervised pretraining with federated fine-tuning to tolerate heterogeneous, sparsely labeled sites [Yan et al. 2023]. Federated transfer learning has also been applied to wearable sensor data [Chen et al. 2020b], and federated tensor factorization and patient-similarity learning to EHR phenotyping [Kim et al. 2017; Lee et al. 2018].

Clinical deployments. Beyond the landmark studies, the corpus contains a steady line of real multi-institution work. In structured EHR data, federated models have predicted hospitalizations from records held by separate custodians [Brisimi et al. 2018], used patient clustering to cope with heterogeneous intensive care populations [Huang et al. 2019], and predicted COVID-19 mortality across the hospitals of one health system [Vaid et al. 2021]. In imaging, federated training has been run across institutions for prostate segmentation [Sarma et al. 2021], breast density classification [Roth et al. 2020], COVID-19 diagnosis on CT [Bai et al. 2021; Dou et al. 2021], and prediction of response to neoadjuvant chemotherapy in triple-negative breast cancer from whole-slide images [Ogier du Terrail et al. 2023]; swarm learning has been applied to histopathology [Saldanha et al. 2022], and weakly supervised federated learning to gigapixel pathology [Lu et al. 2022]. The Personal Health Train trained a model on more than 20,000 lung cancer patients without moving their data [Deist et al. 2020], building on earlier distributed survival models in non-small cell lung cancer [Jochems et al. 2017]. At network scale, the OHDSI LEGEND study compared first-line antihypertensive classes across millions of patients in observational databases from several countries [Suchard et al. 2019], using a standardized framework for patient-level prediction [Reps et al. 2018], and web services made distributed logistic and Cox regression available to clinical researchers [Jiang et al. 2013; Lu et al. 2015], with one-shot algorithms extending the approach to time-to-event outcomes [Duan et al. 2020b]. Most other pooling papers in the corpus, including several influential ones, evaluate on a single dataset partitioned to simulate sites [Guo et al. 2021; Li et al. 2019; Sheller et al. 2019]. That distinction matters in Section 7.

Personalized federated learning. A more radical response is to stop looking for one model. Federated multi-task learning [Smith et al. 2017], meta-learning based personalization [Fallah et al. 2020], Moreau-envelope regularization [T. Dinh et al. 2020], Ditto [Li et al. 2021a], shared representations with local heads (FedRep) [Collins et al. 2021], mixtures of distributions [Marfoq et al. 2021], attention-weighted collaboration among similar silos [Huang et al. 2021a] and clustered federated learning [Ghosh et al. 2020] all learn a model per site while sharing statistical strength. Tan et al. [2023] survey the area. Theory on shared representations explains why this works when sites share a low-dimensional structure [Baxter 2000; Du et al. 2021; Maurer et al. 2016; Tripuraneni et al. 2020].

Serverless and decentralized learning. Federated learning in its standard form assumes a central server that all sites trust to aggregate. Many hospital networks lack such an entity, or cannot agree on who should play it. Decentralized SGD replaces the server with neighbour averaging over a communication graph, and can match centralized rates when the graph mixes well [Koloskova et al. 2020; Lian et al. 2017; Nedić and Ozdaglar 2009]. Gossip algorithms make this robust to asynchrony

[Boyd et al. 2006; Xiao and Boyd 2004]. Personalized decentralized learning over a similarity graph was introduced by Vanhaesebrouck et al. [2017] and extended with differential privacy by Bellet et al. [2018]. Cyffers and Bellet showed that decentralization itself amplifies privacy, because each party sees only what reaches it through the graph [Cyffers and Bellet 2022]. In healthcare, Swarm Learning used a blockchain to coordinate peer-to-peer training without a central server across four clinical use cases [Warnat-Herresthal et al. 2021], and ProxyFL shares only differentially private proxy models among peers [Kalra et al. 2023]. Split learning partitions the network itself between parties [Gupta and Raskar 2018; Vepakomma et al. 2018], and vertical federated learning addresses the case where sites hold different features for the same patients [Yang et al. 2019]. Peer-to-peer training has been shown for brain segmentation [Roy et al. 2019] and tumor segmentation in hospital networks [Tedeschini et al. 2022], and split learning has been evaluated on clinical imaging tasks [Poirot et al. 2019]. These healthcare demonstrations are few and mostly simulated, a gap that Section 8 addresses.

Privacy technologies. Sharing parameters or gradients is not private by default. Gradients can be inverted to reconstruct training images [Geiping et al. 2020; Zhu et al. 2019], and trained models leak membership [Carlini et al. 2022; Shokri et al. 2017]. Differential privacy (DP) [Dwork et al. 2006; Dwork and Roth 2014] bounds what any output reveals about an individual and has been adapted to deep learning [Abadi et al. 2016; Mironov 2017] and to federated training at the user or site level [McMahan et al. 2018; Wei et al. 2020]. Secure aggregation reveals only the sum of updates [Bonawitz et al. 2017]; secure multi-party computation [Evans et al. 2018] and homomorphic encryption [Gentry 2009] allow computation on encrypted data, with multiparty homomorphic encryption demonstrated for federated analytics in precision medicine [Froelicher et al. 2021]. The costs are well known. DP degrades accuracy, more so for small datasets, high-dimensional models and under-represented groups [Bagdasaryan et al. 2019; Jayaraman and Evans 2019; Tramèr and Boneh 2021]. Cryptographic methods protect the computation but not what the final model reveals. Robust aggregation protects against malicious participants [Blanchard et al. 2017]. In healthcare, end-to-end privacy-preserving training combining secure aggregation and DP was demonstrated on pediatric chest radiographs [Kaissis et al. 2021], DP has been studied for medical imaging [Ziller et al. 2021] and for federated pathology [Adnan et al. 2022] and EHR models [Choudhury et al. 2019], and cryptographic systems support distributed cohort exploration [Raisaro et al. 2019] and genome-wide association studies [Cho et al. 2018]. The threat is concrete. Gradient inversion has been demonstrated on medical federated training, with risk governed by batch size and the number of local steps [Hatamizadeh et al. 2023]; deep networks can re-identify patients from chest radiographs [Packhäuser et al. 2022]; face-recognition software can identify research participants from MRI [Schwarz et al. 2019]; and re-identification attacks on health data have a long record [El Emam et al. 2011].

6.3 P2 Proxy: borrowing from related tasks

Transfer learning. Pretraining on a large source and fine-tuning on a small target is the most widely used remedy [Pan and Yang 2010]. Its value in medical imaging is more nuanced than often assumed. Raghu et al. found that ImageNet pretraining gave little performance benefit on large medical imaging tasks and that much smaller architectures performed comparably, with the benefit concentrated in faster convergence and in the smallest data regimes [Raghu et al. 2019]. Domain adaptation theory gives the reason: the target risk is bounded by the source risk plus a divergence between domains and the error of the best joint hypothesis [Ben-David et al. 2010]. When the proxy is far from the target, these terms dominate. Earlier empirical studies on medical images found that fine-tuning a pretrained network performed at least as well as training from scratch and was more

robust to reductions in training data [Shin et al. 2016; Tajbakhsh et al. 2016], and the specialist-level results on skin and retinal images were themselves fine-tuned ImageNet models trained on large labeled sets [Esteva et al. 2017; Gulshan et al. 2016]. Medical sources make the proxy closer: 3D pretraining on aggregated segmentation datasets [Chen et al. 2019], self-supervised Models Genesis [Zhou et al. 2021] and the RadImageNet database [Mei et al. 2022] all aim at this, and domain adaptation methods target the remaining shift directly [Guan and Liu 2022; Kamnitsas et al. 2017].

Self-supervised and foundation models. Self-supervised pretraining by contrastive learning [Chen et al. 2020a] or masked reconstruction [He et al. 2022] removes the need for source labels and allows pretraining on in-domain unlabeled data. Azizi et al. showed that self-supervised pretraining on unlabeled medical images improved top-1 accuracy by 6.7% on dermatology classification and mean AUC by 1.1% on chest X-ray classification over supervised ImageNet pretraining [Azizi et al. 2021]. RETFound, pretrained on 1.6 million unlabeled retinal images, adapted to ocular and systemic disease detection with fewer labels than comparison models [Zhou et al. 2023]. Vision-language pretraining [Radford et al. 2021; Wang et al. 2022], large language models that encode clinical knowledge [Singhal et al. 2023] or extract clinical information from few examples [Agrawal et al. 2022], and EHR foundation models [Rasmy et al. 2021; Steinberg et al. 2021] extend the idea across modalities; Moor et al. [2023] argue for generalist medical models, and Bommasani et al. [2021] discuss their risks. Medical image-text pretraining [Huang et al. 2021b; Zhang et al. 2022] and contrastive pretraining on chest radiographs [Sowrirajan et al. 2021] improve label efficiency, and a model trained only on radiographs paired with their reports detected several pathologies at expert level without explicit labels [Tiu et al. 2022]. Computational pathology has seen the fastest growth, with general-purpose slide and tile encoders [Chen et al. 2024; Xu et al. 2024], vision-language models [Lu et al. 2024], and a foundation model that reported clinical-grade detection of rare cancers [Vorontsov et al. 2024]; segmentation [Ma et al. 2024] and cancer imaging biomarkers in limited-data settings [Pai et al. 2024] follow the same pattern. For EHR, sequence models pretrained on large patient histories [Choi et al. 2016; Li et al. 2020b] and a health-system scale language model trained on clinical notes [Jiang et al. 2023] transfer to many prediction tasks, although a review of EHR foundation models found evaluations narrow and rarely tied to clinical value [Wornow et al. 2023]. Clinical and biomedical language models [Alsentzer et al. 2019; Lee et al. 2020] and general large language models [Nori et al. 2023] bring textual knowledge, and transcriptomic foundation models pretrained on tens of millions of single cells transfer to settings with few samples [Cui et al. 2024; Theodoris et al. 2023]. Parameter-efficient fine-tuning [Hu et al. 2022] reduces the number of parameters to adapt, which lowers $n^\star$ directly.

Tabular data. Much of healthcare is tabular, where deep learning has a weaker record. Tree ensembles still outperform deep networks on typical medium-sized tabular benchmarks [Grinsztajn et al. 2022]. Two recent lines are directly about small data. TabPFN is a transformer trained only on synthetic datasets drawn from a prior over structural causal models, which performs Bayesian inference in a single forward pass [Hollmann et al. 2023]; its second version outperformed an ensemble of strong baselines tuned for four hours, in 2.8 seconds, on datasets of up to 10,000 samples [Hollmann et al. 2025]. In the language of our taxonomy, it converts a prior (P3) into a proxy (P2) by sampling. TabLLM serializes rows into text and uses a large language model's knowledge for few-shot classification [Hegselmann et al. 2023], drawing on prior knowledge absorbed during pretraining. Both point to a general principle: in the small regime, the quality of the prior matters more than the flexibility of the model.

Few-shot and meta-learning. Meta-learning learns an inductive bias from a distribution of related tasks, so that a new task can be learned from few examples [Baxter 2000; Finn et al. 2017; Hospedales

et al. 2022; Snell et al. 2017; Wang et al. 2020]. Its assumptions are demanding: many related training tasks drawn from the same distribution as the target task. In healthcare that distribution is often what is missing. Medical applications exist for clinical risk prediction from limited EHR [Zhang et al. 2019], rare skin conditions [Mahajan et al. 2020; Prabhu et al. 2019] and volumetric segmentation from few annotated scans [Roy et al. 2020], and facial-phenotype models for rare genetic syndromes [Gurovich et al. 2019; Hsieh et al. 2022] show how a proxy trained on many syndromes can support conditions with only a handful of known patients.

6.4 P3 Prior: knowledge that is not data

Bayesian and hierarchical modeling. Bayesian inference makes the prior explicit [Gelman et al. 2013]. Hierarchical models are the canonical way to share strength across groups, and their frequentist counterpart, shrinkage, dominates separate estimation under squared error in the Gaussian model with three or more groups [Efron and Morris 1973; Stein 1956]. Penalized regression [Hastie et al. 2009; Tibshirani 1996] is the prior in disguise, and its use is recommended for small clinical prediction datasets, although it does not replace adequate sample size [Riley et al. 2020; van Smeden et al. 2019]. The clinical evidence is more cautious than the theory. Shrinkage improved small-sample prognostic models in early comparisons [Steyerberg et al. 2000], but later studies showed that it does not guarantee better performance and is least reliable precisely when samples are smallest, because the amount of shrinkage is itself estimated with large uncertainty [Riley et al. 2021; Van Calster et al. 2020]. This is the same problem that the blueprint of Section 8 faces when it estimates the heterogeneity τ^2, and Proposition 8.1 quantifies it.

Mechanistic and physics-informed models. When the system has known dynamics, as in pharmacokinetics, cardiovascular hemodynamics or tumor growth, embedding them as constraints or residuals dramatically reduces the data needed [Karniadakis et al. 2021; Raissi et al. 2019]. von Rueden et al. [2023] classify how knowledge can enter learning: through the training data, the hypothesis set, the learning algorithm or the final hypothesis. Medical examples include physics-informed networks for cardiac activation mapping [Sahli Costabal et al. 2020] and arterial blood pressure [Kissas et al. 2020], biologically informed networks for sparse experimental data [Lagergren et al. 2020], and neural ordinary differential equations for pharmacokinetics [Lu et al. 2021a]. Human concept learning from one example, formalized as probabilistic program induction [Lake et al. 2015], is the limiting case of what a strong prior can do.

Ontologies and knowledge graphs. Medical codes are organized in hierarchies. GRAM uses these ontologies to give rare codes representations close to their ancestors, improving prediction for codes that appear only a few times [Choi et al. 2017a]. This directly targets the long tail created by E3 and O1. KAME and GAMENet extend the idea with knowledge-graph attention and drug-interaction graphs [Ma et al. 2018; Shang et al. 2019], health knowledge graphs can be learned from records [Rotmensch et al. 2017], and the Human Phenotype Ontology [Köhler et al. 2021] supplies the structured vocabulary that rare-disease models rely on.

Causal knowledge. Causal structure tells a model which relations should be stable across sites and time. Invariance-based learning [Arjovsky et al. 2019], causal representation learning [Pearl 2009; Schölkopf et al. 2021] and causal analysis of imaging datasets [Castro et al. 2020] target T1 and T2 scarcity by excluding features whose relation to the outcome is site-specific. Graphical models of the shift itself can identify which relations will transport [Subbaswamy et al. 2019]. The empirical record is sobering, however: under careful model selection, domain generalization methods did not consistently beat plain empirical risk minimization, either on general benchmarks [Gulrajani and Lopez-Paz 2021] or in clinical settings [Zhang et al. 2021].

6.5 P4 Unlabeled: information in the inputs

Semi-supervised methods such as consistency regularization with pseudo-labels [Sohn et al. 2020] use unlabeled inputs to shape the decision boundary. Cheplygina et al. [2019] review semi-supervised, multiple-instance and transfer learning in medical imaging, and Tajbakhsh et al. [2020] review solutions for imperfect segmentation datasets. Weak supervision builds noisy labels from heuristics, knowledge bases and existing records and combines them with a generative label model [Ratner et al. 2017]. As Proposition 4.1 and the causal analysis of Schölkopf et al. [2012] imply, these methods help when the input distribution carries information about the label mechanism. They are effective against E1 and much less so against O-type scarcity.

In medical image segmentation, semi-supervised learning has been developed through self-training [Bai et al. 2017], uncertainty-aware and transformation-consistent self-ensembling [Li et al. 2021c; Yu et al. 2019], and contrastive pretraining on the unlabeled part of the same dataset [Chaitanya et al. 2020; Taleb et al. 2020], with few-shot segmentation trained from superpixels without annotation [Ouyang et al. 2020]; Krishnan et al. [2022] review self-supervision in medicine. In computational pathology, attention-based multiple-instance learning [Ilse et al. 2018] made slide-level labels sufficient: a system trained on tens of thousands of slides with only diagnostic labels reached clinical-grade performance [Campanella et al. 2019], and CLAM made the approach data-efficient [Lu et al. 2021b]. Large chest radiograph datasets were labeled by mining radiology reports [Irvin et al. 2019; Wang et al. 2017], which trades label poverty (E1) for measurable label noise (E2). In EHR data, anchor-based phenotyping [Halpern et al. 2016], learning from noisy labels [Agarwal et al. 2016] and weak supervision of clinical text [Fries et al. 2019; Wang et al. 2019] play the same role.

6.6 P5 Procurement: acquiring the right new data

Procurement is the most neglected channel in machine learning papers and the most important one in clinical research. Sample size calculations [Riley et al. 2020] turn smallness into a design problem before data collection starts. Active learning chooses which examples to label next to maximize information [Settles 2009], and has been widely studied in medical imaging [Budd et al. 2021]. Representative methods select annotation regions by uncertainty and representativeness [Yang et al. 2017], use learned feature distances [Smailagic et al. 2018], or exploit Bayesian uncertainty [Gal et al. 2017; Nath et al. 2021]. Learning curves fitted on pilot data can predict how many labeled examples a target accuracy needs [Cho et al. 2015; Figueroa et al. 2012], and Balki et al. [2019] review sample-size methods for machine learning in imaging; for continuous outcomes the clinical criteria have their own form [Riley et al. 2019a]. Procurement is the smallest channel in our corpus (15 works), which is at odds with its importance. Data valuation methods such as Data Shapley [Ghorbani and Zou 2019] estimate which records or sources contribute most, which could guide both acquisition and the fair crediting of contributors in a consortium.

6.7 P6 Re-expression: augmentation and synthetic data

Data augmentation encodes invariances and is standard practice [Shorten and Khoshgoftaar 2019; Zhang et al. 2018]. Oversampling methods such as SMOTE [Chawla et al. 2002] interpolate minority examples. Deep generative models, from GANs [Goodfellow et al. 2014] to diffusion models [Ho et al. 2020; Kazerouni et al. 2023], can produce realistic medical images, and tabular and EHR generators include medGAN [Choi et al. 2017b] and CTGAN [Xu et al. 2019]. In a widely cited study, GAN-based augmentation improved liver lesion classification from 78.6% to 85.7% sensitivity and from 88.4% to 92.4% specificity on a dataset of 182 lesions [Frid-Adar et al. 2018]. Similar results have been reported for brain MRI synthesis used for both augmentation and anonymization

[Shin et al. 2018], for segmentation [Bowles et al. 2018], for 3D generation with diffusion models [Khader et al. 2023], and for learned spatial and appearance transformations that make one-shot segmentation possible [Chaitanya et al. 2021; Zhao et al. 2019]; Yi et al. [2019] and Chlap et al. [2021] review the area. A large study found that diffusion-based augmentation improved fairness and robustness of medical classifiers under distribution shift [Ktena et al. 2024]; in our terms, the information came from the generator's training data, not from the act of sampling. For structured records, EHR generators [Baowaly et al. 2019; Zhang et al. 2020], anonymization-oriented synthesis [Beaulieu-Jones et al. 2019; Yoon et al. 2020], diffusion and language-model tabular generators [Borisov et al. 2023; Kotelnikov et al. 2023], and the rule-based Synthea simulator [Walonoski et al. 2018] have been proposed and evaluated [Goncalves et al. 2020]. Rebalancing deserves a warning: in clinical risk prediction, class imbalance corrections produced badly calibrated risks without improving discrimination [van den Goorbergh et al. 2022], despite their popularity [Fernández et al. 2018].

Synthetic data are also promoted as a privacy technology [Chen et al. 2021a; Jordon et al. 2022]. Here the evidence is sobering. Stadler et al. found that synthetic data either fail to protect against linkage and attribute inference or lose much of their utility, and do not provide a better privacy-utility trade-off than traditional anonymization [Stadler et al. 2022]. Formally private generators such as PATE-GAN [Jordon et al. 2019], built on the PATE framework [Papernot et al. 2017], provide guarantees at the cost of fidelity. And recursive training on generated data causes models to lose the tails of the distribution [Shumailov et al. 2024], which in healthcare are exactly the rare cases and minority groups that motivated the effort. Proposition 4.1 summarizes the position: synthetic data are a useful vehicle for information that already exists (in a pretrained generator, or in another site's data under a privacy guarantee), but not a source of it.

Taken together, the six channels cover a wide range of techniques, but the evidence behind them is uneven. Some channels have been tested at multi-site scale, others mainly on internal splits of a single dataset. The next section reads the reported results through the taxonomy to separate the two.

7 WHAT THE EVIDENCE SHOWS, AND WHAT IT DOES NOT

We read the evidence at two levels: the shape of the coded corpus, which shows where the literature has worked, and the quantitative findings it reports, which show what that work achieved. All counts below come from the coding of Section 5 and are subject to the threats to validity listed there.

7.1 Where the literature has worked

Figure 8 profiles the corpus. Two thirds of the 290 works (204) appeared in or after 2019. Of the 206 works evaluated on health data, almost half (100) use imaging, while structured EHR (35), tabular registries (36), clinical text (9), omics (4), physiological signals (2) and multimodal data (20) share the rest; the remaining 84 are general methods that health applications build on. Most works propose a method (152). Only 18 have a multi-institution study as their primary contribution, and every one of them is a pooling study. No proxy, prior, unlabeled, procurement or re-expression work in the corpus has a multi-institution study as its main result, although several report external validation alongside a method.

Figure 9 sets the coverage of each cell of the taxonomy against the judged potential of Figure 5. Coverage is very uneven. Label poverty (E1, 100 works) and custodial fragmentation (C1, 97 works) dominate, and they are served by different channels: E1 mainly by proxy (56) and unlabeled data (33), C1 almost only by pooling (93). Temporal obsolescence (T1) and measurement poverty (E2) are targeted by 8 works each, and stratum sparsity (O3) by 10, although these are the classes that

Table 7. Representative solution families by provenance channel, with the scarcity classes they target, the main assumption, and the main limitation in healthcare.

Family	Representative work	Targets	Key assumption	Main limitation
P1 Pooling				
Common data models	OHDSI/OMOP [Hripcsak et al. 2015; Overhage et al. 2012]	C2	semantic harmonization is feasible	harmonization cost; no joint fitting by itself
Distributed statistics	GLORE [Wu et al. 2012], ODAL [Duan et al. 2020a], DataSHIELD [Gaye et al. 2014]	C1, O2	low-dimensional parametric model	limited model classes
Federated learning	FedAvg [McMahan et al. 2017], EXAM [Dayan et al. 2021], [Pati et al. 2022]	C1, O1, O2	trusted server; moderate heterogeneity	non-i.i.d. drift [Hsieh et al. 2020]; leakage without DP [Geiping et al. 2020]
Personalized FL	Ditto [Li et al. 2021a], FedRep [Collins et al. 2021], IFCA [Ghosh et al. 2020]	C1, T2	shared low-dim. structure	small sites may still overfit local heads
Decentralized learning	D-PSGD [Lian et al. 2017], Swarm [Warnat-Herresthal et al. 2021], ProxyFL [Kalra et al. 2023]	C1	connected graph	slow mixing on sparse graphs
P2 Proxy				
Transfer learning	[Pan and Yang 2010; Raghu et al. 2019]	E1, E3	source close to target	negative transfer; limited gain on large tasks
Self-supervised and foundation models	[Azizi et al. 2021; Moor et al. 2023; Zhou et al. 2023]	E1, E3, T2	pretraining data cover target	bias inherited from pretraining; compute
Tabular priors	TabPFN [Hollmann et al. 2025], TabLLM [Hegselmann et al. 2023]	E3, O2	synthetic or textual prior matches reality	limited size and feature range; opacity
Meta-learning	MAML [Finn et al. 2017], ProtoNets [Snell et al. 2017]	O1, E1	many related tasks available	task distribution itself scarce
P3 Prior				
Hierarchical Bayes, shrinkage	[Efron and Morris 1973; Gelman et al. 2013]	O1 to O3, C1	exchangeability across groups	misspecified hyperpriors
Physics-informed	[Karniadakis et al. 2021; Raissi et al. 2019]	O1, E3	known mechanism	mechanism often partial in medicine
Ontology-guided	GRAM [Choi et al. 2017a]	E3, O1	ontology reflects similarity	ontologies built for billing
Causal and invariant	IRM [Arjovsky et al. 2019], [Castro et al. 2020]	T1, T2	stable causal mechanisms	needs diverse environments
P4 Unlabeled, P5 Procurement				
Semi- and weak supervision	FixMatch [Sohn et al. 2020], Snorkel [Ratner et al. 2017]	E1	cluster or manifold assumption	confirmation bias
Active learning, design	[Budd et al. 2021; Riley et al. 2020; Settles 2009]	E1, O2	labeling oracle available	cold start; sampling bias
P6 Re-expression				
Augmentation, SMOTE	[Chawla et al. 2002; Shorten and Khoshgoftaar 2019]	E1, O2	invariances known	no new information (Proposition 4.1)
Synthetic data	[Frid-Adar et al. 2018; Jordon et al. 2019; Xu et al. 2019]	C1, C2, O2	generator faithful	privacy-utility trade-off [Stadler et al. 2022]; tail loss [Shumailov et al. 2024]

the external-validation failures of Section 1 point to. Ten cells judged at least moderately promising contain two works or fewer; they are outlined in red. The systematic search of Section 5.4 confirms seven of them in the wider literature; the exceptions are dimensional excess addressed by pooling or by unlabeled data, which the corpus under-covers, and stratum sparsity addressed by pooling, where the wider literature borrows across strata mainly in trial design. Four of them lie in the T1 row and four in the procurement column. Channels are also rarely combined. Of the 257 remedy works, 48 have a secondary channel, mostly self-supervised pretraining that mixes proxy and unlabeled information (16) and personalized pooling that mixes pooling with a prior or a shared representation (11). No work in the corpus combines pooling with procurement.

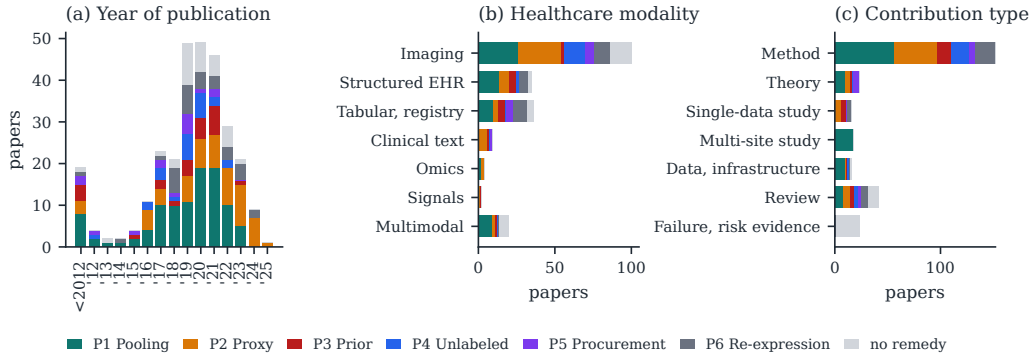


Fig. 8. Profile of the coded corpus (290 works), colored by primary provenance channel. (a) Year of publication; the drop after 2022 reflects indexing and citation lag in a curated corpus, not a trend. (b) Data modality, for the 206 works evaluated on health data. (c) Primary contribution type.

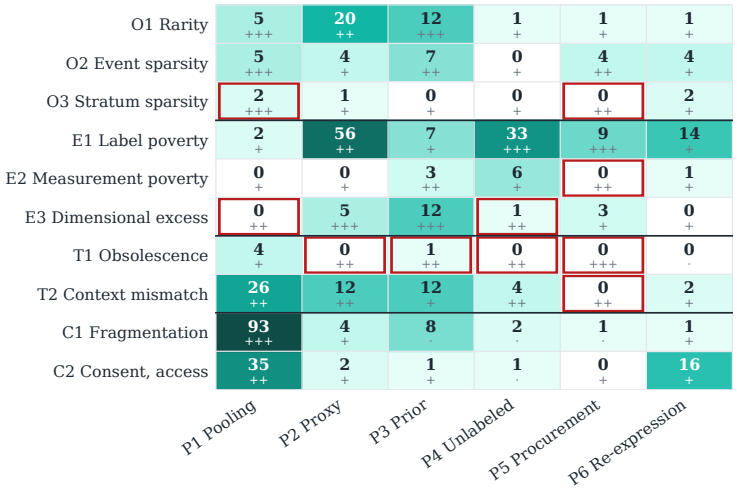


Fig. 9. Gap map. Each cell gives the number of corpus works that target a scarcity class (rows) through a channel, as primary or secondary channel (columns); shading is on a log scale. The small marks below each count repeat the judged potential of Figure 5 (+ weak to + + + strong, · none). Red outlines mark cells judged at least moderately promising that contain two works or fewer.

7.2 Datasets and evaluation metrics

To see what the literature is evaluated on, we extracted from each of the 206 corpus works evaluated on health data the datasets it uses, their size and number of sites, and its headline result, from the work's abstract or, where the abstract was not available, its own page, with a verbatim quote supporting every field; the online supplement lists every work. 154 works report an evaluation on data, 44 are reviews, guidance, theory or simulation studies, and for 8 no abstract could be reached. Entries reflect what the abstracts state: a dataset or number they omit may still be in the paper.

Table 8. Public datasets named in at least two of the 154 empirical corpus works on health data (from the abstracts; every work is listed in the online supplement).

Public dataset	Works
MIMIC-III	5
CheXpert	5
TCGA	4
LIDC-IDRI / LUNA16	3
ChestX-ray8/14	3
ImageNet	2
EyePACS / Kaggle DR	2
BraTS	2
Medical Segmentation Decathlon	2
Messidor	2
i2b2/n2c2	2
LiTS	2
Kaggle	2
MIMIC-CXR	2
ADNI	2

Datasets. Of the 154 empirical works, 33 evaluate only on named public datasets, 19 combine public and institutional data, 72 use institutional or unnamed data only, and 30 name no data in the abstract. Reuse is low (Table 8): MIMIC-III [Johnson et al. 2016] and CheXpert [Irvin et al. 2019] appear in five works each and TCGA in four, and no other dataset in more than three, so few results can be compared across works. Where a size is stated (62 works), the median of each work's largest dataset is 11,467 records, patients or images (interquartile range 287 to 224,316), and only 19 state fewer than 1,000. The data on which remedies for small data are evaluated are therefore mostly large, often pooled or public collections, not the few dozen to few hundred patients per hospital of Section 2. Sites are rarely reported: 24 abstracts state the number of institutions, 16 of them two or more, which echoes the observation of Section 7.3 that pooling is the only channel tested across institutions.

Metrics. Table 9 defines the metrics in use and counts how often each appears in the headline result of an abstract. Only 68 of the 154 empirical works give any number there. The area under the ROC curve is the most frequent named metric (20 works), followed by accuracy (12) and the Dice coefficient for segmentation (7); 42 report a percentage or relative change against a baseline. No headline result reports calibration, log-loss or a concordance index, although calibration is what decides whether a small hospital can act on a predicted risk [Van Calster et al. 2019]. A remedy for small data should be judged at least by discrimination and calibration at each site, against that site working alone.

7.3 What the reported results show

Table 10 collects quantitative results reported in the literature. They are not directly comparable, since tasks, metrics and baselines differ, and we report them as the original authors did. Read through the taxonomy and together with the corpus profile, they support six observations.

1. Pooling works when the custodial gap is real, and is the only channel tested across institutions. The largest gains in Table 10 come from federated pooling in settings that are siloed-small in the sense of Definition 2.5: rare tumors, a new disease, many modest sites. This is what the matrix in Figure 5 predicts. The gains are reported relative to single-site or public-data baselines, and are largest at the

Table 9. Evaluation metrics: definition, direction, and the number of the 154 empirical corpus works whose abstract reports the metric in its headline result. $s(x)$: predicted score; p_i: predicted probability; TP, FP, TN, FN: true and false positives and negatives; A, B: predicted and reference segmentations; T_i: survival time.

Metric	Definition	Better	Works
AUROC	$\Pr\{s(x^+) > s(x^-)\}$ for a random positive and negative	higher; 0.5 is chance	20
Accuracy, balanced accuracy	share of correct predictions; mean of the per-class recalls	higher	12
Sensitivity, specificity	$\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$; $\mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$	higher	4
Precision, AUPRC	$\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$; area under the precision-recall curve	higher; the baseline is the prevalence	4
F1	$2\,\mathrm{precision}\cdot\mathrm{recall}/(\mathrm{precision}+\mathrm{recall})$	higher	2
Dice, IoU	$2\lvert A\cap B\rvert/(\lvert A\rvert+\lvert B\rvert)$; $\lvert A\cap B\rvert/\lvert A\cup B\rvert$	higher; 1 is perfect overlap	7
Calibration	Brier score $n^{-1}\sum_i(p_i-y_i)^2$; calibration slope; expected calibration error	Brier and error lower; slope 1	0
Log-loss	$-n^{-1}\sum_i[\,y_i\ln p_i+(1-y_i)\ln(1-p_i)\,]$	lower	0
Concordance index	$\Pr\{s_i > s_j \mid T_i < T_j\}$ over comparable pairs	higher; 0.5 is chance	0
MSE, MAE	mean squared or absolute error	lower	1
Privacy budget	ϵ of Equation (5)	lower is more private	0
Relative change	(method − baseline)/baseline, in percent	depends on the metric	42

Table 10. Selected quantitative findings, as reported by the original authors.

Channel	Study	Reported finding
P1	Sheller et al. [2020]	Federated training of brain tumor segmentation reached 99% of the model quality of centrally pooled data.
P1	Dayan et al. [2021]	Across 20 institutions, federated EXAM improved average AUC by 16% and generalisability by 38% over single-site models.
P1	Pati et al. [2022]	71 sites, 6,314 patients: 33% better delineation of the surgically targetable tumor and 23% of the whole tumor than a publicly trained model.
P2	Azizi et al. [2021]	Self-supervised pretraining: +6.7% top-1 accuracy in dermatology, +1.1% mean AUC on chest X-rays.
P2	Raghu et al. [2019]	ImageNet transfer gave little benefit on large medical imaging tasks; small models performed comparably.
P2, P3	Hollmann et al. [2025]	TabPFN outperformed baselines tuned for 4 h, in 2.8 s, on datasets up to 10,000 samples.
P6	Frid-Adar et al. [2018]	GAN augmentation on 182 liver lesions: sensitivity 78.6% to 85.7%, specificity 88.4% to 92.4%.
P6	Stadler et al. [2022]	Synthetic data gave no better privacy-utility trade-off than traditional anonymization.
T2	Wong et al. [2021]	A widely deployed sepsis model: external AUC 0.63, missing 67% of sepsis cases.
T2	Zech et al. [2018]	Pneumonia CNNs identified the source hospital; external performance was significantly lower in 3 of 5 comparisons.
O2, T1	Roberts et al. [2021]	Of 62 COVID-19 imaging models reviewed in detail, none was of potential clinical use.
all	Varoquaux [2018]	With about 100 samples, cross-validation error bars are around ±10%.
C	Rocher et al. [2019]	99.98% of Americans re-identifiable from 15 demographic attributes.
C	Bagdasaryan et al. [2019]	DP training reduces accuracy more for under-represented groups.

smallest sites. Pooling is also the only channel whose evidence includes multi-institution studies as

primary results (18 of 102 pooling works); the other 84 evaluate on simulated partitions, public benchmarks or theory.

2. Proxy information helps in proportion to its relevance. Self-supervised pretraining on in-domain unlabeled data outperforms generic ImageNet transfer [Azizi et al. 2021; Raghu et al. 2019], and the strongest small-data results on tabular problems come from models whose prior was designed for small tabular problems [Hollmann et al. 2025]. Generic scale is less valuable than relevant structure. Foundation models are the fastest-growing part of the proxy channel, but their evaluation is still mostly on curated benchmarks [Wornow et al. 2023], so how much they reduce σ at a specific hospital is largely unmeasured.

3. Re-expression helps modestly and does not travel. Augmentation gains are real but usually reported on internal test sets, and the privacy case for synthetic data is weak [Stadler et al. 2022]. Rebalancing by resampling harms calibration in clinical risk models [van den Goorbergh et al. 2022]. Where generative augmentation does help under shift [Ktena et al. 2024], the generator was trained on much larger data, so the gain comes through the proxy channel. All of this follows from Proposition 4.1.

4. Evaluation is itself a small-data problem. The most consistent finding of the meta-research is that small studies produce over-optimistic and unstable estimates. Cross-validation on 100 samples has error bars near ±10% [Varoquaux 2018]; reported accuracy in some imaging applications falls as studies get larger [Varoquaux and Cheplygina 2022]; under-powered studies inflate effect sizes generally [Button et al. 2013; Ioannidis 2005]; and most COVID-19 models could not be used [Roberts et al. 2021; Wynants et al. 2020]. Among imaging AI studies, only 6% of 516 performed external validation [Kim et al. 2019]; systematic comparisons with clinicians found few studies with out-of-sample validation and frequent risk of bias [Liu et al. 2019; Nagendran et al. 2020]; externally validated algorithms often lose performance [Yu et al. 2022]; and in clinical prediction, machine learning showed no performance benefit over logistic regression in comparisons at low risk of bias [Christodoulou et al. 2019]. A method that "works on small data" has often only been evaluated on small data. External, site-held-out validation and reporting standards such as TRIPOD+AI [Collins et al. 2024] address this directly.

5. Privacy costs are paid by the smallest. DP noise is calibrated to sensitivity, which scales inversely with sample size, and its accuracy cost falls disproportionately on small datasets and minority groups [Bagdasaryan et al. 2019; Tramèr and Boneh 2021]. A privacy mechanism designed without attention to small sites can cancel the benefit of pooling for exactly the participants who need it. Section 8 designs around this.

6. Research effort follows convenient data, not severe scarcity. The corpus profile shows two dense regions, label poverty in imaging and custodial fragmentation through pooling, and three nearly empty rows: temporal obsolescence, measurement poverty and stratum sparsity. The first two are as empty in the systematic search as in the corpus. Procurement, which clinical research treats as the first remedy, is the smallest channel and is never combined with pooling. The empty cells of Figure 9 are not evidence that those remedies fail; they are untested, and they are where new work is most likely to change practice.

7.4 Limitations by channel

Pooling assumes a trusted coordinator, roughly exchangeable sites, and the governance to run a consortium; the first and third are often missing. Model updates leak information without formal protection [Geiping et al. 2020; Zhu et al. 2019], and formal protection costs accuracy. Heterogeneity

can make a global model worse than a local one for some sites [Hsieh et al. 2020]. **Proxy** methods inherit the biases and blind spots of the source, including demographic shortcuts [Gichoya et al. 2022], and their benefit is hard to predict before trying. **Prior** knowledge is incomplete for most diseases and, when wrong, biases the model in a way that more local data corrects only slowly. **Unlabeled** data help only under assumptions on the data distribution that are rarely tested. **Procurement** is slow, expensive and subject to the same governance as any data collection. **Re-expression** cannot exceed the information in its inputs. Across all channels, uncertainty is under-reported: modern networks are poorly calibrated [Guo et al. 2017], and calibration is the first property to fail under small samples and shift.

What the gaps imply. The six observations and the limitations above point to the same place. Pooling is the only strong remedy for custodial scarcity (observation 1), but in its standard form it needs a trusted server, assumes away heterogeneity, and lets privacy costs fall on the smallest participants (observation 5). Proxy and prior information are most useful as a starting point rather than a substitute (observation 2), and re-expression cannot carry the load (observation 3). A method for siloed hospitals should therefore combine personalized pooling with privacy spent where it hurts least, on top of a public representation, and be evaluated honestly (observation 4). It should also leave room for the neglected channels, in particular procurement inside a consortium (observation 6). The next section derives such a method.

8 A BLUEPRINT FOR SPARSE NETWORKS OF SILOED HOSPITALS

We now turn to the setting that motivates this paper: a set of hospitals, each too small to learn a task reliably, that do not share patient data and are connected only by a sparse set of bilateral agreements. In the terms of Section 2, the task is siloed-small (Definition 2.5); in the terms of Section 3, its dominant scarcity is custodial fragmentation (C1), usually combined with event or stratum sparsity (O2, O3); and by Figure 5, the only channel that addresses it strongly is pooling. We ask what a theoretically grounded form of pooling looks like under the constraints of this setting, derive it in a model that can be solved exactly, and then show how the other channels of Section 4 fit around it. This section gives the design argument and its main results; a companion paper [Simões et al. 2026a] turns it into a method for realistic model heads, analyzes its privacy in full, and evaluates it against federated and decentralized baselines.

8.1 Problem statement

There are K hospitals, the vertices of an undirected connected graph $\mathcal{G} = (V, E)$. An edge means that two hospitals have an agreement allowing them to exchange privatized model summaries; $\mathcal{G}$ is sparse, with $|E| = O(K)$, and there is no vertex that every hospital trusts. Hospital k holds n_k labeled records from P_k and wants a model that performs well on its own population. We require:

(R1) no patient-level record leaves its hospital;
(R2) every released quantity satisfies (ϵ, δ)-differential privacy with respect to the patients of the releasing hospital;
(R3) no central server: computation uses only messages along edges of $\mathcal{G}$;
(R4) the smallest hospitals benefit at least as much as the largest, in relative terms;
(R5) heterogeneity across hospitals is modeled rather than assumed away.

Requirements (R4) and (R5) exclude a single global model: by observation 5 of Section 7 and the non-i.i.d. results of Section 6.2, a global model serves the typical site, not the smallest or most atypical.

Why existing data-exchange agreements are not enough. Hospitals already exchange data under protocols, but those protocols do not remove the problem. Most of them serve primary use: a patient's record follows the patient to the next point of care. Training a model on the records of many hospitals is secondary use, which needs its own legal basis, ethics approval and data-access permit, and frequently excludes the export of patient-level records altogether [European Parliament and Council of the European Union 2016, 2025]. Where agreements for secondary use exist, they are typically bilateral, slow to negotiate and limited to a subset of hospitals and data, so the network they form is the sparse graph $\mathcal{G}$ assumed here: GHB is designed to run over existing agreements, not to replace them. Even where pooling is lawful, a central copy of identifiable records is not the least invasive option, given how easily such records are re-identified [Rocher et al. 2019], and data minimization favours releasing one privatized low-dimensional summary per hospital. Existing infrastructures that move analyses to the data rather than data to a centre, such as the OHDSI network [Hripcsak et al. 2015], DataSHIELD [Gaye et al. 2014] and the Personal Health Train [Beyan et al. 2020; Deist et al. 2020], are complementary: they provide the channels, and GHB specifies what to compute through them, how to combine the results and what it costs in privacy. Finally, even unrestricted pooling yields one model for all hospitals, which Section 8.7 shows to be worse than local estimation for atypical and large hospitals; the personalization of GHB is useful whether or not privacy binds. The converse also holds. When a network already has a trusted environment in which pooling is lawful, such as a national registry or a secure processing environment under the European Health Data Space, requirement (R3) is moot, and the hierarchical borrowing of Section 8.2 can run inside that environment without gossip or privatized releases.

8.2 Hierarchical borrowing with an imperfect prior

We analyze the core of the problem in the simplest model that keeps its structure. Each hospital has a parameter $\theta_k \in \mathbb{R}$ (one coordinate of a model head; the vector case is coordinatewise), and an asymptotically normal local estimate:

$$\theta_k = \mu + u_k, \quad u_k \sim \mathcal{N}(0, \tau^2); \qquad \hat{\theta}_k = \theta_k + e_k, \quad e_k \sim \mathcal{N}(0, v_k), \quad v_k = \sigma^2/n_k, \tag{10}$$

with all u_k, e_k independent. Here τ^2 is the between-hospital heterogeneity, the quantity that random-effects meta-analysis estimates [DerSimonian and Laird 1986], and σ^2 absorbs the inverse Fisher information per patient. This is the classical setting of empirical Bayes [Efron and Morris 1973; Gelman et al. 2013]. What is new here is that the population mean μ must be estimated privately and without a server, so hospital k only has access to some estimate $\tilde{\mu}$ with error.

Proposition 8.1 (Borrowing with an imperfect prior). *Let $\tilde{\mu}$ be any estimate of μ that is independent of (u_k, e_k) and has mean squared error $M = \mathbb{E}(\tilde{\mu} - \mu)^2$. Among estimators $\tilde{\theta}_k = \alpha\hat{\theta}_k + (1-\alpha)\tilde{\mu}$, the risk $R_k(\alpha) = \mathbb{E}(\tilde{\theta}_k - \theta_k)^2$ is minimized at*

$$\alpha_k^\star = \frac{\tau^2 + M}{\tau^2 + M + v_k}, \qquad R_k^\star = \frac{v_k\,(\tau^2 + M)}{v_k + \tau^2 + M} \;\le\; \min\{v_k,\ \tau^2 + M\}.$$

Equivalently, the effective sample size of hospital k becomes

$$n_k^{\text{eff}} = \frac{\sigma^2}{R_k^\star} = n_k + \underbrace{\frac{\sigma^2}{\tau^2 + M}}_{\text{borrowed patients } B}. \tag{11}$$

Proof. $\tilde{\theta}_k - \theta_k = \alpha e_k + (1-\alpha)(\tilde{\mu} - \mu) - (1-\alpha)u_k$. The three terms are mutually independent, and the first and last have zero mean, so the cross terms vanish; the middle term has second moment M. Hence $R_k(\alpha) = \alpha^2 v_k + (1-\alpha)^2(\tau^2 + M)$. This is a convex quadratic in α, minimized at $\alpha_k^\star$;

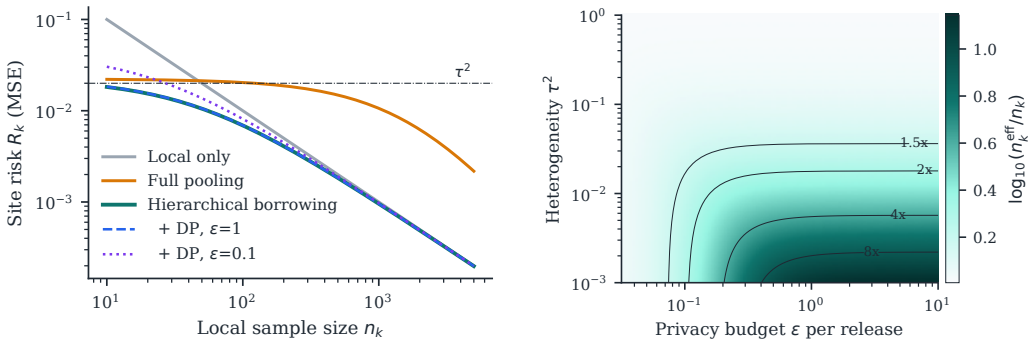


Fig. 10. Left: risk of hospital k as a function of its size n_k, with $K = 12$, other hospitals of 200 patients, $\sigma^2 = 1$, $\tau^2 = 0.02$ and $\delta = 10^{-5}$. Full pooling is excellent for small sites until it hits the heterogeneity floor τ^2, and then harms large ones; hierarchical borrowing (Proposition 8.1), with τ^2 and M known, is never worse than either in expected risk. Privacy at $\epsilon = 1$ is almost free; at $\epsilon = 0.1$ it reduces but does not remove the gain. Right: relative gain in effective sample size n_k^{eff}/n_k for a hospital with $n_k = 50$, as a function of heterogeneity and privacy budget, from Equations (11) and (12).

substituting gives $R_k^\star$. The bound follows because $R_k^\star$ is half the harmonic mean of v_k and $\tau^2 + M$. Finally $\sigma^2/R_k^\star = \sigma^2/v_k + \sigma^2/(\tau^2 + M) = n_k + B$. □

Two remarks fix the scope of Proposition 8.1. First, the proof uses only first and second moments and independence, so it does not need the normality of Equation (10); normality matters only for reading $R_k^\star$ as a posterior variance. Second, R_k is an expectation over the site effect u_k as well as over the data: it is the risk of a hospital drawn from the population of hospitals, not of a particular one. For an unbiased $\tilde{\mu}$ and a given u_k, the risk is $\alpha^2 v_k + (1-\alpha)^2(M + u_k^2)$, which exceeds v_k when $|u_k|$ is large compared with τ. A genuinely atypical hospital can therefore lose by borrowing. This is why τ^2 is estimated from the data rather than assumed small, and why the structured prior of Section 8.4 lets similarity decide which hospitals borrow from which.

Equation (11) is the central quantity of the blueprint. Collaboration gives every hospital almost the same number B of *virtual patients*, regardless of its size; B depends on k only through the leave-one-out error M, which varies little across hospitals when there are many of them. A hospital with 30 patients and one with 3,000 both gain B; the first may double or triple its effective sample, the second barely notices. Requirement (R4) is therefore satisfied by construction. B is limited by heterogeneity: as $\tau^2 \to \infty$, $B \to 0$ and each hospital should act alone; as $\tau^2 \to 0$ with $M \to 0$, $B \to \infty$ and full pooling is optimal. Hierarchical borrowing interpolates between the two, using the data to decide where. With τ^2 and M known it never does worse than local estimation, which is not true of a global model (Figure 10); when both are estimated from the privatized releases the loss is small in simulation, but no longer zero (Section 8.7).

8.3 The cost of privacy

Proposition 8.1 holds for any estimate $\tilde{\mu}$. Requirements (R2) and (R3) constrain how that estimate can be obtained, and the next two results quantify what each constraint costs. Hospital k needs $\tilde{\mu}$ from the others, and each release must be private (R2). Note what does not need privatizing: hospital k uses its own estimate $\hat{\theta}_k$ without noise, since that never leaves the hospital. Privacy affects only M.

Proposition 8.2 (Privatized population estimate). *Suppose $\hat{\theta}_j$ is the mean of per-patient contributions clipped to $[-C, C]$, so that replacing one patient changes $\hat{\theta}_j$ by at most $\Delta_j = 2C/n_j$, and that clipping introduces no bias. Each hospital j releases $\hat{\theta}_j + \xi_j$ with $\xi_j \sim \mathcal{N}(0, s_j^2)$ and $s_j = \Delta_j\sqrt{2\ln(1.25/\delta)}/\epsilon$, which is (ϵ, δ)-differentially private for $\epsilon \in (0, 1)$ by the Gaussian mechanism [Dwork and Roth 2014]. For the leave-one-out average $\tilde{\mu} = \frac{1}{K-1}\sum_{j\neq k}(\hat{\theta}_j + \xi_j)$,*

$$M = \frac{1}{(K-1)^2}\sum_{j\neq k}\Big(\tau^2 + \frac{\sigma^2}{n_j} + \frac{8C^2\ln(1.25/\delta)}{n_j^2\,\epsilon^2}\Big). \tag{12}$$

Proof. The terms $u_j + e_j + \xi_j$ are independent with zero mean and variances $\tau^2 + \sigma^2/n_j + s_j^2$, and $s_j^2 = 4C^2 \cdot 2\ln(1.25/\delta)/(n_j^2\epsilon^2)$. Independence from (u_k, e_k) holds because k is excluded. □

Three consequences follow. First, privacy noise is divided by $(K-1)$, so it vanishes as the network grows, while the local estimate remains noise-free. Second, the privacy term scales as n_j^{-2} while the sampling term scales as n_j^{-1}: privacy is cheap for large contributors and expensive for tiny ones. This is the formal version of observation 5 in Section 7. The remedy is to let small hospitals release jointly: with secure aggregation [Bonawitz et al. 2017] among neighbours, noise can be calibrated to the sensitivity of the sum, which is set by the total size of the group rather than by each member. Third, inverse-variance weighting of the releases, rather than equal weights, reduces M further and is what a practical implementation should use; Equation (12) is the simplest case.

Unknown heterogeneity and the scope of the guarantee. Proposition 8.1 takes τ^2 as known. It can be estimated from the releases by the method of moments of random-effects meta-analysis [DerSimonian and Laird 1986], which is post-processing; better, each hospital can integrate τ out over its posterior given the releases, weighting by the marginal likelihood of its own data, so that a hospital whose data conflict with the network borrows little. The privacy guarantee holds under explicit conditions: a patient held by several hospitals is protected at the composed level, site sizes are public, a vector head is released as one vector, estimators that are not clipped means need their own sensitivity, and repeated refits compose. For likelihood-based heads the natural release is the privatized score at the published model, inverted by public moment matching; a wrong anchor then makes the release noisier rather than biased. The companion paper [Simões et al. 2026a] develops each of these points.

8.4 Computation on a sparse graph without a server

The leave-one-out average needs every hospital's release. Without a server, the hospitals compute it by gossip over $\mathcal{G}$ [Boyd et al. 2006; Xiao and Boyd 2004]. Let W be a symmetric, doubly stochastic matrix supported on the edges of $\mathcal{G}$, for instance Metropolis-Hastings weights $W_{kl} = 1/(1 + \max(d_k, d_l))$ for $(k, l) \in E$, where d_k is the degree of k. Starting from the privatized releases $x_j^0 = \hat{\theta}_j + \xi_j$, each round sets $x^{t+1} = Wx^t$.

Proposition 8.3 (Free post-processing and convergence). *(i) Every message exchanged during gossip is a function of the privatized releases only, so the whole gossip phase satisfies the same (ϵ, δ) guarantee as the initial release, with no additional privacy cost. (ii) Let $\lambda = \max\{|\lambda_2(W)|, |\lambda_K(W)|\}$. If $\mathcal{G}$ is connected and W has a positive diagonal, then $\lambda < 1$ and*

$$\big\|x^t - \bar{x}\mathbf{1}\big\|_2 \;\le\; \lambda^t\,\big\|x^0 - \bar{x}\mathbf{1}\big\|_2,$$

so that the estimate held by any hospital after t rounds has root mean squared error at most $\sqrt{M_{\text{all}}} + \lambda^t\sqrt{\mathbb{E}\|x^0 - \bar{x}\mathbf{1}\|_2^2}$ as an estimate of μ, where $M_{\text{all}} = K^{-2}\sum_j(\tau^2 + \sigma^2/n_j + s_j^2)$ is the mean squared error

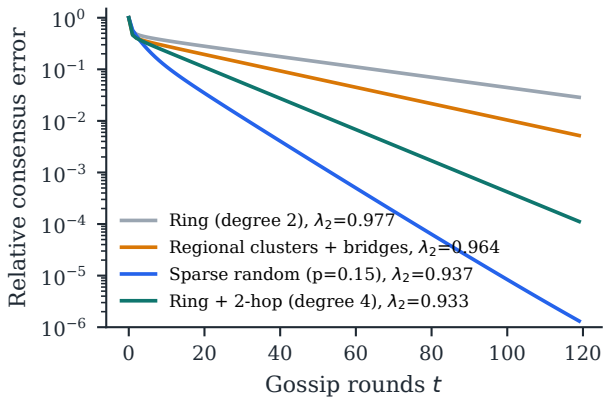


Fig. 11. Gossip consensus error for 24 hospitals on four sparse topologies with Metropolis-Hastings weights. Rates follow the second largest eigenvalue modulus λ of the gossip matrix (Proposition 8.3).

of the full average. (iii) Hospital k obtains its leave-one-out estimate as $\tilde{\mu}_k^t = (Kx_k^t - x_k^0)/(K-1)$. It converges to the leave-one-out average of Proposition 8.2, and differs from it by $K(x_k^t - \bar{x})/(K-1)$, whose root mean square is at most $\frac{K}{K-1}\lambda^t\sqrt{\mathbb{E}\|x^0 - \bar{x}\mathbf{1}\|_2^2}$.

Proof. (i) is the post-processing property of differential privacy [Dwork and Roth 2014]: the gossip iterates are a deterministic function of the released values, and each hospital's subsequent computations on its own data are never released. (ii) Since W is symmetric and doubly stochastic, $\mathbf{1}$ is an eigenvector with eigenvalue 1 and $x^t - \bar{x}\mathbf{1} = W^t(x^0 - \bar{x}\mathbf{1})$ lies in the orthogonal complement of $\mathbf{1}$, on which $\|W\|_2 = \lambda$. Connectivity and a positive diagonal make W irreducible and aperiodic, so $\lambda < 1$ [Xiao and Boyd 2004]. The error bound follows from the triangle inequality in L^2, since each coordinate deviation is bounded by the norm of the whole vector. (iii) Since $\sum_{j\neq k} x_j^0 = K\bar{x} - x_k^0$, the leave-one-out average is $(K\bar{x} - x_k^0)/(K-1)$, and subtracting it from $\tilde{\mu}_k^t$ leaves $K(x_k^t - \bar{x})/(K-1)$, which is bounded as in (ii). □

Part (iii) also bounds the one departure from Proposition 8.1 that gossip introduces: after finitely many rounds, $\tilde{\mu}_k^t$ still contains a fraction of hospital k's own release, so it is not exactly independent of (u_k, e_k). The residual dependence decays as λ^t, and a hospital should use $\tilde{\mu}_k^t$ only once λ^t is small; Section 8.7 shows what happens otherwise.

The number of rounds needed for accuracy γ is about $\log(1/\gamma)/(1-\lambda)$, so the spectral gap $1-\lambda$ of the collaboration graph determines the communication cost. The spectral gap is related to how well the graph is connected through Cheeger-type inequalities [Chung 1997]. Figure 11 shows this for 24 hospitals under four realistic topologies. A chain of bilateral agreements arranged in a ring mixes slowly; dense regional clusters joined by single bridges are not much better, because the bridges are bottlenecks; a few additional random or two-hop links triple the spectral gap, which after a hundred rounds lowers the consensus error by orders of magnitude. The practical lesson is that *which* agreements a network signs matters more than *how many*: adding a handful of long-range edges is worth more than densifying regions. Privacy amplification by decentralization [Cyffers and Bellet 2022] gives a further benefit on sparse graphs, because each hospital observes only what reaches it through its neighbours.

Structured heterogeneity. The exchangeable model (10) treats all other hospitals as equally relevant. When similarity is structured, for instance by region or case mix, the prior can follow a similarity graph with weights $a_{kl} \geq 0$, which may differ from the communication graph. A Gaussian Markov random field prior [Rue and Held 2005] with graph Laplacian L gives the maximum a posteriori problem

$$\min_{\theta} \sum_{k} \frac{(\theta_k - \hat{\theta}_k)^2}{2v_k} + \frac{\beta}{2} \sum_{k<l} a_{kl}(\theta_k - \theta_l)^2 + \frac{\gamma}{2} \sum_{k} (\theta_k - \mu_0)^2, \tag{13}$$

whose solution satisfies the linear system $(V^{-1} + \beta L + \gamma I)\boldsymbol{\theta} = V^{-1}\hat{\boldsymbol{\theta}} + \gamma\mu_0 \mathbf{1}$ with $V = \mathrm{diag}(v_k)$. The Jacobi iteration

$$\theta_k \leftarrow \frac{\hat{\theta}_k / v_k + \beta \sum_l a_{kl}\theta_l + \gamma\mu_0}{1/v_k + \beta d_k + \gamma}, \qquad d_k = \textstyle\sum_l a_{kl},$$

uses only neighbour values and converges for any starting point, because the system matrix is strictly diagonally dominant. This generalizes the personalized peer-to-peer objective of Vanhaesebrouck et al. [2017] with an explicit anchor μ_0, which can come from published knowledge (channel P3). To respect (R2), neighbours exchange smoothed values computed from privatized releases, and each hospital applies the final update with its own noise-free $\hat{\theta}_k$ locally without releasing it, so Proposition 8.3(i) applies.

8.5 From the Gaussian model to deep models

The analysis applies directly to parametric models, and in particular to the last layer of a network on top of a fixed representation. That observation suggests the architecture in Figure 12, which we call *Graph Hierarchical Borrowing* (GHB). It uses each provenance channel where it is strongest and keeps the privatized, shared part of the model small.

L0 Public representation (P2). A foundation encoder f_{ϕ_0} pretrained on public or licensed data for the modality [Moor et al. 2023; Rasmy et al. 2021; Zhou et al. 2023]. It is available to all hospitals without any privacy cost.

L1 Local self-supervised adaptation (P4). Each hospital adapts the encoder to its own unlabeled data with low-rank adapters [Hu et al. 2022] and a self-supervised objective [Chen et al. 2020a; He et al. 2022]. Adapters may be averaged by privatized gossip, but need not be.

L2 Hierarchical heads (P1). Each hospital fits a low-dimensional head $\theta_k \in \mathbb{R}^p$ on the adapted features, with an estimate of its covariance, and borrows strength by Propositions 8.1, 8.2 and 8.3 or by Equation (13).

L3 Knowledge anchor (P3). The hyperprior mean μ_0 comes from published models or clinical knowledge, so that even a hospital with no neighbours starts from a sensible place.

L4 Site calibration. Each hospital calibrates its predictions by conformal prediction on local held-out data [Angelopoulos and Bates 2023; Shafer and Vovk 2008], or by federated conformal prediction when the local calibration set is too small [Lu et al. 2023].

L5 Targeted procurement (P5). Posterior variances from L2 identify which patients or subgroups would be most informative to label next [Settles 2009].

Synthetic data (P6) is deliberately absent from the training path. It is useful for testing pipelines and for teaching, but by Proposition 4.1 it cannot add information, and its privacy case is weak [Stadler et al. 2022].

Two features of GHB deserve emphasis. Keeping the privatized component low-dimensional is not a convenience but a necessity: the norm of Gaussian-mechanism noise grows with $\sqrt{p}$, and private learning is far more accurate on good fixed features than end to end [Tramèr and Boneh

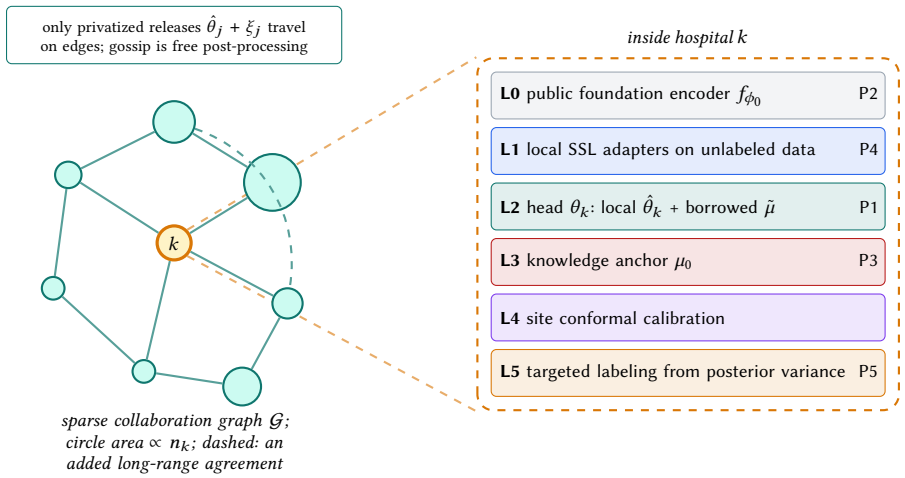


Fig. 12. Graph Hierarchical Borrowing (GHB). Left: hospitals of very different sizes connected by a sparse set of agreements, without a central server. Right: the model held by each hospital, with the provenance channel used at each layer. Only the low-dimensional head participates in privatized exchange.

2021]. And the split between a public representation and a private head mirrors the theory of shared representations, in which a common representation reduces the per-task sample complexity to that of the head [Collins et al. 2021; Du et al. 2021; Maurer et al. 2016; Tripuraneni et al. 2020]. Only privatized summaries of the heads cross a hospital's boundary, and everything else is computed from them or stays local. The companion paper [Simões et al. 2026a] gives the full protocol and its privacy boundary. There, the network releases two privatized sums over all of its patients, computed by secure summation without a server, so that the noise is calibrated to the whole network rather than to the smallest hospital. A lighter variant, GHB-lite, in which every hospital releases on its own, follows the analysis of this section most closely.

8.6 Why this is a principled default

Table 11 compares GHB with the alternatives on requirements (R1) to (R5). Centralized pooling violates (R1). Federated averaging violates (R3) and, under heterogeneity, (R4) and (R5). Personalized federated learning satisfies (R4) and (R5) but still needs a server. Existing decentralized approaches satisfy (R3); Swarm Learning does not provide formal differential privacy by default, while ProxyFL does but shares full proxy models. Synthetic data sharing satisfies (R1) and (R3) only at the cost of utility or privacy. Distributed meta-analysis is the closest classical relative and satisfies most requirements, but does not exploit learned representations or graph structure. The comparison is one of design properties, not of measured performance: the GHB row records what holds by construction under the assumptions of Propositions 8.1, 8.2 and 8.3, while the other rows describe each method in its standard form, and most of them can be extended to satisfy more requirements.

The argument for GHB rests on four results that are each well established, combined in a way that, to our knowledge, has not been set out for this setting. None of the four is new, and the case for GHB is that it is the simplest combination that meets all five requirements at once, not that it outperforms the alternatives; that has to be shown empirically. Separate estimation of three or more group means is inadmissible under squared error, and empirical Bayes shrinkage toward a common mean dominates it in the Gaussian model with equal variances [Efron and Morris 1973; Stein 1956]; Proposition 8.1 gives the best combination of a local and a population estimate in closed

Table 11. Approaches for sparse, siloed hospital networks against requirements (R1) to (R5), as design properties of each method in its standard form. ✓ satisfied; ○ partially or optionally; ✗ not satisfied. Entries for GHB hold by construction under the assumptions of Section 8 and of the companion paper [Simões et al. 2026a]; they have been checked in simulation and on real cancer networks (Section 8.7).

Approach	(R1) local data	(R2) formal DP	(R3) no server	(R4) small sites gain	(R5) heterogeneity
Centralized pooling	✗	✗	✗	○	○
Federated averaging [McMahan et al. 2017]	✓	○	✗	✗	✗
Personalized FL [Collins et al. 2021; Li et al. 2021a]	✓	○	✗	✓	✓
Swarm Learning [Warnat-Herresthal et al. 2021]	✓	✗	✓	○	✗
ProxyFL [Kalra et al. 2023]	✓	✓	✓	○	○
Synthetic data sharing [Jordon et al. 2019]	✓	○	✓	○	✗
Distributed meta-analysis [DerSimonian and Laird 1986; Duan et al. 2020a]	✓	○	○	✓	✓
GHB (this paper)	✓	✓	✓	✓	✓

form when the population estimate is itself imperfect. Differential privacy composes and is closed under post-processing [Dwork and Roth 2014], which makes the gossip phase free (Proposition 8.3). Gossip converges at a rate set by the spectral gap [Boyd et al. 2006; Xiao and Boyd 2004], which turns network design into a measurable quantity. And shared representations reduce the problem to a low-dimensional one [Maurer et al. 2016; Tripuraneni et al. 2020], which is what keeps privacy noise small. Each component can be replaced by a stronger one, such as secure aggregation within regions, or nonlinear heads, without changing the architecture; integrating τ out over its posterior (Section 8.3) is one such replacement, and releasing only network-wide sums by secure summation, so that the privacy noise is calibrated to the whole network rather than to each hospital, is another, which the companion paper adopts after it proved decisive on real data.

8.7 What simulation and real data show

The propositions take τ^2 and M as known, the model as Gaussian and every participant as honest. The companion paper [Simões et al. 2026a] tests what is lost when these assumptions fail, in eighteen Monte Carlo simulations whose main results Figure 13 summarizes. In a network of 24 hospitals of 16 to 479 patients, hierarchical borrowing with every unknown estimated from the privatized releases stays at or below local estimation for every hospital across four orders of magnitude of heterogeneity, while a single global model becomes up to 382 times worse for its worst hospital. Every hospital gains about $B \approx 46$ virtual patients, which multiplies the effective sample size of hospitals with 60 patients or fewer by 2.28 and of those with 300 or more by 1.08. With logistic-regression heads, against federated averaging with fine-tuning, decentralized DP-SGD and clustered federated learning at the same privacy budget, with their hyperparameters tuned on the true loss, the borrowing hospital's worst case stays within a factor 1.06 of its own local error in every private setting except the most heterogeneous, while DP-FedAvg [McMahan et al. 2018] with fine-tuning leaves some hospital, usually a large one, 1.39 to 2.96 times worse off, and the other iterative methods also leave some hospital worse off than alone. Adversarial simulations show the

limits: a small hospital that is genuinely unlike the others loses by borrowing, because it cannot tell from its own data that it is atypical.

Real data tell a more demanding story. On networks of a few to a few dozen small hospitals, per-hospital releases at $\epsilon = 1$ carry almost no information, and the lighter design was no better than a hospital that regularizes well alone. The design that the companion paper settles on releases only network sums, and integrates the unknown heterogeneity out against each hospital's own evidence. It was developed on twelve cancer networks from The Cancer Genome Atlas and evaluated on five networks held out from its development. There it reduced held-out log-loss to 0.925 of local estimation at $\epsilon = 1$ and 0.909 at $\epsilon = 10$. That was lower than every federated and personalized method tuned on the development networks, the best of which reached 0.976 and 0.919, and lower than or level with each of them tuned on the test folds, which no hospital can do (0.939 and 0.911). That lead comes from fitting related networks together: a single network fitted alone gives the method no more than a well-regularized hospital. On the public FLamby benchmark [Ogier du Terrail et al. 2022], with four to six hospitals per network, the method improved on local estimation and was the most accurate in some settings but not in others, and in simulated worlds whose hospitals share their base rates, which the method deliberately does not borrow, tuned federated baselines were more accurate. With a frozen public encoder [Tan and Le 2019] and one softmax head per hospital, the same design extends to images: on Camelyon17 histopathology from five hospitals [Bándi et al. 2019], its balanced accuracy at $\epsilon = 10$ was 0.896, against 0.885 for a hospital alone and 0.895 for the best literature method tuned on development networks; it led, but by small margins, and each literature method tuned on the test patches was ahead. Collaboration between small hospitals therefore pays under privacy, but by percent rather than by multiples, and only when the privacy noise is calibrated to the network rather than to each hospital.

Limits. GHB protects patients against honest-but-curious neighbours and any observer of the messages; it does not exclude a hospital that lies, and stronger adversaries call for robust aggregation [Blanchard et al. 2017]. Its privacy guarantee is per release, so refits compose and should be accounted with zero-concentrated or Rényi differential privacy [Bun and Steinke 2016; Mironov 2017]. The Gaussian analysis is asymptotic, and the simulations test internal consistency against adversaries we chose. The real-data evidence of the companion paper [Simões et al. 2026a] comes from one main source of networks, The Cancer Genome Atlas, with one outcome and an eight-coefficient head; its margins under weak privacy are small, and on the cross-silo benchmark FLamby [Ogier du Terrail et al. 2022] the method leads only in some settings. Evaluation on more networks, outcomes and larger heads is the first item of future work. GHB answers the question posed at the start of this section only under these assumptions, and relaxing each of them, together with the measurement problems raised in Sections 2 and 7, defines most of the open problems that follow.

9 OPEN PROBLEMS AND RESEARCH AGENDA

The problems below follow the structure of the paper. The first two concern measuring smallness and evaluating models under it (Sections 2 and 7), the next five arise from the blueprint (Section 8), and the last four concern channels and cells whose potential is not yet matched by evidence (Sections 6 and 7).

(1) **Smallness-aware reporting.** Papers should report n^{eff}, σ and the scarcity profile of Table 4, alongside the TRIPOD+AI items [Collins et al. 2024]. This requires agreed estimators for each factor of Equation (2), especially ρ_S when the deployment population is not yet observed.

(2) **Federated evaluation.** The evaluation of a model is itself small-data (Section 7). Protocols for privacy-preserving external validation across silos, including calibration and subgroup performance, are as important as protocols for training.

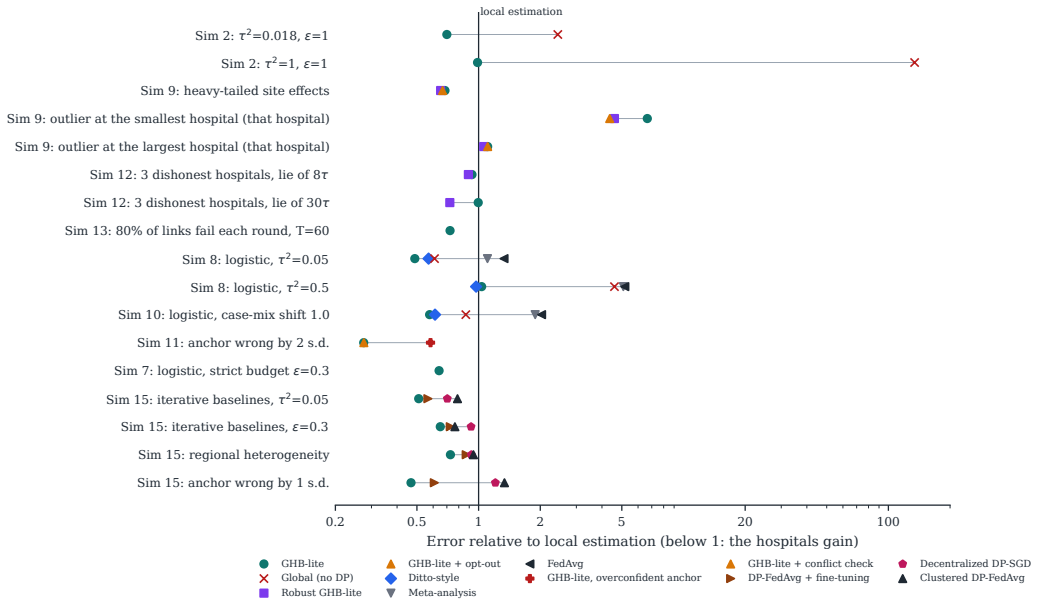


Fig. 13. Summary of the simulation study of the companion paper [Simões et al. 2026a]. Each row is a scenario and each marker a method; the horizontal position is the method's error relative to local estimation (mean risk R_k/v_k in the Gaussian simulations, mean excess log-loss divided by that of local estimation in the logistic ones). Left of the vertical line, the hospitals gain. Rows marked "that hospital" report the outlying hospital alone.

(3) **Private estimation of heterogeneity.** The value of collaboration is governed by τ^2 (Equation (11)), yet hospitals rarely know it before joining a consortium. Private, few-round estimators of τ^2 would let a hospital predict its benefit before committing. They would also close the gap that remains in the companion simulation study [Simões et al. 2026a]: with the true τ^2, GHB-lite would reach 0.0158 at $\epsilon = 0.3$, against 0.0175 with τ integrated out over its posterior and 0.0269 for local estimation.

(4) **Topology design.** Since mixing depends on the spectral gap (Proposition 8.3), which agreements to sign is an optimization problem with legal and economic constraints. Choosing a few long-range links that maximize the gap is a concrete, under-studied question.

(5) **Small-site privacy.** DP noise scales as n_j^{-2} (Equation (12)). Mechanisms that let small hospitals release jointly, through secure aggregation or shuffling, without a trusted party, would remove the main fairness problem of private pooling.

(6) **Incentives and valuation.** Large hospitals gain little from hierarchical borrowing and contribute most. Data valuation [Ghorbani and Zou 2019] adapted to privatized, decentralized settings could support fair credit, and regulation such as the European Health Data Space [European Parliament and Council of the European Union 2025] will shape what is feasible.

(7) **Shift-aware borrowing.** Proposition 8.1 treats heterogeneity as random. In practice it has structure, through case mix, devices and time. Combining causal invariance [Arjovsky et al. 2019; Castro et al. 2020] with hierarchical borrowing could separate the part of τ^2 that is transportable from the part that is site-specific.

(8) **Priors for clinical tabular data.** TabPFN shows that a well-designed synthetic prior can beat tuned baselines on small tables [Hollmann et al. 2025]. Priors built from clinical knowledge, such as plausible effect directions, monotonicity and interactions, are a natural next step and a principled use of synthetic data that does not conflict with Proposition 4.1, because the information comes from the prior, not from the samples.

(9) **Language models as priors, carefully.** Large language models carry medical knowledge [Singhal et al. 2023] and can act as priors for tabular and few-shot tasks [Hegselmann et al. 2023]. Measuring how much information they actually contribute about a specific local θ, as opposed to population-level generalities, is open.

(10) **Intrinsically small problems.** For conditions where the world supply is below $n^\star$, the right question is how to combine global pooling, mechanistic knowledge and patient-level "n of 1" inference [Estrin 2014; Hekler et al. 2019]. This needs methods that are explicit about which part of the answer comes from which channel.

(11) **The empty cells.** The gap map (Figure 9) lists ten combinations judged promising that the corpus barely touches. Three stand out: methods that down-weight obsolete data in a principled way (T1), which today is handled by ad hoc retraining windows; procurement inside a consortium, where the posterior variance of Section 8 tells every hospital which patients to label next and nobody has studied the joint design (P5 with P1); and pooling aimed at small subgroups rather than small sites (O3 with P1), for which Equation (11) applies with strata in place of hospitals, and which the wider literature has developed for trial design but not for prediction models.

10 CONCLUSION

Small data are the normal condition of clinical machine learning, not an exception to it. Treating smallness as a measurable, task-relative property, through the effective sample size and the smallness index, makes it possible to say how small a problem is and why. The Scarcity-Provenance taxonomy turns that diagnosis into a choice of remedy: identify which term of σ is degraded, then import information from a channel that can address it, keeping in mind that re-expression adds none. Coding 290 works with it shows where the field has looked so far: at label poverty, and at custodial fragmentation through pooling, with very little on obsolete data or noisy measurements, in the corpus and in the wider literature alike, and with multi-institution evidence only for pooling. For the common case of many small hospitals that cannot share data and have no central coordinator, hierarchical borrowing with privatized releases and gossip gives every hospital nearly the same number of virtual patients, pays the privacy cost only in the shared estimate, and runs over whatever collaboration graph exists. On real cancer networks, the gain is a few percent rather than a multiple, and it appears only when the privacy noise is calibrated to the network as a whole; there, the companion paper's method was more accurate than every federated and personalized method tuned as a consortium could tune it, and on a public benchmark it was the most accurate in some settings but not in others. The model in which these results are exact is simple. Simulation in the companion paper shows that its conclusions, that collaboration should be personalized, that privacy should be spent at the population level, and that network topology matters, survive the estimation of every unknown quantity and the move to logistic models, and that with a privatized score release and the heterogeneity integrated out most of the gain survives even a strict privacy budget; whether they hold beyond the networks, outcomes and model heads tested so far is the main question this paper leaves open.

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REFERENCES

Martin Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. 2016. Deep Learning with Differential Privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 308-318. https://doi.org/10.1145/2976749.2978318

Mohammed Adnan, Shivam Kalra, Jesse C. Cresswell, Graham W. Taylor, and Hamid R. Tizhoosh. 2022. Federated learning and differential privacy for medical image analysis. *Scientific Reports* 12 (2022), 1953. https://doi.org/10.1038/s41598-022-05539-7

Vibhu Agarwal, Tanya Podchiyska, Juan M. Banda, Veena Goel, Tiffany I. Leung, Evan P. Minty, Timothy E. Sweeney, Elsie Gyang, and Nigam H. Shah. 2016. Learning statistical models of phenotypes using noisy labeled training data. *Journal of the American Medical Informatics Association* 23, 6 (2016), 1166-1173. https://doi.org/10.1093/jamia/ocw028

Monica Agrawal, Stefan Hegselmann, Hunter Lang, Yoon Kim, and David Sontag. 2022. Large language models are few-shot clinical information extractors. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 1998-2022. https://doi.org/10.18653/v1/2022.emnlp-main.130

Emily Alsentzer, John Murphy, William Boag, Wei-Hung Weng, Di Jindi, Tristan Naumann, and Matthew McDermott. 2019. Publicly Available Clinical BERT Embeddings. In *Proceedings of the 2nd Clinical Natural Language Processing Workshop*. 72-78.

Anastasios N. Angelopoulos and Stephen Bates. 2023. Conformal Prediction: A Gentle Introduction. *Foundations and Trends in Machine Learning* 16, 4 (2023), 494-591. https://doi.org/10.1561/2200000101

Dana Angluin and Philip Laird. 1988. Learning From Noisy Examples. *Machine Learning* 2, 4 (1988), 343-370. https://doi.org/10.1007/BF00116829

Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. 2019. Invariant Risk Minimization. *arXiv preprint arXiv:1907.02893* (2019).

Shekoofeh Azizi, Basil Mustafa, Fiona Ryan, Zachary Beaver, Jan Freyberg, Jonathan Deaton, et al. 2021. Big Self-Supervised Models Advance Medical Image Classification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 3478-3488.

Eugene Bagdasaryan, Omid Poursaeed, and Vitaly Shmatikov. 2019. Differential Privacy Has Disparate Impact on Model Accuracy. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32.

Wenjia Bai, Ozan Oktay, Matthew Sinclair, Hideaki Suzuki, Martin Rajchl, Giacomo Tarroni, Ben Glocker, Andrew King, Paul M. Matthews, and Daniel Rueckert. 2017. Semi-supervised Learning for Network-Based Cardiac MR Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI) (LNCS, Vol. 10434)*. 253-260.

Xiang Bai, Hanchen Wang, Liya Ma, Yongchao Xu, Jiefeng Gan, Ziwei Fan, et al. 2021. Advancing COVID-19 diagnosis with privacy-preserving collaboration in artificial intelligence. *Nature Machine Intelligence* 3 (2021), 1081-1089. https://doi.org/10.1038/s42256-021-00421-z

Indranil Balki, Afsaneh Amirabadi, Jacob Levman, Anne L. Martel, Ziga Emersic, Blaz Meden, et al. 2019. Sample-Size Determination Methodologies for Machine Learning in Medical Imaging Research: A Systematic Review. *Canadian Association of Radiologists Journal* 70, 4 (2019), 344-353. https://doi.org/10.1016/j.carj.2019.06.002

Péter Bándi, Oscar Geessink, Quirine Manson, Marcory Van Dijk, Maschenka Balkenhol, Meyke Hermsen, Babak Ehteshami Bejnordi, Byungjae Lee, Kyunghyun Paeng, Aoxiao Zhong, et al. 2019. From Detection of Individual Metastases to Classification of Lymph Node Status at the Patient Level: The CAMELYON17 Challenge. *IEEE Transactions on Medical Imaging* 38, 2 (2019), 550-560. https://doi.org/10.1109/TMI.2018.2867350

Jineta Banerjee, Jaclyn N. Taroni, Robert J. Allaway, Deepashree Venkatesh Prasad, Justin Guinney, and Casey Greene. 2023. Machine learning in rare disease. *Nature Methods* 20 (2023), 803-814.

Mrinal Kanti Baowaly, Chia-Ching Lin, Chao-Lin Liu, and Kuan-Ta Chen. 2019. Synthesizing electronic health records using improved generative adversarial networks. *Journal of the American Medical Informatics Association* 26, 3 (2019), 228-241. https://doi.org/10.1093/jamia/ocy142

Peter L. Bartlett and Shahar Mendelson. 2002. Rademacher and Gaussian Complexities: Risk Bounds and Structural Results. *Journal of Machine Learning Research* 3 (2002), 463-482.

Jonathan Baxter. 2000. A Model of Inductive Bias Learning. *Journal of Artificial Intelligence Research* 12 (2000), 149-198. https://doi.org/10.1613/jair.731

Brett K. Beaulieu-Jones, Zhiwei Steven Wu, Chris Williams, Ran Lee, Sanjeev P. Bhavnani, James Brian Byrd, and Casey S. Greene. 2019. Privacy-Preserving Generative Deep Neural Networks Support Clinical Data Sharing. *Circulation:*

Cardiovascular Quality and Outcomes 12, 7 (2019), e005122. https://doi.org/10.1161/CIRCOUTCOMES.118.005122

Aurélien Bellet, Rachid Guerraoui, Mahsa Taziki, and Marc Tommasi. 2018. Personalized and Private Peer-to-Peer Machine Learning. In *Proceedings of the 21st International Conference on Artificial Intelligence and Statistics (AISTATS) (PMLR, Vol. 84)*.

Shai Ben-David, John Blitzer, Koby Crammer, Alex Kulesza, Fernando Pereira, and Jennifer Wortman Vaughan. 2010. A theory of learning from different domains. *Machine Learning* 79 (2010), 151-175. https://doi.org/10.1007/s10994-009-5152-4

Oya Beyan, Ananya Choudhury, Johan van Soest, Oliver Kohlbacher, Lukas Zimmermann, Holger Stenzhorn, et al. 2020. Distributed Analytics on Sensitive Medical Data: The Personal Health Train. *Data Intelligence* 2, 1-2 (2020), 96-107. https://doi.org/10.1162/dint_a_00032

Peva Blanchard, El Mahdi El Mhamdi, Rachid Guerraoui, and Julien Stainer. 2017. Machine Learning with Adversaries: Byzantine Tolerant Gradient Descent. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30.

Rishi Bommasani, Drew A. Hudson, Ehsan Adeli, Russ Altman, Simran Arora, et al. 2021. On the Opportunities and Risks of Foundation Models. *arXiv preprint arXiv:2108.07258* (2021).

Keith Bonawitz, Vladimir Ivanov, Ben Kreuter, Antonio Marcedone, H. Brendan McMahan, Sarvar Patel, Daniel Ramage, Aaron Segal, and Karn Seth. 2017. Practical Secure Aggregation for Privacy-Preserving Machine Learning. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 1175-1191. https://doi.org/10.1145/3133956.3133982

Vadim Borisov, Kathrin Seßler, Tobias Leemann, Martin Pawelczyk, and Gjergji Kasneci. 2023. Language Models are Realistic Tabular Data Generators. In *International Conference on Learning Representations (ICLR)*.

Christopher Bowles, Liang Chen, Ricardo Guerrero, Paul Bentley, Roger Gunn, Alexander Hammers, David Alexander Dickie, Maria Valdés Hernández, Joanna Wardlaw, and Daniel Rueckert. 2018. GAN Augmentation: Augmenting Training Data using Generative Adversarial Networks. *arXiv preprint arXiv:1810.10863* (2018).

Stephen Boyd, Arpita Ghosh, Balaji Prabhakar, and Devavrat Shah. 2006. Randomized Gossip Algorithms. *IEEE Transactions on Information Theory* 52, 6 (2006), 2508-2530. https://doi.org/10.1109/TIT.2006.874516

Sandra Brasil, Carlota Pascoal, Rita Francisco, Vanessa dos Reis Ferreira, Paula A. Videira, and Gonçalo Valadão. 2019. Artificial Intelligence (AI) in Rare Diseases: Is the Future Brighter? *Genes* 10, 12 (2019), 978. https://doi.org/10.3390/genes10120978

Theodora S. Brisimi, Ruidi Chen, Theofanie Mela, Alex Olshevsky, Ioannis Ch. Paschalidis, and Wei Shi. 2018. Federated learning of predictive models from federated Electronic Health Records. *International Journal of Medical Informatics* 112 (2018), 59-67. https://doi.org/10.1016/j.ijmedinf.2018.01.007

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, et al. 2020. Language Models are Few-Shot Learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Samuel Budd, Emma C. Robinson, and Bernhard Kainz. 2021. A survey on active learning and human-in-the-loop deep learning for medical image analysis. *Medical Image Analysis* 71 (2021), 102062. https://doi.org/10.1016/j.media.2021.102062

Peter Bühlmann and Sara van de Geer. 2011. *Statistics for High-Dimensional Data: Methods, Theory and Applications*. Springer.

Mark Bun and Thomas Steinke. 2016. Concentrated Differential Privacy: Simplifications, Extensions, and Lower Bounds. In *Theory of Cryptography (TCC 2016-B) (Lecture Notes in Computer Science, Vol. 9985)*. Springer, 635-658. https://doi.org/10.1007/978-3-662-53641-4_24

Katherine S. Button, John P. A. Ioannidis, Claire Mokrysz, Brian A. Nosek, Jonathan Flint, Emma S. J. Robinson, and Marcus R. Munafò. 2013. Power failure: why small sample size undermines the reliability of neuroscience. *Nature Reviews Neuroscience* 14 (2013), 365-376. https://doi.org/10.1038/nrn3475

Gabriele Campanella, Matthew G. Hanna, Luke Geneslaw, Allen Miraflor, Vitor Werneck Krauss Silva, Klaus J. Busam, Edi Brogi, Victor E. Reuter, David S. Klimstra, and Thomas J. Fuchs. 2019. Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. *Nature Medicine* 25 (2019), 1301-1309. https://doi.org/10.1038/s41591-019-0508-1

Nicholas Carlini, Steve Chien, Milad Nasr, Shuang Song, Andreas Terzis, and Florian Tramèr. 2022. Membership Inference Attacks From First Principles. In *IEEE Symposium on Security and Privacy (S&P)*. 1897-1914. https://doi.org/10.1109/SP46214.2022.9833649

Daniel C. Castro, Ian Walker, and Ben Glocker. 2020. Causality matters in medical imaging. *Nature Communications* 11 (2020), 3673. https://doi.org/10.1038/s41467-020-17478-w

Krishna Chaitanya, Ertunc Erdil, Neerav Karani, and Ender Konukoglu. 2020. Contrastive learning of global and local features for medical image segmentation with limited annotations. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Krishna Chaitanya, Neerav Karani, Christian F. Baumgartner, Ertunc Erdil, Anton Becker, Olivio Donati, and Ender Konukoglu. 2021. Semi-supervised task-driven data augmentation for medical image segmentation. *Medical Image Analysis* 68 (2021), 101934. https://doi.org/10.1016/j.media.2020.101934

Ken Chang, Niranjan Balachandar, Carson Lam, Darvin Yi, James Brown, Andrew Beers, Bruce Rosen, Daniel L. Rubin, and Jayashree Kalpathy-Cramer. 2018. Distributed deep learning networks among institutions for medical imaging. *Journal of the American Medical Informatics Association* 25, 8 (2018), 945-954. https://doi.org/10.1093/jamia/ocy017

Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. 2002. SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research* 16 (2002), 321-357. https://doi.org/10.1613/jair.953

Irene Y. Chen, Emma Pierson, Sherri Rose, Shalmali Joshi, Kadija Ferryman, and Marzyeh Ghassemi. 2021b. Ethical Machine Learning in Healthcare. *Annual Review of Biomedical Data Science* 4 (2021), 123-144. https://doi.org/10.1146/annurev-biodatasci-092820-114757

Richard J. Chen, Tong Ding, Ming Y. Lu, Drew F. K. Williamson, Guillaume Jaume, Andrew H. Song, et al. 2024. Towards a general-purpose foundation model for computational pathology. *Nature Medicine* 30 (2024), 850-862. https://doi.org/10.1038/s41591-024-02857-3

Richard J. Chen, Ming Y. Lu, Tiffany Y. Chen, Drew F. K. Williamson, and Faisal Mahmood. 2021a. Synthetic data in machine learning for medicine and healthcare. *Nature Biomedical Engineering* 5 (2021), 493-497. https://doi.org/10.1038/s41551-021-00751-8

Sihong Chen, Kai Ma, and Yefeng Zheng. 2019. Med3D: Transfer Learning for 3D Medical Image Analysis. *arXiv preprint arXiv:1904.00625* (2019).

Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. 2020a. A Simple Framework for Contrastive Learning of Visual Representations. In *Proceedings of the 37th International Conference on Machine Learning (ICML) (PMLR, Vol. 119)*. 1597-1607.

Yiqiang Chen, Xin Qin, Jindong Wang, Chaohui Yu, and Wen Gao. 2020b. FedHealth: A Federated Transfer Learning Framework for Wearable Healthcare. *IEEE Intelligent Systems* 35, 4 (2020), 83-93. https://doi.org/10.1109/MIS.2020.2988604

Veronika Cheplygina, Marleen de Bruijne, and Josien P. W. Pluim. 2019. Not-so-supervised: A survey of semi-supervised, multi-instance, and transfer learning in medical image analysis. *Medical Image Analysis* 54 (2019), 280-296. https://doi.org/10.1016/j.media.2019.03.009

Phillip Chlap, Hang Min, Nym Vandenberg, Jason Dowling, Lois Holloway, and Annette Haworth. 2021. A review of medical image data augmentation techniques for deep learning applications. *Journal of Medical Imaging and Radiation Oncology* 65, 5 (2021), 545-563. https://doi.org/10.1111/1754-9485.13261

Hyunghoon Cho, David J. Wu, and Bonnie Berger. 2018. Secure genome-wide association analysis using multiparty computation. *Nature Biotechnology* 36 (2018), 547-551. https://doi.org/10.1038/nbt.4108

Junghwan Cho, Kyewook Lee, Ellie Shin, Garry Choy, and Synho Do. 2015. How much data is needed to train a medical image deep learning system to achieve necessary high accuracy? *arXiv preprint arXiv:1511.06348* (2015).

Edward Choi, Mohammad Taha Bahadori, Andy Schuetz, Walter F. Stewart, and Jimeng Sun. 2016. Doctor AI: Predicting Clinical Events via Recurrent Neural Networks. In *Proceedings of the 1st Machine Learning for Healthcare Conference (MLHC) (PMLR, Vol. 56)*. 301-318.

Edward Choi, Mohammad Taha Bahadori, Le Song, Walter F. Stewart, and Jimeng Sun. 2017a. GRAM: Graph-based Attention Model for Healthcare Representation Learning. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 787-795. https://doi.org/10.1145/3097983.3098126

Edward Choi, Siddharth Biswal, Bradley Malin, Jon Duke, Walter F. Stewart, and Jimeng Sun. 2017b. Generating Multi-label Discrete Patient Records using Generative Adversarial Networks. In *Proceedings of the 2nd Machine Learning for Healthcare Conference (MLHC) (PMLR, Vol. 68)*. 286-305.

Olivia Choudhury, Aris Gkoulalas-Divanis, Theodoros Salonidis, Issa Sylla, Yoonyoung Park, Grace Hsu, and Amar Das. 2019. Differential Privacy-enabled Federated Learning for Sensitive Health Data. *arXiv preprint arXiv:1910.02578* (2019).

Evangelia Christodoulou, Jie Ma, Gary S. Collins, Ewout W. Steyerberg, Jan Y. Verbakel, and Ben Van Calster. 2019. A systematic review shows no performance benefit of machine learning over logistic regression for clinical prediction models. *Journal of Clinical Epidemiology* 110 (2019), 12-22. https://doi.org/10.1016/j.jclinepi.2019.02.004

Fan R. K. Chung. 1997. *Spectral Graph Theory*. CBMS Regional Conference Series in Mathematics, Vol. 92. American Mathematical Society.

I. Glenn Cohen and Michelle M. Mello. 2018. HIPAA and Protecting Health Information in the 21st Century. *JAMA* 320, 3 (2018), 231-232. https://doi.org/10.1001/jama.2018.5630

Gary S. Collins, Karel G. M. Moons, Paula Dhiman, Richard D. Riley, Andrew L. Beam, Ben Van Calster, et al. 2024. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 385 (2024), e078378. https://doi.org/10.1136/bmj-2023-078378

Liam Collins, Hamed Hassani, Aryan Mokhtari, and Sanjay Shakkottai. 2021. Exploiting Shared Representations for Personalized Federated Learning. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (PMLR, Vol. 139)*. 2089-2099.

Thomas M. Cover and Joy A. Thomas. 2006. *Elements of Information Theory* (2nd ed.). Wiley-Interscience.

Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang. 2024. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods* 21 (2024), 1470-1480. https://doi.org/10.1038/s41592-024-02201-0

Edwige Cyffers and Aurélien Bellet. 2022. Privacy Amplification by Decentralization. In *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS) (PMLR, Vol. 151)*.

Ittai Dayan, Holger R. Roth, Aoxiao Zhong, Ahmed Harouni, Amilcare Gentili, Anas Z. Abidin, et al. 2021. Federated learning for predicting clinical outcomes in patients with COVID-19. *Nature Medicine* 27 (2021), 1735-1743. https://doi.org/10.1038/s41591-021-01506-3

Thomas P. A. Debray, Yvonne Vergouwe, Hendrik Koffijberg, Daan Nieboer, Ewout W. Steyerberg, and Karel G. M. Moons. 2015. A new framework to enhance the interpretation of external validation studies of clinical prediction models. *Journal of Clinical Epidemiology* 68, 3 (2015), 279-289. https://doi.org/10.1016/j.jclinepi.2014.06.018

Timo M. Deist, Frank J. W. M. Dankers, Priyanka Ojha, et al. 2020. Distributed learning on 20 000+ lung cancer patients - The Personal Health Train. *Radiotherapy and Oncology* 144 (2020), 189-200. https://doi.org/10.1016/j.radonc.2019.11.019

Rebecca DerSimonian and Nan Laird. 1986. Meta-analysis in clinical trials. *Controlled Clinical Trials* 7, 3 (1986), 177-188. https://doi.org/10.1016/0197-2456(86)90046-2

Qi Dou, Tiffany Y. So, Meirui Jiang, Quande Liu, Varut Vardhanabhuti, Georgios Kaissis, et al. 2021. Federated deep learning for detecting COVID-19 lung abnormalities in CT: a privacy-preserving multinational validation study. *npj Digital Medicine* 4 (2021), 60. https://doi.org/10.1038/s41746-021-00431-6

Simon S. Du, Wei Hu, Sham M. Kakade, Jason D. Lee, and Qi Lei. 2021. Few-Shot Learning via Learning the Representation, Provably. In *International Conference on Learning Representations (ICLR)*.

Rui Duan, Mary Regina Boland, Zixuan Liu, et al. 2020a. Learning from electronic health records across multiple sites: A communication-efficient and privacy-preserving distributed algorithm. *Journal of the American Medical Informatics Association* 27, 3 (2020), 376-385. https://doi.org/10.1093/jamia/ocz199

Rui Duan, Chongliang Luo, Martijn J. Schuemie, et al. 2020b. Learning from local to global: An efficient distributed algorithm for modeling time-to-event data. *Journal of the American Medical Informatics Association* 27, 7 (2020), 1028-1036. https://doi.org/10.1093/jamia/ocaa044

Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. 2006. Calibrating Noise to Sensitivity in Private Data Analysis. In *Theory of Cryptography Conference (TCC) (LNCS, Vol. 3876)*. 265-284. https://doi.org/10.1007/11681878_14

Cynthia Dwork and Aaron Roth. 2014. The Algorithmic Foundations of Differential Privacy. *Foundations and Trends in Theoretical Computer Science* 9, 3-4 (2014), 211-407. https://doi.org/10.1561/0400000042

Bradley Efron and Carl Morris. 1973. Stein's Estimation Rule and Its Competitors: An Empirical Bayes Approach. *J. Amer. Statist. Assoc.* 68, 341 (1973), 117-130. https://doi.org/10.1080/01621459.1973.10481350

Khaled El Emam, Elizabeth Jonker, Luk Arbuckle, and Bradley Malin. 2011. A Systematic Review of Re-Identification Attacks on Health Data. *PLoS ONE* 6, 12 (2011), e28071. https://doi.org/10.1371/journal.pone.0028071

Andre Esteva, Brett Kuprel, Roberto A. Novoa, Justin Ko, Susan M. Swetter, Helen M. Blau, and Sebastian Thrun. 2017. Dermatologist-level classification of skin cancer with deep neural networks. *Nature* 542 (2017), 115-118. https://doi.org/10.1038/nature21056

Deborah Estrin. 2014. Small data, where n = me. *Commun. ACM* 57, 4 (2014), 32-34. https://doi.org/10.1145/2580944

European Parliament and Council of the European Union. 2016. Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). Official Journal of the European Union, L 119.

European Parliament and Council of the European Union. 2025. Regulation (EU) 2025/327 on the European Health Data Space. Official Journal of the European Union, L series.

David Evans, Vladimir Kolesnikov, and Mike Rosulek. 2018. A Pragmatic Introduction to Secure Multi-Party Computation. *Foundations and Trends in Privacy and Security* 2, 2-3 (2018), 70-246. https://doi.org/10.1561/3300000019

Alireza Fallah, Aryan Mokhtari, and Asuman Ozdaglar. 2020. Personalized Federated Learning with Theoretical Guarantees: A Model-Agnostic Meta-Learning Approach. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Julian J. Faraway and Nicole H. Augustin. 2018. When small data beats big data. *Statistics & Probability Letters* 136 (2018), 142-145. https://doi.org/10.1016/j.spl.2018.02.031

Alberto Fernández, Salvador García, Francisco Herrera, and Nitesh V. Chawla. 2018. SMOTE for Learning from Imbalanced Data: Progress and Challenges, Marking the 15-year Anniversary. *Journal of Artificial Intelligence Research* 61 (2018), 863-905. https://doi.org/10.1613/jair.1.11192

Rosa L. Figueroa, Qing Zeng-Treitler, Sasikiran Kandula, and Long H. Ngo. 2012. Predicting sample size required for classification performance. *BMC Medical Informatics and Decision Making* 12 (2012), 8. https://doi.org/10.1186/1472-6947-12-8

Samuel G. Finlayson, Adarsh Subbaswamy, Karandeep Singh, John Bowers, Annabel Kupke, Jonathan Zittrain, Isaac S. Kohane, and Suchi Saria. 2021. The Clinician and Dataset Shift in Artificial Intelligence. *New England Journal of Medicine*

385, 3 (2021), 283-286. https://doi.org/10.1056/NEJMc2104626

Chelsea Finn, Pieter Abbeel, and Sergey Levine. 2017. Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML) (PMLR, Vol. 70)*. 1126-1135.

Maayan Frid-Adar, Idit Diamant, Eyal Klang, Michal Amitai, Jacob Goldberger, and Hayit Greenspan. 2018. GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification. *Neurocomputing* 321 (2018), 321-331. https://doi.org/10.1016/j.neucom.2018.09.013

Jason A. Fries, Paroma Varma, Vincent S. Chen, Ke Xiao, Heliodoro Tejeda, Priyanka Saha, et al. 2019. Weakly supervised classification of aortic valve malformations using unlabeled cardiac MRI sequences. *Nature Communications* 10 (2019), 3111. https://doi.org/10.1038/s41467-019-11012-3

David Froelicher, Juan R. Troncoso-Pastoriza, Jean Louis Raisaro, Michel A. Cuendet, Joao Sa Sousa, Hyunghoon Cho, Bonnie Berger, Jacques Fellay, and Jean-Pierre Hubaux. 2021. Truly privacy-preserving federated analytics for precision medicine with multiparty homomorphic encryption. *Nature Communications* 12 (2021), 5910. https://doi.org/10.1038/s41467-021-25972-y

Yarin Gal, Riashat Islam, and Zoubin Ghahramani. 2017. Deep Bayesian Active Learning with Image Data. In *Proceedings of the 34th International Conference on Machine Learning (ICML) (PMLR, Vol. 70)*. 1183-1192.

Amadou Gaye, Yannick Marcon, Julia Isaeva, Philippe LaFlamme, Andrew Turner, Elinor M. Jones, et al. 2014. DataSHIELD: taking the analysis to the data, not the data to the analysis. *International Journal of Epidemiology* 43, 6 (2014), 1929-1944. https://doi.org/10.1093/ije/dyu188

Jonas Geiping, Hartmut Bauermeister, Hannah Dröge, and Michael Moeller. 2020. Inverting Gradients - How easy is it to break privacy in federated learning?. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. 2013. *Bayesian Data Analysis* (3rd ed.). CRC Press.

Craig Gentry. 2009. Fully Homomorphic Encryption Using Ideal Lattices. In *Proceedings of the 41st Annual ACM Symposium on Theory of Computing (STOC)*. 169-178. https://doi.org/10.1145/1536414.1536440

Amirata Ghorbani and James Zou. 2019. Data Shapley: Equitable Valuation of Data for Machine Learning. In *Proceedings of the 36th International Conference on Machine Learning (ICML) (PMLR, Vol. 97)*. 2242-2251.

Avishek Ghosh, Jichan Chung, Dong Yin, and Kannan Ramchandran. 2020. An Efficient Framework for Clustered Federated Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Judy Wawira Gichoya, Imon Banerjee, Ananth Reddy Bhimireddy, John L. Burns, Leo Anthony Celi, et al. 2022. AI recognition of patient race in medical imaging: a modelling study. *The Lancet Digital Health* 4, 6 (2022), e406-e414. https://doi.org/10.1016/S2589-7500(22)00063-2

Andre Goncalves, Priyadip Ray, Braden Soper, Jennifer Stevens, Linda Coyle, and Ana Paula Sales. 2020. Generation and evaluation of synthetic patient data. *BMC Medical Research Methodology* 20 (2020), 108. https://doi.org/10.1186/s12874-020-00977-1

Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 2014. Generative Adversarial Nets. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 27.

Trisha Greenhalgh and Richard Peacock. 2005. Effectiveness and efficiency of search methods in systematic reviews of complex evidence: audit of primary sources. *BMJ* 331, 7524 (2005), 1064-1065. https://doi.org/10.1136/bmj.38636.593461.68

Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. 2022. Why do tree-based models still outperform deep learning on typical tabular data?. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, Vol. 35.

Hao Guan and Mingxia Liu. 2022. Domain Adaptation for Medical Image Analysis: A Survey. *IEEE Transactions on Biomedical Engineering* 69, 3 (2022), 1173-1185. https://doi.org/10.1109/TBME.2021.3117407

Ishaan Gulrajani and David Lopez-Paz. 2021. In Search of Lost Domain Generalization. In *International Conference on Learning Representations (ICLR)*.

Varun Gulshan, Lily Peng, Marc Coram, Martin C. Stumpe, Derek Wu, Arunachalam Narayanaswamy, et al. 2016. Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs. *JAMA* 316, 22 (2016), 2402-2410. https://doi.org/10.1001/jama.2016.17216

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML) (PMLR, Vol. 70)*. 1321-1330.

Pengfei Guo, Puyang Wang, Jinyuan Zhou, Shanshan Jiang, and Vishal M. Patel. 2021. Multi-institutional Collaborations for Improving Deep Learning-based Magnetic Resonance Image Reconstruction Using Federated Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2423-2432.

Otkrist Gupta and Ramesh Raskar. 2018. Distributed learning of deep neural network over multiple agents. *Journal of Network and Computer Applications* 116 (2018), 1-8. https://doi.org/10.1016/j.jnca.2018.05.003

Yaron Gurovich, Yair Hanani, Omri Bar, Guy Nadav, Nicole Fleischer, Dekel Gelbman, et al. 2019. Identifying facial phenotypes of genetic disorders using deep learning. *Nature Medicine* 25 (2019), 60-64. https://doi.org/10.1038/s41591-018-0279-0

Alon Halevy, Peter Norvig, and Fernando Pereira. 2009. The Unreasonable Effectiveness of Data. *IEEE Intelligent Systems* 24, 2 (2009), 8-12. https://doi.org/10.1109/MIS.2009.36

Yoni Halpern, Steven Horng, Youngduck Choi, and David Sontag. 2016. Electronic medical record phenotyping using the anchor and learn framework. *Journal of the American Medical Informatics Association* 23, 4 (2016), 731-740. https://doi.org/10.1093/jamia/ocw011

Trevor Hastie, Robert Tibshirani, and Jerome Friedman. 2009. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction (2nd ed.)*. Springer.

Ali Hatamizadeh, Hongxu Yin, Pavlo Molchanov, Andriy Myronenko, Wenqi Li, Prerna Dogra, et al. 2023. Do Gradient Inversion Attacks Make Federated Learning Unsafe? *IEEE Transactions on Medical Imaging* 42, 7 (2023), 2044-2056. https://doi.org/10.1109/TMI.2023.3239391

Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. 2022. Masked Autoencoders Are Scalable Vision Learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 16000-16009.

Stefan Hegselmann, Alejandro Buendia, Hunter Lang, Monica Agrawal, Xiaoyi Jiang, and David Sontag. 2023. TabLLM: Few-shot Classification of Tabular Data with Large Language Models. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS) (PMLR, Vol. 206)*. 5549-5581.

Eric B. Hekler, Predrag Klasnja, Guillaume Chevance, Natalie M. Golaszewski, Dana Lewis, and Ida Sim. 2019. Why we need a small data paradigm. *BMC Medicine* 17 (2019), 133. https://doi.org/10.1186/s12916-019-1366-x

Joel Hestness, Sharan Narang, Newsha Ardalani, Gregory Diamos, Heewoo Jun, Hassan Kianinejad, Md. Mostofa Ali Patwary, Yang Yang, and Yanqi Zhou. 2017. Deep Learning Scaling is Predictable, Empirically. *arXiv preprint arXiv:1712.00409* (2017).

Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Noah Hollmann, Samuel Müller, Katharina Eggensperger, and Frank Hutter. 2023. TabPFN: A Transformer That Solves Small Tabular Classification Problems in a Second. In *International Conference on Learning Representations (ICLR)*.

Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. 2025. Accurate predictions on small data with a tabular foundation model. *Nature* 637 (2025), 319-326. https://doi.org/10.1038/s41586-024-08328-6

Timothy Hospedales, Antreas Antoniou, Paul Micaelli, and Amos Storkey. 2022. Meta-Learning in Neural Networks: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 9 (2022), 5149-5169. https://doi.org/10.1109/TPAMI.2021.3079209

George Hripcsak, Jon D. Duke, Nigam H. Shah, Christian G. Reich, Vojtech Huser, Martijn J. Schuemie, et al. 2015. Observational Health Data Sciences and Informatics (OHDSI): Opportunities for Observational Researchers. *Studies in Health Technology and Informatics* 216 (2015), 574-578.

Kevin Hsieh, Amar Phanishayee, Onur Mutlu, and Phillip Gibbons. 2020. The Non-IID Data Quagmire of Decentralized Machine Learning. In *Proceedings of the 37th International Conference on Machine Learning (ICML) (PMLR, Vol. 119)*. 4387-4398.

Tzung-Chien Hsieh, Aviram Bar-Haim, Shahida Moosa, Nadja Ehmke, Karen W. Gripp, Jean Tori Pantel, et al. 2022. GestaltMatcher facilitates rare disease matching using facial phenotype descriptors. *Nature Genetics* 54 (2022), 349-357. https://doi.org/10.1038/s41588-021-01010-x

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations (ICLR)*.

Li Huang, Andrew L. Shea, Huining Qian, Aditya Masurkar, Hao Deng, and Dianbo Liu. 2019. Patient clustering improves efficiency of federated machine learning to predict mortality and hospital stay time using distributed electronic medical records. *Journal of Biomedical Informatics* 99 (2019), 103291. https://doi.org/10.1016/j.jbi.2019.103291

Shih-Cheng Huang, Liyue Shen, Matthew P. Lungren, and Serena Yeung. 2021b. GLoRIA: A Multimodal Global-Local Representation Learning Framework for Label-efficient Medical Image Recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 3942-3951.

Yutao Huang, Lingyang Chu, Zirui Zhou, Lanjun Wang, Jiangchuan Liu, Jian Pei, and Yong Zhang. 2021a. Personalized Cross-Silo Federated Learning on Non-IID Data. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35. 7865-7873.

Maximilian Ilse, Jakub Tomczak, and Max Welling. 2018. Attention-based Deep Multiple Instance Learning. In *Proceedings of the 35th International Conference on Machine Learning (ICML) (PMLR, Vol. 80)*. 2127-2136.

John P. A. Ioannidis. 2005. Why Most Published Research Findings Are False. *PLoS Medicine* 2, 8 (2005), e124. https://doi.org/10.1371/journal.pmed.0020124

Jeremy Irvin, Pranav Rajpurkar, Michael Ko, Yifan Yu, Silviana Ciurea-Ilcus, Chris Chute, et al. 2019. CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison. In *Proceedings of the AAAI Conference on*

Artificial Intelligence, Vol. 33. 590-597.

Bargav Jayaraman and David Evans. 2019. Evaluating Differentially Private Machine Learning in Practice. In *28th USENIX Security Symposium*. 1895-1912.

Lavender Yao Jiang, Xujin Chris Liu, Nima Pour Nejatian, Mustafa Nasir-Moin, Duo Wang, Anas Abidin, et al. 2023. Health system-scale language models are all-purpose prediction engines. *Nature* 619 (2023), 357-362. https://doi.org/10.1038/s41586-023-06160-y

Meirui Jiang, Zirui Wang, and Qi Dou. 2022. HarmoFL: Harmonizing Local and Global Drifts in Federated Learning on Heterogeneous Medical Images. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 1087-1095.

Wenchao Jiang, Pinghao Li, Shuang Wang, Yuan Wu, Meng Xue, Lucila Ohno-Machado, and Xiaoqian Jiang. 2013. WebGLORE: a web service for Grid LOgistic REgression. *Bioinformatics* 29, 24 (2013), 3238-3240. https://doi.org/10.1093/bioinformatics/btt559

Arthur Jochems, Timo M. Deist, Issam El Naqa, et al. 2017. Developing and Validating a Survival Prediction Model for NSCLC Patients Through Distributed Learning Across 3 Countries. *International Journal of Radiation Oncology, Biology, Physics* 99, 2 (2017), 344-352. https://doi.org/10.1016/j.ijrobp.2017.04.021

Arthur Jochems, Timo M. Deist, Johan van Soest, et al. 2016. Distributed learning: Developing a predictive model based on data from multiple hospitals without data leaving the hospital - A real life proof of concept. *Radiotherapy and Oncology* 121, 3 (2016), 459-467. https://doi.org/10.1016/j.radonc.2016.10.002

Alistair E. W. Johnson, Tom J. Pollard, Lu Shen, Li-wei H. Lehman, Mengling Feng, Mohammad Ghassemi, Benjamin Moody, Peter Szolovits, Leo Anthony Celi, and Roger G. Mark. 2016. MIMIC-III, a freely accessible critical care database. *Scientific Data* 3 (2016), 160035. https://doi.org/10.1038/sdata.2016.35

Michael I. Jordan, Jason D. Lee, and Yun Yang. 2019. Communication-Efficient Distributed Statistical Inference. *J. Amer. Statist. Assoc.* 114, 526 (2019), 668-681. https://doi.org/10.1080/01621459.2018.1429274

James Jordon, Lukasz Szpruch, Florimond Houssiau, Mirko Bottarelli, Giovanni Cherubin, Carsten Maple, Samuel N. Cohen, and Adrian Weller. 2022. Synthetic Data - what, why and how? *arXiv preprint arXiv:2205.03257* (2022).

James Jordon, Jinsung Yoon, and Mihaela van der Schaar. 2019. PATE-GAN: Generating Synthetic Data with Differential Privacy Guarantees. In *International Conference on Learning Representations (ICLR)*.

Peter Kairouz, H. Brendan McMahan, Brendan Avent, Aurélien Bellet, Mehdi Bennis, et al. 2021. Advances and Open Problems in Federated Learning. *Foundations and Trends in Machine Learning* 14, 1-2 (2021), 1-210. https://doi.org/10.1561/2200000083

Georgios Kaissis, Alexander Ziller, Jonathan Passerat-Palmbach, Théo Ryffel, Dmitrii Usynin, Andrew Trask, et al. 2021. End-to-end privacy preserving deep learning on multi-institutional medical imaging. *Nature Machine Intelligence* 3 (2021), 473-484. https://doi.org/10.1038/s42256-021-00337-8

Georgios A. Kaissis, Marcus R. Makowski, Daniel Rückert, and Rickmer F. Braren. 2020. Secure, privacy-preserving and federated machine learning in medical imaging. *Nature Machine Intelligence* 2 (2020), 305-311. https://doi.org/10.1038/s42256-020-0186-1

Shivam Kalra, Junfeng Wen, Jesse C. Cresswell, Maksims Volkovs, and H. R. Tizhoosh. 2023. Decentralized federated learning through proxy model sharing. *Nature Communications* 14, 1, Article 2899 (2023). https://doi.org/10.1038/s41467-023-38569-4

Konstantinos Kamnitsas, Christian Baumgartner, Christian Ledig, Virginia Newcombe, Joanna Simpson, Andrew Kane, et al. 2017. Unsupervised Domain Adaptation in Brain Lesion Segmentation with Adversarial Networks. In *Information Processing in Medical Imaging (IPMI) (LNCS, Vol. 10265)*. 597-609.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. Scaling Laws for Neural Language Models. *arXiv preprint arXiv:2001.08361* (2020).

Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank Reddi, Sebastian Stich, and Ananda Theertha Suresh. 2020. SCAFFOLD: Stochastic Controlled Averaging for Federated Learning. In *Proceedings of the 37th International Conference on Machine Learning (ICML) (PMLR, Vol. 119)*. 5132-5143.

George Em Karniadakis, Ioannis G. Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang. 2021. Physics-informed machine learning. *Nature Reviews Physics* 3 (2021), 422-440. https://doi.org/10.1038/s42254-021-00314-5

Amirhossein Kazerouni, Ehsan Khodapanah Aghdam, Moein Heidari, Reza Azad, Mohsen Fayyaz, Ilker Hacihaliloglu, and Dorit Merhof. 2023. Diffusion models in medical imaging: A comprehensive survey. *Medical Image Analysis* 88 (2023), 102846. https://doi.org/10.1016/j.media.2023.102846

Firas Khader, Gustav Müller-Franzes, Soroosh Tayebi Arasteh, Tianyu Han, Christoph Haarburger, Maximilian Schulze-Hagen, et al. 2023. Denoising diffusion probabilistic models for 3D medical image generation. *Scientific Reports* 13 (2023), 7303. https://doi.org/10.1038/s41598-023-34341-2

Dong Wook Kim, Hye Young Jang, Kyung Won Kim, Youngbin Shin, and Seong Ho Park. 2019. Design Characteristics of Studies Reporting the Performance of Artificial Intelligence Algorithms for Diagnostic Analysis of Medical Images: Results from Recently Published Papers. *Korean Journal of Radiology* 20, 3 (2019), 405-410. https://doi.org/10.3348/kjr.2019.0025

Yejin Kim, Jimeng Sun, Hwanjo Yu, and Xiaoqian Jiang. 2017. Federated Tensor Factorization for Computational Phenotyping. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 887-895. https://doi.org/10.1145/3097983.3098118

Leslie Kish. 1965. *Survey Sampling*. John Wiley & Sons, New York.

Georgios Kissas, Yibo Yang, Eileen Hwuang, Walter R. Witschey, John A. Detre, and Paris Perdikaris. 2020. Machine learning in cardiovascular flows modeling: Predicting arterial blood pressure from non-invasive 4D flow MRI data using physics-informed neural networks. *Computer Methods in Applied Mechanics and Engineering* 358 (2020), 112623. https://doi.org/10.1016/j.cma.2019.112623

Barbara Kitchenham and Stuart Charters. 2007. *Guidelines for performing Systematic Literature Reviews in Software Engineering*. Technical Report EBSE 2007-001. Keele University and Durham University.

Rob Kitchin and Tracey P. Lauriault. 2015. Small data in the era of big data. *GeoJournal* 80, 4 (2015), 463-475. https://doi.org/10.1007/s10708-014-9601-7

Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, et al. 2021. WILDS: A Benchmark of in-the-Wild Distribution Shifts. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (PMLR, Vol. 139)*. 5637-5664.

Sebastian Köhler, Michael Gargano, Nicolas Matentzoglu, Leigh C. Carmody, David Lewis-Smith, Nicole A. Vasilevsky, et al. 2021. The Human Phenotype Ontology in 2021. *Nucleic Acids Research* 49, D1 (2021), D1207-D1217. https://doi.org/10.1093/nar/gkaa1043

Anastasia Koloskova, Nicolas Loizou, Sadra Boreiri, Martin Jaggi, and Sebastian U. Stich. 2020. A Unified Theory of Decentralized SGD with Changing Topology and Local Updates. In *Proceedings of the 37th International Conference on Machine Learning (ICML) (PMLR, Vol. 119)*. 5381-5393.

Akim Kotelnikov, Dmitry Baranchuk, Ivan Rubachev, and Artem Babenko. 2023. TabDDPM: Modelling Tabular Data with Diffusion Models. In *Proceedings of the 40th International Conference on Machine Learning (ICML) (PMLR, Vol. 202)*. 17564-17579.

Rayan Krishnan, Pranav Rajpurkar, and Eric J. Topol. 2022. Self-supervised learning in medicine and healthcare. *Nature Biomedical Engineering* 6 (2022), 1346-1352. https://doi.org/10.1038/s41551-022-00914-1

Ira Ktena, Olivia Wiles, Isabela Albuquerque, Sylvestre-Alvise Rebuffi, Ryutaro Tanno, Abhijit Guha Roy, et al. 2024. Generative models improve fairness of medical classifiers under distribution shifts. *Nature Medicine* 30 (2024), 1166-1173. https://doi.org/10.1038/s41591-024-02838-6

John H. Lagergren, John T. Nardini, Ruth E. Baker, Matthew J. Simpson, and Kevin B. Flores. 2020. Biologically-informed neural networks guide mechanistic modeling from sparse experimental data. *PLoS Computational Biology* 16, 12 (2020), e1008462. https://doi.org/10.1371/journal.pcbi.1008462

Brenden M. Lake, Ruslan Salakhutdinov, and Joshua B. Tenenbaum. 2015. Human-level concept learning through probabilistic program induction. *Science* 350, 6266 (2015), 1332-1338. https://doi.org/10.1126/science.aab3050

Dong-Hyun Lee. 2013. Pseudo-Label: The Simple and Efficient Semi-Supervised Learning Method for Deep Neural Networks. In *ICML Workshop on Challenges in Representation Learning*.

Junghye Lee, Jimeng Sun, Fei Wang, Shuang Wang, Chi-Hyuck Jun, and Xiaoqian Jiang. 2018. Privacy-Preserving Patient Similarity Learning in a Federated Environment: Development and Analysis. *JMIR Medical Informatics* 6, 2 (2018), e20. https://doi.org/10.2196/medinform.7744

Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. 2020. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics* 36, 4 (2020), 1234-1240. https://doi.org/10.1093/bioinformatics/btz682

Tian Li, Shengyuan Hu, Ahmad Beirami, and Virginia Smith. 2021a. Ditto: Fair and Robust Federated Learning Through Personalization. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (PMLR, Vol. 139)*. 6357-6368.

Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. 2020c. Federated Optimization in Heterogeneous Networks. In *Proceedings of Machine Learning and Systems (MLSys)*, Vol. 2. 429-450.

Wenqi Li, Fausto Milletarì, Daguang Xu, Nicola Rieke, Jonny Hancox, Wentao Zhu, et al. 2019. Privacy-Preserving Federated Brain Tumour Segmentation. In *Machine Learning in Medical Imaging (MLMI) (LNCS, Vol. 11861)*. 133-141.

Xuhong Li, Yves Grandvalet, and Franck Davoine. 2018. Explicit Inductive Bias for Transfer Learning with Convolutional Networks. In *Proceedings of the 35th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 80)*. 2825–2834.

Xiaoxiao Li, Yufeng Gu, Nicha Dvornek, Lawrence H. Staib, Pamela Ventola, and James S. Duncan. 2020a. Multi-site fMRI analysis using privacy-preserving federated learning and domain adaptation: ABIDE results. *Medical Image Analysis* 65 (2020), 101765. https://doi.org/10.1016/j.media.2020.101765

Xiaoxiao Li, Meirui Jiang, Xiaofei Zhang, Michael Kamp, and Qi Dou. 2021b. FedBN: Federated Learning on Non-IID Features via Local Batch Normalization. In *International Conference on Learning Representations (ICLR)*.

Xiaomeng Li, Lequan Yu, Hao Chen, Chi-Wing Fu, Lei Xing, and Pheng-Ann Heng. 2021c. Transformation-Consistent Self-Ensembling Model for Semisupervised Medical Image Segmentation. *IEEE Transactions on Neural Networks and Learning Systems* 32, 2 (2021), 523-534. https://doi.org/10.1109/TNNLS.2020.2995319

Yikuan Li, Shishir Rao, José Roberto Ayala Solares, Abdelaali Hassaine, Rema Ramakrishnan, Dexter Canoy, Yajie Zhu, Kazem Rahimi, and Gholamreza Salimi-Khorshidi. 2020b. BEHRT: Transformer for Electronic Health Records. *Scientific Reports* 10 (2020), 7155. https://doi.org/10.1038/s41598-020-62922-y

Xiangru Lian, Ce Zhang, Huan Zhang, Cho-Jui Hsieh, Wei Zhang, and Ji Liu. 2017. Can Decentralized Algorithms Outperform Centralized Algorithms? A Case Study for Decentralized Parallel Stochastic Gradient Descent. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30.

Tao Lin, Lingjing Kong, Sebastian U. Stich, and Martin Jaggi. 2020. Ensemble Distillation for Robust Model Fusion in Federated Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Quande Liu, Cheng Chen, Jing Qin, Qi Dou, and Pheng-Ann Heng. 2021. FedDG: Federated Domain Generalization on Medical Image Segmentation via Episodic Learning in Continuous Frequency Space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 1013-1023.

Xiaoxuan Liu, Livia Faes, Aditya U. Kale, Siegfried K. Wagner, Dun Jack Fu, Alice Bruynseels, et al. 2019. A comparison of deep learning performance against health-care professionals in detecting diseases from medical imaging: a systematic review and meta-analysis. *The Lancet Digital Health* 1, 6 (2019), e271-e297. https://doi.org/10.1016/S2589-7500(19)30123-2

Charles Lu, Yaodong Yu, Sai Praneeth Karimireddy, Michael I. Jordan, and Ramesh Raskar. 2023. Federated Conformal Predictors for Distributed Uncertainty Quantification. In *Proceedings of the 40th International Conference on Machine Learning (ICML) (PMLR, Vol. 202)*.

Chia-Lun Lu, Shuang Wang, Zhanglong Ji, Yuan Wu, Li Xiong, Xiaoqian Jiang, and Lucila Ohno-Machado. 2015. WebDISCO: a web service for distributed cox model learning without patient-level data sharing. *Journal of the American Medical Informatics Association* 22, 6 (2015), 1212-1219. https://doi.org/10.1093/jamia/ocv083

James Lu, Kaiwen Deng, Xinyuan Zhang, Gengbo Liu, and Yuanfang Guan. 2021a. Neural-ODE for pharmacokinetics modeling and its advantage to alternative machine learning models in predicting new dosing regimens. *iScience* 24, 7 (2021), 102804. https://doi.org/10.1016/j.isci.2021.102804

Ming Y. Lu, Bowen Chen, Drew F. K. Williamson, Richard J. Chen, Ivy Liang, Tong Ding, et al. 2024. A visual-language foundation model for computational pathology. *Nature Medicine* 30 (2024), 863-874. https://doi.org/10.1038/s41591-024-02856-4

Ming Y. Lu, Richard J. Chen, Dehan Kong, Jana Lipkova, Rajendra Singh, Drew F. K. Williamson, Tiffany Y. Chen, and Faisal Mahmood. 2022. Federated learning for computational pathology on gigapixel whole slide images. *Medical Image Analysis* 76 (2022), 102298. https://doi.org/10.1016/j.media.2021.102298

Ming Y. Lu, Drew F. K. Williamson, Tiffany Y. Chen, Richard J. Chen, Matteo Barbieri, and Faisal Mahmood. 2021b. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nature Biomedical Engineering* 5 (2021), 555-570. https://doi.org/10.1038/s41551-020-00682-w

Fenglong Ma, Quanzeng You, Houping Xiao, Radha Chitta, Jing Zhou, and Jing Gao. 2018. KAME: Knowledge-based Attention Model for Diagnosis Prediction in Healthcare. In *Proceedings of the 27th ACM International Conference on Information and Knowledge Management (CIKM)*. 743-752. https://doi.org/10.1145/3269206.3271701

Jun Ma, Yuting He, Feifei Li, Lin Han, Chenyu You, and Bo Wang. 2024. Segment anything in medical images. *Nature Communications* 15 (2024), 654. https://doi.org/10.1038/s41467-024-44824-z

Kushagra Mahajan, Monika Sharma, and Lovekesh Vig. 2020. Meta-DermDiagnosis: Few-Shot Skin Disease Identification Using Meta-Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. IEEE, 3142-3151. https://doi.org/10.1109/CVPRW50498.2020.00373

Joshua C. Mandel, David A. Kreda, Kenneth D. Mandl, Isaac S. Kohane, and Rachel B. Ramoni. 2016. SMART on FHIR: a standards-based, interoperable apps platform for electronic health records. *Journal of the American Medical Informatics Association* 23, 5 (2016), 899-908. https://doi.org/10.1093/jamia/ocv189

Othmane Marfoq, Giovanni Neglia, Aurélien Bellet, Laetitia Kameni, and Richard Vidal. 2021. Federated Multi-Task Learning under a Mixture of Distributions. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 34.

Andreas Maurer, Massimiliano Pontil, and Bernardino Romera-Paredes. 2016. The Benefit of Multitask Representation Learning. *Journal of Machine Learning Research* 17, 81 (2016), 1-32.

Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS) (PMLR, Vol. 54)*. 1273-1282.

H. Brendan McMahan, Daniel Ramage, Kunal Talwar, and Li Zhang. 2018. Learning Differentially Private Recurrent Language Models. In *International Conference on Learning Representations (ICLR)*.

Xueyan Mei, Zelong Liu, Philip M. Robson, Brett Marinelli, Mingqian Huang, Amish Doshi, et al. 2022. RadImageNet: An Open Radiologic Deep Learning Research Dataset for Effective Transfer Learning. *Radiology: Artificial Intelligence* 4, 5

(2022), e210315. https://doi.org/10.1148/ryai.210315

Ilya Mironov. 2017. Rényi Differential Privacy. In *IEEE 30th Computer Security Foundations Symposium (CSF)*. 263-275. https://doi.org/10.1109/CSF.2017.11

Mehryar Mohri, Gary Sivek, and Ananda Theertha Suresh. 2019. Agnostic Federated Learning. In *Proceedings of the 36th International Conference on Machine Learning (ICML) (PMLR, Vol. 97)*. 4615-4625.

Michael Moor, Oishi Banerjee, Zahra Shakeri Hossein Abad, Harlan M. Krumholz, Jure Leskovec, Eric J. Topol, and Pranav Rajpurkar. 2023. Foundation models for generalist medical artificial intelligence. *Nature* 616 (2023), 259-265. https://doi.org/10.1038/s41586-023-05881-4

Satoshi Morita, Peter F. Thall, and Peter Müller. 2008. Determining the Effective Sample Size of a Parametric Prior. *Biometrics* 64, 2 (2008), 595–602. https://doi.org/10.1111/j.1541-0420.2007.00888.x

Myura Nagendran, Yang Chen, Christopher A. Lovejoy, Anthony C. Gordon, Matthieu Komorowski, Hugh Harvey, Eric J. Topol, John P. A. Ioannidis, Gary S. Collins, and Mahiben Maruthappu. 2020. Artificial intelligence versus clinicians: systematic review of design, reporting standards, and claims of deep learning studies. *BMJ* 368 (2020), m689. https://doi.org/10.1136/bmj.m689

Vishwesh Nath, Dong Yang, Bennett A. Landman, Daguang Xu, and Holger R. Roth. 2021. Diminishing Uncertainty Within the Training Pool: Active Learning for Medical Image Segmentation. *IEEE Transactions on Medical Imaging* 40, 10 (2021), 2534-2547. https://doi.org/10.1109/TMI.2020.3048055

Angelia Nedić and Asuman Ozdaglar. 2009. Distributed Subgradient Methods for Multi-Agent Optimization. *IEEE Trans. Automat. Control* 54, 1 (2009), 48-61. https://doi.org/10.1109/TAC.2008.2009515

Stéphanie Nguengang Wakap, Deborah M. Lambert, Annie Olry, Charlotte Rodwell, Charlotte Gueydan, Valérie Lanneau, Daniel Murphy, Yann Le Cam, and Ana Rath. 2020. Estimating cumulative point prevalence of rare diseases: analysis of the Orphanet database. *European Journal of Human Genetics* 28 (2020), 165-173. https://doi.org/10.1038/s41431-019-0508-0

Dinh C. Nguyen, Quoc-Viet Pham, Pubudu N. Pathirana, Ming Ding, Aruna Seneviratne, Zihuai Lin, Octavia Dobre, and Won-Joo Hwang. 2022. Federated Learning for Smart Healthcare: A Survey. *Comput. Surveys* 55, 3 (2022), 60:1-60:37. https://doi.org/10.1145/3501296

Harsha Nori, Nicholas King, Scott Mayer McKinney, Dean Carignan, and Eric Horvitz. 2023. Capabilities of GPT-4 on Medical Challenge Problems. *arXiv preprint arXiv:2303.13375* (2023).

Luke Oakden-Rayner, Jared Dunnmon, Gustavo Carneiro, and Christopher Ré. 2020. Hidden stratification causes clinically meaningful failures in machine learning for medical imaging. In *Proceedings of the ACM Conference on Health, Inference, and Learning (CHIL)*. 151-159. https://doi.org/10.1145/3368555.3384468

Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. 2019. Dissecting racial bias in an algorithm used to manage the health of populations. *Science* 366, 6464 (2019), 447-453. https://doi.org/10.1126/science.aax2342

Jean Ogier du Terrail, Samy-Safwan Ayed, Edwige Cyffers, Felix Grimberg, Chaoyang He, Regis Loeb, et al. 2022. FLamby: Datasets and Benchmarks for Cross-Silo Federated Learning in Realistic Healthcare Settings. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, Vol. 35.

Jean Ogier du Terrail, Armand Leopold, Clément Joly, Constance Béguier, Mathieu Andreux, Charles Maussion, et al. 2023. Federated learning for predicting histological response to neoadjuvant chemotherapy in triple-negative breast cancer. *Nature Medicine* 29 (2023), 135-146. https://doi.org/10.1038/s41591-022-02155-w

Cheng Ouyang, Carlo Biffi, Chen Chen, Turkay Kart, Huaqi Qiu, and Daniel Rueckert. 2020. Self-supervision with Superpixels: Training Few-Shot Medical Image Segmentation Without Annotation. In *European Conference on Computer Vision (ECCV) (LNCS, Vol. 12374)*. 762-780.

J. Marc Overhage, Patrick B. Ryan, Christian G. Reich, Abraham G. Hartzema, and Paul E. Stang. 2012. Validation of a common data model for active safety surveillance research. *Journal of the American Medical Informatics Association* 19, 1 (2012), 54-60. https://doi.org/10.1136/amiajnl-2011-000376

Kai Packhäuser, Sebastian Gündel, Nicolas Münster, Christopher Syben, Vincent Christlein, and Andreas Maier. 2022. Deep learning-based patient re-identification is able to exploit the biometric nature of medical chest X-ray data. *Scientific Reports* 12 (2022), 14851. https://doi.org/10.1038/s41598-022-19045-3

Matthew J. Page, Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, et al. 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 372 (2021), n71. https://doi.org/10.1136/bmj.n71

Suraj Pai, Dennis Bontempi, Ibrahim Hadzic, Vasco Prudente, Mateo Sokač, Tafadzwa L. Chaunzwa, et al. 2024. Foundation model for cancer imaging biomarkers. *Nature Machine Intelligence* 6 (2024), 354-367. https://doi.org/10.1038/s42256-024-00807-9

Sinno Jialin Pan and Qiang Yang. 2010. A Survey on Transfer Learning. *IEEE Transactions on Knowledge and Data Engineering* 22, 10 (2010), 1345-1359. https://doi.org/10.1109/TKDE.2009.191

Nicolas Papernot, Martín Abadi, Úlfar Erlingsson, Ian Goodfellow, and Kunal Talwar. 2017. Semi-supervised Knowledge Transfer for Deep Learning from Private Training Data. In *International Conference on Learning Representations (ICLR)*.

Sarthak Pati, Ujjwal Baid, Brandon Edwards, Micah Sheller, Shih-Han Wang, G. Anthony Reina, et al. 2022. Federated learning enables big data for rare cancer boundary detection. *Nature Communications* 13 (2022), 7346. https://doi.org/10.1038/s41467-022-33407-5

Judea Pearl. 2009. *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.

Peter Peduzzi, John Concato, Elizabeth Kemper, Theodore R. Holford, and Alvan R. Feinstein. 1996. A simulation study of the number of events per variable in logistic regression analysis. *Journal of Clinical Epidemiology* 49, 12 (1996), 1373-1379. https://doi.org/10.1016/S0895-4356(96)00236-3

Maarten G. Poirot, Praneeth Vepakomma, Ken Chang, Jayashree Kalpathy-Cramer, Rajiv Gupta, and Ramesh Raskar. 2019. Split Learning for collaborative deep learning in healthcare. *arXiv preprint arXiv:1912.12115* (2019).

Viraj Prabhu, Anitha Kannan, Murali Ravuri, Manish Chaplain, David Sontag, and Xavier Amatriain. 2019. Few-Shot Learning for Dermatological Disease Diagnosis. In *Proceedings of the 4th Machine Learning for Healthcare Conference (MLHC) (PMLR, Vol. 106)*. 532-552.

W. Nicholson Price and I. Glenn Cohen. 2019. Privacy in the age of medical big data. *Nature Medicine* 25 (2019), 37-43. https://doi.org/10.1038/s41591-018-0272-7

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, et al. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (PMLR, Vol. 139)*. 8748-8763.

Maithra Raghu, Chiyuan Zhang, Jon Kleinberg, and Samy Bengio. 2019. Transfusion: Understanding Transfer Learning for Medical Imaging. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32.

Jean Louis Raisaro, Juan Ramón Troncoso-Pastoriza, Mickaël Misbach, João Sá Sousa, Sylvain Pradervand, Edoardo Missiaglia, et al. 2019. MedCo: Enabling Secure and Privacy-Preserving Exploration of Distributed Clinical and Genomic Data. *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 16, 4 (2019), 1328-1341. https://doi.org/10.1109/TCBB.2018.2854776

Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J. Comput. Phys.* 378 (2019), 686-707. https://doi.org/10.1016/j.jcp.2018.10.045

Pranav Rajpurkar, Emma Chen, Oishi Banerjee, and Eric J. Topol. 2022. AI in health and medicine. *Nature Medicine* 28 (2022), 31-38. https://doi.org/10.1038/s41591-021-01614-0

Laila Rasmy, Yang Xiang, Ziqian Xie, Cui Tao, and Degui Zhi. 2021. Med-BERT: pretrained contextualized embeddings on large-scale structured electronic health records for disease prediction. *npj Digital Medicine* 4 (2021), 86. https://doi.org/10.1038/s41746-021-00455-y

Alexander Ratner, Stephen H. Bach, Henry Ehrenberg, Jason Fries, Sen Wu, and Christopher Ré. 2017. Snorkel: Rapid Training Data Creation with Weak Supervision. *Proceedings of the VLDB Endowment* 11, 3 (2017), 269-282. https://doi.org/10.14778/3157794.3157797

Sarunas J. Raudys and Anil K. Jain. 1991. Small Sample Size Effects in Statistical Pattern Recognition: Recommendations for Practitioners. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 13, 3 (1991), 252–264. https://doi.org/10.1109/34.75512

Jenna M. Reps, Martijn J. Schuemie, Marc A. Suchard, Patrick B. Ryan, and Peter R. Rijnbeek. 2018. Design and implementation of a standardized framework to generate and evaluate patient-level prediction models using observational healthcare data. *Journal of the American Medical Informatics Association* 25, 8 (2018), 969-975. https://doi.org/10.1093/jamia/ocy032

Nicola Rieke, Jonny Hancox, Wenqi Li, Fausto Milletarì, Holger R. Roth, Shadi Albarqouni, et al. 2020. The future of digital health with federated learning. *npj Digital Medicine* 3 (2020), 119. https://doi.org/10.1038/s41746-020-00323-1

Richard D. Riley, Joie Ensor, Kym I. E. Snell, Frank E. Harrell, Glen P. Martin, Johannes B. Reitsma, Karel G. M. Moons, Gary Collins, and Maarten van Smeden. 2020. Calculating the sample size required for developing a clinical prediction model. *BMJ* 368 (2020), m441. https://doi.org/10.1136/bmj.m441

Richard D. Riley, Paul C. Lambert, and Ghada Abo-Zaid. 2010. Meta-analysis of individual participant data: rationale, conduct, and reporting. *BMJ* 340 (2010), c221. https://doi.org/10.1136/bmj.c221

Richard D. Riley, Kym I. E. Snell, Joie Ensor, Danielle L. Burke, Frank E. Harrell, Karel G. M. Moons, and Gary S. Collins. 2019a. Minimum sample size for developing a multivariable prediction model: Part I - Continuous outcomes. *Statistics in Medicine* 38, 7 (2019), 1262-1275. https://doi.org/10.1002/sim.7993

Richard D. Riley, Kym I. E. Snell, Joie Ensor, Danielle L. Burke, Frank E. Harrell, Karel G. M. Moons, and Gary S. Collins. 2019b. Minimum sample size for developing a multivariable prediction model: PART II - binary and time-to-event outcomes. *Statistics in Medicine* 38, 7 (2019), 1276-1296. https://doi.org/10.1002/sim.7992

Richard D. Riley, Kym I. E. Snell, Glen P. Martin, Rebecca Whittle, Lucinda Archer, Matthew Sperrin, and Gary S. Collins. 2021. Penalization and shrinkage methods produced unreliable clinical prediction models especially when sample size was small. *Journal of Clinical Epidemiology* 132 (2021), 88-96. https://doi.org/10.1016/j.jclinepi.2020.12.005

Michael Roberts, Derek Driggs, Matthew Thorpe, Julian Gilbey, Michael Yeung, Stephan Ursprung, et al. 2021. Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. *Nature Machine Intelligence* 3 (2021), 199-217. https://doi.org/10.1038/s42256-021-00307-0

Luc Rocher, Julien M. Hendrickx, and Yves-Alexandre de Montjoye. 2019. Estimating the success of re-identifications in incomplete datasets using generative models. *Nature Communications* 10 (2019), 3069. https://doi.org/10.1038/s41467-019-10933-3

Holger R. Roth, Ken Chang, Praveer Singh, Nir Neumark, Wenqi Li, Vikash Gupta, et al. 2020. Federated Learning for Breast Density Classification: A Real-World Implementation. In *Domain Adaptation and Representation Transfer, and Distributed and Collaborative Learning (DART/DCL) (LNCS, Vol. 12444)*. 181-191.

Maya Rotmensch, Yoni Halpern, Abdulhakim Tlimat, Steven Horng, and David Sontag. 2017. Learning a Health Knowledge Graph from Electronic Medical Records. *Scientific Reports* 7 (2017), 5994. https://doi.org/10.1038/s41598-017-05778-z

Abhijit Guha Roy, Shayan Siddiqui, Sebastian Pölsterl, Nassir Navab, and Christian Wachinger. 2019. BrainTorrent: A Peer-to-Peer Environment for Decentralized Federated Learning. *arXiv preprint arXiv:1905.06731* (2019).

Abhijit Guha Roy, Shayan Siddiqui, Sebastian Pölsterl, Nassir Navab, and Christian Wachinger. 2020. 'Squeeze & excite' guided few-shot segmentation of volumetric images. *Medical Image Analysis* 59 (2020), 101587. https://doi.org/10.1016/j.media.2019.101587

Håvard Rue and Leonhard Held. 2005. *Gaussian Markov Random Fields: Theory and Applications*. Chapman and Hall/CRC.

Francisco Sahli Costabal, Yibo Yang, Paris Perdikaris, Daniel E. Hurtado, and Ellen Kuhl. 2020. Physics-Informed Neural Networks for Cardiac Activation Mapping. *Frontiers in Physics* 8 (2020), 42. https://doi.org/10.3389/fphy.2020.00042

Oliver Lester Saldanha, Philip Quirke, Nicholas P. West, Jacqueline A. James, Maurice B. Loughrey, Heike I. Grabsch, et al. 2022. Swarm learning for decentralized artificial intelligence in cancer histopathology. *Nature Medicine* 28 (2022), 1232-1239. https://doi.org/10.1038/s41591-022-01768-5

Karthik V. Sarma, Stephanie Harmon, Thomas Sanford, Holger R. Roth, Ziyue Xu, Jesse Tetreault, et al. 2021. Federated learning improves site performance in multicenter deep learning without data sharing. *Journal of the American Medical Informatics Association* 28, 6 (2021), 1259-1264. https://doi.org/10.1093/jamia/ocaa341

Julia Schaefer, Moritz Lehne, Josef Schepers, Fabian Prasser, and Sylvia Thun. 2020. The use of machine learning in rare diseases: a scoping review. *Orphanet Journal of Rare Diseases* 15 (2020), 145. https://doi.org/10.1186/s13023-020-01424-6

Bernhard Schölkopf, Dominik Janzing, Jonas Peters, Eleni Sgouritsa, Kun Zhang, and Joris Mooij. 2012. On Causal and Anticausal Learning. In *Proceedings of the 29th International Conference on Machine Learning (ICML)*. 1255-1262.

Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. 2021. Toward Causal Representation Learning. *Proc. IEEE* 109, 5 (2021), 612-634. https://doi.org/10.1109/JPROC.2021.3058954

Christopher G. Schwarz, Walter K. Kremers, Terry M. Therneau, Richard R. Sharp, Jeffrey L. Gunter, Prashanthi Vemuri, et al. 2019. Identification of Anonymous MRI Research Participants with Face-Recognition Software. *New England Journal of Medicine* 381, 17 (2019), 1684-1686. https://doi.org/10.1056/NEJMc1908881

Burr Settles. 2009. *Active Learning Literature Survey*. Technical Report 1648. University of Wisconsin-Madison, Department of Computer Sciences.

Glenn Shafer and Vladimir Vovk. 2008. A Tutorial on Conformal Prediction. *Journal of Machine Learning Research* 9 (2008), 371-421.

Shai Shalev-Shwartz and Shai Ben-David. 2014. *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press.

Junyuan Shang, Cao Xiao, Tengfei Ma, Hongyan Li, and Jimeng Sun. 2019. GAMENet: Graph Augmented MEmory Networks for Recommending Medication Combination. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 33. 1126-1133.

Micah J. Sheller, Brandon Edwards, G. Anthony Reina, Jason Martin, Sarthak Pati, Aikaterini Kotrotsou, et al. 2020. Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. *Scientific Reports* 10 (2020), 12598. https://doi.org/10.1038/s41598-020-69250-1

Micah J. Sheller, G. Anthony Reina, Brandon Edwards, Jason Martin, and Spyridon Bakas. 2019. Multi-institutional Deep Learning Modeling Without Sharing Patient Data: A Feasibility Study on Brain Tumor Segmentation. In *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries (BrainLes) (LNCS, Vol. 11383)*. 92-104.

Hoo-Chang Shin, Holger R. Roth, Mingchen Gao, Le Lu, Ziyue Xu, Isabella Nogues, Jianhua Yao, Daniel Mollura, and Ronald M. Summers. 2016. Deep Convolutional Neural Networks for Computer-Aided Detection: CNN Architectures, Dataset Characteristics and Transfer Learning. *IEEE Transactions on Medical Imaging* 35, 5 (2016), 1285-1298. https://doi.org/10.1109/TMI.2016.2528162

Hoo-Chang Shin, Neil A. Tenenholtz, Jameson K. Rogers, Christopher G. Schwarz, Matthew L. Senjem, Jeffrey L. Gunter, Katherine P. Andriole, and Mark Michalski. 2018. Medical Image Synthesis for Data Augmentation and Anonymization Using Generative Adversarial Networks. In *Simulation and Synthesis in Medical Imaging (SASHIMI) (LNCS, Vol. 11037)*.

1-11.
Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. 2017. Membership Inference Attacks Against Machine Learning Models. In *IEEE Symposium on Security and Privacy (S&P)*. 3-18. https://doi.org/10.1109/SP.2017.41
Connor Shorten and Taghi M. Khoshgoftaar. 2019. A survey on Image Data Augmentation for Deep Learning. *Journal of Big Data* 6 (2019), 60. https://doi.org/10.1186/s40537-019-0197-0
Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. 2024. AI models collapse when trained on recursively generated data. *Nature* 631 (2024), 755-759. https://doi.org/10.1038/s41586-024-07566-y
Sancho Amaral Simões, João P. Vilela, and Pedro Henriques Abreu. 2026a. Borrowing Strength Without Borrowing Data: Private, Serverless and Tuning-Free Collaboration with Graph Hierarchical Borrowing. (2026). Companion paper, under review.
Sancho Amaral Simões, João P. Vilela, and Pedro Henriques Abreu. 2026b. When Every Row Counts: Verifier-Guided Agentic Synthesis for Small Tabular Data. (2026). Under review.
Karan Singhal, Shekoofeh Azizi, Tao Tu, S. Sara Mahdavi, Jason Wei, Hyung Won Chung, et al. 2023. Large language models encode clinical knowledge. *Nature* 620 (2023), 172-180. https://doi.org/10.1038/s41586-023-06291-2
Asim Smailagic, Pedro Costa, Hae Young Noh, Devesh Walawalkar, Kartik Khandelwal, Adrian Galdran, et al. 2018. MedAL: Accurate and Robust Deep Active Learning for Medical Image Analysis. In *17th IEEE International Conference on Machine Learning and Applications (ICMLA)*. 481-488.
Virginia Smith, Chao-Kai Chiang, Maziar Sanjabi, and Ameet Talwalkar. 2017. Federated Multi-Task Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30.
Jake Snell, Kevin Swersky, and Richard Zemel. 2017. Prototypical Networks for Few-shot Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30.
Kihyuk Sohn, David Berthelot, Nicholas Carlini, Zizhao Zhang, Han Zhang, Colin A. Raffel, Ekin Dogus Cubuk, Alexey Kurakin, and Chun-Liang Li. 2020. FixMatch: Simplifying Semi-Supervised Learning with Consistency and Confidence. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.
Hari Sowrirajan, Jingbo Yang, Andrew Y. Ng, and Pranav Rajpurkar. 2021. MoCo Pretraining Improves Representation and Transferability of Chest X-ray Models. In *Proceedings of the Fourth Conference on Medical Imaging with Deep Learning (MIDL) (PMLR, Vol. 143)*.
Theresa Stadler, Bristena Oprisanu, and Carmela Troncoso. 2022. Synthetic Data - Anonymisation Groundhog Day. In *31st USENIX Security Symposium*. 1451-1468.
Charles Stein. 1956. Inadmissibility of the Usual Estimator for the Mean of a Multivariate Normal Distribution. In *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability*, Vol. 1. 197-206.
Ethan Steinberg, Ken Jung, Jason A. Fries, Conor K. Corbin, Stephen R. Pfohl, and Nigam H. Shah. 2021. Language models are an effective representation learning technique for electronic health record data. *Journal of Biomedical Informatics* 113 (2021), 103637. https://doi.org/10.1016/j.jbi.2020.103637
Ewout W. Steyerberg, Marinus J. C. Eijkemans, Frank E. Harrell, and J. Dik F. Habbema. 2000. Prognostic modelling with logistic regression analysis: a comparison of selection and estimation methods in small data sets. *Statistics in Medicine* 19, 8 (2000), 1059-1079.
Adarsh Subbaswamy, Peter Schulam, and Suchi Saria. 2019. Preventing Failures Due to Dataset Shift: Learning Predictive Models That Transport. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS) (PMLR, Vol. 89)*. 3118-3127.
Marc A. Suchard, Martijn J. Schuemie, Harlan M. Krumholz, Seng Chan You, RuiJun Chen, Nicole Pratt, et al. 2019. Comprehensive comparative effectiveness and safety of first-line antihypertensive drug classes: a systematic, multinational, large-scale analysis. *The Lancet* 394, 10211 (2019), 1816-1826. https://doi.org/10.1016/S0140-6736(19)32317-7
Chen Sun, Abhinav Shrivastava, Saurabh Singh, and Abhinav Gupta. 2017. Revisiting Unreasonable Effectiveness of Data in Deep Learning Era. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. 843-852.
Latanya Sweeney. 2002. k-anonymity: A model for protecting privacy. *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems* 10, 5 (2002), 557-570. https://doi.org/10.1142/S0218488502001648
Canh T. Dinh, Nguyen Tran, and Josh Nguyen. 2020. Personalized Federated Learning with Moreau Envelopes. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.
Nima Tajbakhsh, Laura Jeyaseelan, Qian Li, Jeffrey N. Chiang, Zhihao Wu, and Xiaowei Ding. 2020. Embracing imperfect datasets: A review of deep learning solutions for medical image segmentation. *Medical Image Analysis* 63 (2020), 101693. https://doi.org/10.1016/j.media.2020.101693
Nima Tajbakhsh, Jae Y. Shin, Suryakanth R. Gurudu, R. Todd Hurst, Christopher B. Kendall, Michael B. Gotway, and Jianming Liang. 2016. Convolutional Neural Networks for Medical Image Analysis: Full Training or Fine Tuning? *IEEE Transactions on Medical Imaging* 35, 5 (2016), 1299-1312. https://doi.org/10.1109/TMI.2016.2535302
Aiham Taleb, Winfried Loetzsch, Noel Danz, Julius Severin, Thomas Gaertner, Benjamin Bergner, and Christoph Lippert. 2020. 3D Self-Supervised Methods for Medical Imaging. In *Advances in Neural Information Processing Systems (NeurIPS)*,

Vol. 33.

Alysa Ziying Tan, Han Yu, Lizhen Cui, and Qiang Yang. 2023. Towards Personalized Federated Learning. *IEEE Transactions on Neural Networks and Learning Systems* 34, 12 (2023), 9587-9603. https://doi.org/10.1109/TNNLS.2022.3160699

Mingxing Tan and Quoc V. Le. 2019. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. In *Proceedings of the 36th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 97)*. 6105-6114.

Bernardo Camajori Tedeschini, Stefano Savazzi, Roman Stoklasa, et al. 2022. Decentralized Federated Learning for Healthcare Networks: A Case Study on Tumor Segmentation. *IEEE Access* 10 (2022), 8693-8708. https://doi.org/10.1109/ACCESS.2022.3141913

Christina V. Theodoris, Ling Xiao, Anant Chopra, Mark D. Chaffin, Zeina R. Al Sayed, Matthew C. Hill, et al. 2023. Transfer learning enables predictions in network biology. *Nature* 618 (2023), 616-624. https://doi.org/10.1038/s41586-023-06139-9

Robert Tibshirani. 1996. Regression Shrinkage and Selection via the Lasso. *Journal of the Royal Statistical Society: Series B* 58, 1 (1996), 267-288.

Ekin Tiu, Ellie Talius, Pujan Patel, Curtis P. Langlotz, Andrew Y. Ng, and Pranav Rajpurkar. 2022. Expert-level detection of pathologies from unannotated chest X-ray images via self-supervised learning. *Nature Biomedical Engineering* 6 (2022), 1399-1406. https://doi.org/10.1038/s41551-022-00936-9

Eric J. Topol. 2019. High-performance medicine: the convergence of human and artificial intelligence. *Nature Medicine* 25 (2019), 44-56. https://doi.org/10.1038/s41591-018-0300-7

Florian Tramèr and Dan Boneh. 2021. Differentially Private Learning Needs Better Features (or Much More Data). In *International Conference on Learning Representations (ICLR)*.

Nilesh Tripuraneni, Michael I. Jordan, and Chi Jin. 2020. On the Theory of Transfer Learning: The Importance of Task Diversity. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Andrius Vabalas, Emma Gowen, Ellen Poliakoff, and Alexander J. Casson. 2019. Machine Learning Algorithm Validation with a Limited Sample Size. *PLOS ONE* 14, 11 (2019), e0224365. https://doi.org/10.1371/journal.pone.0224365

Akhil Vaid, Suraj K. Jaladanki, Jie Xu, Shelly Teng, Arvind Kumar, Samuel Lee, et al. 2021. Federated Learning of Electronic Health Records to Improve Mortality Prediction in Hospitalized Patients With COVID-19: Machine Learning Approach. *JMIR Medical Informatics* 9, 1 (2021), e24207. https://doi.org/10.2196/24207

Ben Van Calster, David J. McLernon, Maarten van Smeden, Laure Wynants, and Ewout W. Steyerberg. 2019. Calibration: the Achilles heel of predictive analytics. *BMC Medicine* 17, 1 (2019), 230. https://doi.org/10.1186/s12916-019-1466-7

Ben Van Calster, Maarten van Smeden, Bavo De Cock, and Ewout W. Steyerberg. 2020. Regression shrinkage methods for clinical prediction models do not guarantee improved performance: Simulation study. *Statistical Methods in Medical Research* 29, 11 (2020), 3166-3178. https://doi.org/10.1177/0962280220921415

Ruben van den Goorbergh, Maarten van Smeden, Dirk Timmerman, and Ben Van Calster. 2022. The harm of class imbalance corrections for risk prediction models: illustration and simulation using logistic regression. *Journal of the American Medical Informatics Association* 29, 9 (2022), 1525-1534. https://doi.org/10.1093/jamia/ocac093

Maarten van Smeden, Karel G. M. Moons, Joris A. H. de Groot, Gary S. Collins, Douglas G. Altman, Marinus J. C. Eijkemans, and Johannes B. Reitsma. 2019. Sample size for binary logistic prediction models: Beyond events per variable criteria. *Statistical Methods in Medical Research* 28, 8 (2019), 2455-2474. https://doi.org/10.1177/0962280218784726

Paul Vanhaesebrouck, Aurélien Bellet, and Marc Tommasi. 2017. Decentralized Collaborative Learning of Personalized Models over Networks. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS) (PMLR, Vol. 54)*. 509-517.

Gaël Varoquaux. 2018. Cross-validation failure: Small sample sizes lead to large error bars. *NeuroImage* 180 (2018), 68-77. https://doi.org/10.1016/j.neuroimage.2017.06.061

Gaël Varoquaux and Veronika Cheplygina. 2022. Machine learning for medical imaging: methodological failures and recommendations for the future. *npj Digital Medicine* 5 (2022), 48. https://doi.org/10.1038/s41746-022-00592-y

Praneeth Vepakomma, Otkrist Gupta, Tristan Swedish, and Ramesh Raskar. 2018. Split learning for health: Distributed deep learning without sharing raw patient data. *arXiv preprint arXiv:1812.00564* (2018).

Laura von Rueden, Sebastian Mayer, Katharina Beckh, Bogdan Georgiev, Sven Giesselbach, Raoul Heese, et al. 2023. Informed Machine Learning - A Taxonomy and Survey of Integrating Prior Knowledge into Learning Systems. *IEEE Transactions on Knowledge and Data Engineering* 35, 1 (2023), 614-633. https://doi.org/10.1109/TKDE.2021.3079836

Eugene Vorontsov, Alican Bozkurt, Adam Casson, George Shaikovski, Michal Zelechowski, Kristen Severson, et al. 2024. A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine* 30, 10 (2024), 2924-2935. https://doi.org/10.1038/s41591-024-03141-0

Jason Walonoski, Mark Kramer, Joseph Nichols, Andre Quina, Chris Moesel, Dylan Hall, Carlton Duffett, Kudakwashe Dube, Thomas Gallagher, and Scott McLachlan. 2018. Synthea: An approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record. *Journal of the American Medical Informatics Association* 25, 3 (2018), 230-238. https://doi.org/10.1093/jamia/ocx079

Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M. Summers. 2017. ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2097-2106.

Yanshan Wang, Sunghwan Sohn, Sijia Liu, Feichen Shen, Liwei Wang, Elizabeth J. Atkinson, Shreyasee Amin, and Hongfang Liu. 2019. A clinical text classification paradigm using weak supervision and deep representation. *BMC Medical Informatics and Decision Making* 19 (2019), 1. https://doi.org/10.1186/s12911-018-0723-6

Yaqing Wang, Quanming Yao, James T. Kwok, and Lionel M. Ni. 2020. Generalizing from a Few Examples: A Survey on Few-shot Learning. *Comput. Surveys* 53, 3 (2020), 63:1-63:34. https://doi.org/10.1145/3386252

Zifeng Wang, Zhenbang Wu, Dinesh Agarwal, and Jimeng Sun. 2022. MedCLIP: Contrastive Learning from Unpaired Medical Images and Text. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 3876-3887. https://doi.org/10.18653/v1/2022.emnlp-main.256

Stefanie Warnat-Herresthal, Hartmut Schultze, Krishnaprasad Lingadahalli Shastry, Sathyanarayanan Manamohan, Saikat Mukherjee, Vishesh Garg, et al. 2021. Swarm Learning for decentralized and confidential clinical machine learning. *Nature* 594 (2021), 265-270. https://doi.org/10.1038/s41586-021-03583-3

Kang Wei, Jun Li, Ming Ding, Chuan Ma, Howard H. Yang, Farhad Farokhi, Shi Jin, Tony Q. S. Quek, and H. Vincent Poor. 2020. Federated Learning With Differential Privacy: Algorithms and Performance Analysis. *IEEE Transactions on Information Forensics and Security* 15 (2020), 3454-3469. https://doi.org/10.1109/TIFS.2020.2988575

Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, et al. 2016. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data* 3 (2016), 160018. https://doi.org/10.1038/sdata.2016.18

Martin J. Willemink, Wojciech A. Koszek, Cailin Hardell, Jie Wu, Dominik Fleischmann, Hugh Harvey, Les R. Folio, Ronald M. Summers, Daniel L. Rubin, and Matthew P. Lungren. 2020. Preparing Medical Imaging Data for Machine Learning. *Radiology* 295, 1 (2020), 4-15. https://doi.org/10.1148/radiol.2020192224

Claes Wohlin. 2014. Guidelines for Snowballing in Systematic Literature Studies and a Replication in Software Engineering. In *Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering (EASE)*. ACM, Article 38, 10 pages. https://doi.org/10.1145/2601248.2601268

Andrew Wong, Erkin Otles, John P. Donnelly, et al. 2021. External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients. *JAMA Internal Medicine* 181, 8 (2021), 1065-1070. https://doi.org/10.1001/jamainternmed.2021.2626

Michael Wornow, Yizhe Xu, Rahul Thapa, Birju Patel, Ethan Steinberg, Scott Fleming, et al. 2023. The shaky foundations of large language models and foundation models for electronic health records. *npj Digital Medicine* 6 (2023), 135. https://doi.org/10.1038/s41746-023-00879-8

Yuan Wu, Xiaoqian Jiang, Jihoon Kim, and Lucila Ohno-Machado. 2012. Grid Binary LOgistic REgression (GLORE): building shared models without sharing data. *Journal of the American Medical Informatics Association* 19, 5 (2012), 758-764. https://doi.org/10.1136/amiajnl-2012-000862

Laure Wynants, Ben Van Calster, Gary S. Collins, Richard D. Riley, Georg Heinze, Ewoud Schuit, et al. 2020. Prediction models for diagnosis and prognosis of covid-19: systematic review and critical appraisal. *BMJ* 369 (2020), m1328. https://doi.org/10.1136/bmj.m1328

Lin Xiao and Stephen Boyd. 2004. Fast linear iterations for distributed averaging. *Systems & Control Letters* 53, 1 (2004), 65-78. https://doi.org/10.1016/j.sysconle.2004.02.022

An Xu, Wenqi Li, Pengfei Guo, Dong Yang, Holger R. Roth, Ali Hatamizadeh, Can Zhao, Daguang Xu, Heng Huang, and Ziyue Xu. 2022. Closing the Generalization Gap of Cross-silo Federated Medical Image Segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 20866-20875.

Hanwen Xu, Naoto Usuyama, Jaspreet Bagga, Sheng Zhang, Rajesh Rao, Tristan Naumann, et al. 2024. A whole-slide foundation model for digital pathology from real-world data. *Nature* 630 (2024), 181-188. https://doi.org/10.1038/s41586-024-07441-w

Jie Xu, Benjamin S. Glicksberg, Chang Su, Peter Walker, Jiang Bian, and Fei Wang. 2021. Federated Learning for Healthcare Informatics. *Journal of Healthcare Informatics Research* 5 (2021), 1-19. https://doi.org/10.1007/s41666-020-00082-4

Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. 2019. Modeling Tabular data using Conditional GAN. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32.

Pengcheng Xu, Xiaobo Ji, Minjie Li, and Wencong Lu. 2023. Small Data Machine Learning in Materials Science. *npj Computational Materials* 9 (2023), 42. https://doi.org/10.1038/s41524-023-01000-z

Rui Yan, Liangqiong Qu, Qingyue Wei, Shih-Cheng Huang, Liyue Shen, Daniel L. Rubin, Lei Xing, and Yuyin Zhou. 2023. Label-Efficient Self-Supervised Federated Learning for Tackling Data Heterogeneity in Medical Imaging. *IEEE Transactions on Medical Imaging* 42, 7 (2023), 1932-1943. https://doi.org/10.1109/TMI.2022.3233574

Lin Yang, Yizhe Zhang, Jianxu Chen, Siyuan Zhang, and Danny Z. Chen. 2017. Suggestive Annotation: A Deep Active Learning Framework for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI) (LNCS, Vol. 10435)*. 399-407.

Qiang Yang, Yang Liu, Tianjian Chen, and Yongxin Tong. 2019. Federated Machine Learning: Concept and Applications. *ACM Transactions on Intelligent Systems and Technology* 10, 2 (2019), 12:1-12:19. https://doi.org/10.1145/3298981

Xin Yi, Ekta Walia, and Paul Babyn. 2019. Generative adversarial network in medical imaging: A review. *Medical Image Analysis* 58 (2019), 101552. https://doi.org/10.1016/j.media.2019.101552

Jinsung Yoon, Lydia N. Drumright, and Mihaela van der Schaar. 2020. Anonymization Through Data Synthesis Using Generative Adversarial Networks (ADS-GAN). *IEEE Journal of Biomedical and Health Informatics* 24, 8 (2020), 2378-2388. https://doi.org/10.1109/JBHI.2020.2980262

Alice C. Yu, Bahram Mohajer, and John Eng. 2022. External Validation of Deep Learning Algorithms for Radiologic Diagnosis: A Systematic Review. *Radiology: Artificial Intelligence* 4, 3 (2022), e210064. https://doi.org/10.1148/ryai.210064

Lequan Yu, Shujun Wang, Xiaomeng Li, Chi-Wing Fu, and Pheng-Ann Heng. 2019. Uncertainty-Aware Self-ensembling Model for Semi-supervised 3D Left Atrium Segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI) (LNCS, Vol. 11765)*. 605-613.

John R. Zech, Marcus A. Badgeley, Manway Liu, Anthony B. Costa, Joseph J. Titano, and Eric Karl Oermann. 2018. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLoS Medicine* 15, 11 (2018), e1002683. https://doi.org/10.1371/journal.pmed.1002683

Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. 2018. mixup: Beyond Empirical Risk Minimization. In *International Conference on Learning Representations (ICLR)*.

Haoran Zhang, Natalie Dullerud, Laleh Seyyed-Kalantari, Quaid Morris, Shalmali Joshi, and Marzyeh Ghassemi. 2021. An empirical framework for domain generalization in clinical settings. In *Proceedings of the ACM Conference on Health, Inference, and Learning (CHIL)*. 279-290. https://doi.org/10.1145/3450439.3451878

Xi Sheryl Zhang, Fengyi Tang, Hiroko H. Dodge, Jiayu Zhou, and Fei Wang. 2019. MetaPred: Meta-Learning for Clinical Risk Prediction with Limited Patient Electronic Health Records. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 2487-2495. https://doi.org/10.1145/3292500.3330779

Yuhao Zhang, Hang Jiang, Yasuhide Miura, Christopher D. Manning, and Curtis P. Langlotz. 2022. Contrastive Learning of Medical Visual Representations from Paired Images and Text. In *Proceedings of the 7th Machine Learning for Healthcare Conference (MLHC) (PMLR, Vol. 182)*.

Ying Zhang and Chen Ling. 2018. A Strategy to Apply Machine Learning to Small Datasets in Materials Science. *npj Computational Materials* 4 (2018), 25. https://doi.org/10.1038/s41524-018-0081-z

Ziqi Zhang, Chao Yan, Diego A. Mesa, Jimeng Sun, and Bradley A. Malin. 2020. Ensuring electronic medical record simulation through better training, modeling, and evaluation. *Journal of the American Medical Informatics Association* 27, 1 (2020), 99-108. https://doi.org/10.1093/jamia/ocz161

Amy Zhao, Guha Balakrishnan, Frédo Durand, John V. Guttag, and Adrian V. Dalca. 2019. Data augmentation using learned transformations for one-shot medical image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 8543-8553.

Yue Zhao, Meng Li, Liangzhen Lai, Naveen Suda, Damon Civin, and Vikas Chandra. 2018. Federated Learning with Non-IID Data. *arXiv preprint arXiv:1806.00582* (2018).

Yukun Zhou, Mark A. Chia, Siegfried K. Wagner, Murat S. Ayhan, Dominic J. Williamson, Robbert R. Struyven, et al. 2023. A foundation model for generalizable disease detection from retinal images. *Nature* 622 (2023), 156-163. https://doi.org/10.1038/s41586-023-06555-x

Zongwei Zhou, Vatsal Sodha, Jiaxuan Pang, Michael B. Gotway, and Jianming Liang. 2021. Models Genesis. *Medical Image Analysis* 67 (2021), 101840. https://doi.org/10.1016/j.media.2020.101840

Ligeng Zhu, Zhijian Liu, and Song Han. 2019. Deep Leakage from Gradients. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32.

Alexander Ziller, Dmitrii Usynin, Rickmer Braren, Marcus Makowski, Daniel Rueckert, and Georgios Kaissis. 2021. Medical imaging deep learning with differential privacy. *Scientific Reports* 11 (2021), 13524. https://doi.org/10.1038/s41598-021-93030-0